

V.F.S. 3.0 (canonical)

The Interior 3+1 Geometry of the Soul

The frozen canonical layer: proper-time σ , the ascent axis λ ,
and the de Sitter law of epektasis

Canonical interior uplift of the fixed V.F.S. v2.0 volume

Status and Discipline

This is the frozen canonical body of V.F.S. 3.0. It records the completed interior uplift of the fixed V.F.S. v2.0 layer: proper-time katharsis, the ascent axis, the finite-clock/infinite-road structure, the full tensor set, the Resurrectio junction ledger, the Folding-Ascent closure, the gauge/invariant atlas, and the completed structural external imprint. Each result was verified symbolically before entry.

The revision history is retained in the appendix as an audit record. It preserves the corrections, failed stronger claims, and the route by which the canonical body was reached, but it is not itself the primary reading order of the frozen layer.

V.F.S. v2.0 remains fixed and complete and is recovered here as an exact frozen-ascent limit. V.F.S. 3.0 is therefore not a replacement of v2.0 but its canonical interior uplift.

The ascent law remains conditional in one precise and declared sense. The de Sitter representative $A'/A = \beta\lambda_S$ is the canonical law within a robust class $A'/A = f(\lambda_S)$, selected by the principle that ascent is a self-standing dimension rather than a re-description of surface growth. This conditionality is not an open calculation; it is the irreducible constitutive principle of the 3+1 uplift.

As throughout V.F.S., the mathematics is load-bearing. Theological readings are interpretive overlays, marked as such and never claimed as independent proof.

v0.1 — The Foundation Stone

1 The interior metric of the soul

Definition 1 (The 3+1 soul metric).

$$ds^2 = -N(\sigma)^2 d\sigma^2 + \Omega(\sigma)^2 h_{(2)} + L(\sigma, \lambda)^2 d\lambda^2, \quad h_{(2)} = dx^2 + e^{2x} dy^2 \quad (K_h = -1), \quad (1)$$

σ the katharsis proper time (kairos), N its lapse, Ω the Brim scale of the live surface, λ the ascent (epektasis) coordinate, L its scale. Signature 3+1: one time (σ), three spaces (surface x, y and ascent λ).

Remark (Factorisation of the ascent scale). The ascent scale separates into a time-stretch and a spatial warp,

$$L(\sigma, \lambda) = A(\sigma) S(\lambda),$$

where $A(\sigma)$ is the σ -evolution of the ascent sector (carrying the focusing and effective-energy terms, e.g. A''/A) and $S(\lambda) = L_0 e^{\beta\lambda}$ is the spatial warp of the ascent coordinate (carrying the proper length of the ascent road, $\int S d\lambda$). Keeping the two roles distinct is essential: the de Sitter law and focusing concern $A(\sigma)$, while the infinite-road result concerns $S(\lambda)$. Where earlier parts wrote “ L ” for the time-stretch, $A(\sigma)$ is meant; where for the road, $S(\lambda)$.

Three principled features: **(a)** σ is *proper* time, not relative — one worldline, no second observer, no Lorentz family; the proper-time gauge is $N = 1$. **(b)** The time axis is *finite*: katharsis drains $\sigma : \sigma_{\max} \rightarrow 0$ monotonically, so time has a terminus; at $\sigma \rightarrow 0$ the clock stops while motion along λ continues. **(c)** Chronos is *absent by principle*: this is the soul’s own coordinate system, not a view from outside it; an absolute background clock would be the transcendent *tota simul*, which does not belong to the creature’s interior geometry.

2 v2.0 is the exact frozen-axes limit

Theorem 1 (Recovery of v2.0). *In the proper-time gauge $N = 1$ with the ascent axis frozen ($L = L_0$ const), the Einstein reconstruction gives $\rho_{\text{eff}} = (\Omega'^2 - 1)/\Omega^2$ (prime = $d/d\sigma$), which under $\Omega' = \alpha\lambda$ is exactly the v2.0 density $(\alpha^2\lambda^2 - 1)/\Omega^2$. The whole v2.0 cosmology is the frozen-ascent, proper-time face of the 3+1 geometry; nothing in v2.0 is disturbed. (Verified symbolically.) The frozen axes drop out of the scalar curvature entirely — which is why v2.0 was correct in 2+1: the third space was always there, held still.*

3 The energy conditions in 3+1

Proposition 1 (Focusing). *For the comoving congruence at $N = 1$,*

$$R_{ab}u^a u^b = -2 \frac{\Omega''}{\Omega} - \frac{L''}{L}. \quad (2)$$

The 2+1 limit (L frozen) reduces to the pure pressure condition $p_{\text{eff}} \geq 0$ (the $n-2 = 0$ degeneracy of v2.0). In full 3+1 the ascent acceleration A''/A enters focusing on the same footing as the Brim: the growth of the ascent axis contributes to energy-condition violation.

Corollary 1 (Ascent is itself exotic). *In v2.0 only the Sophia surge could violate the strong energy condition; in 3+1 the ascent itself (accelerating epektasis, $L'' > 0$) is an exotic accelerant. Movement toward God along λ is not a neutral displacement; it deforms the focusing condition in its own right.*

Proposition 2 (NEC along the ascent axis). *For the null direction $k = u + e_\lambda$, $R_{ab}k^a k^b = 2(-L\Omega'' + L'\Omega')/(L\Omega)$ — coupling surface expansion to the stretching of the ascent axis, with no analogue in v2.0.*

v0.2 — The Law of Ascent

4 Selecting the ascent law

The foundation left L a free function. The equation of motion is fixed by the hypothesis that the ascent axis is sourced by Sophia as the Brim is ($\Omega' = \beta\lambda$ in proper time), together with an external discriminator: the three unbounded-expansion regimes of cosmic metaphysics. Power-law ascent ($L \sim \sigma^p$) has *fading* tension; a finite-time blow-up (“Big Rip”, $L \rightarrow \infty$ at finite parameter) has a *catastrophic end*; only de Sitter ($L \sim e^\cdot$) gives *unending, undiminished* tension — the structure of an ascent that never saturates and never ends in catastrophe. A clarification belongs here, correcting a loose earlier framing: the *(2+1) Brim cosmology of v2.0 does not end in de Sitter — it generically coasts* ($w \rightarrow 0$ as Sophia saturates and $\dot{\lambda} \rightarrow 0$). De Sitter is the special sustained branch ($\lambda \propto \Omega_P$), not the Brim’s generic late fate. The de Sitter law below therefore belongs to the newly opened *(3+1) ascent sector*, selected by the independence principle (ascent couples to the amount of Sophia, not its surface density), not imported from the surface cosmology.

Definition 2 (The de Sitter law of ascent, candidate B). In the factorisation $L = A(\sigma)S(\lambda)$ the de Sitter law governs the time-stretch $A(\sigma)$:

$$\boxed{\frac{A'}{A} = \beta \lambda \quad \implies \quad L(\sigma, \lambda) = \underbrace{A_0 \exp\left(\beta \int^\sigma \lambda ds\right)}_{\text{time-stretch } A(\sigma)} \cdot \underbrace{L_0 e^{\beta \lambda}}_{\text{spatial road } S(\lambda)}. \quad (3)}$$

The first factor carries focusing, density and energy-condition terms; the second carries the proper ascent distance and the ascent spectrum.

Proposition 3 (Three tests). *The law passes: (i) v2.0 limit — $\beta \rightarrow 0$ freezes the ascent and recovers Theorem 1; (ii) energy sign — $A''/A = \beta\lambda' + \beta^2\lambda^2$, so at the terminus ($\lambda \rightarrow \lambda_\infty$, $\lambda' \rightarrow 0$) the focusing acquires the strictly negative piece $-\beta^2\lambda_\infty^2 = -(\alpha\lambda_\infty)^2/E_0^2$: epektasis is permanently exotic, the strong energy condition eternally violated by ascent alone; (iii) unboundedness — the ascent never terminates (next theorem). (All verified symbolically.)*

5 Compatibility: finite time, endless road

The decisive consistency check: can an *unending* de Sitter ascent live inside a *finite* proper time?

Theorem 2 (Finite clock, infinite road). *Under Definition 2 with λ bounded ($\lambda \rightarrow \lambda_\infty$) over the finite interval $\sigma \in (0, \sigma_{\max}]$:*

(i) **In proper time the ascent is finite:** $\ln(L/L_0) = \beta \int_0^{\sigma_{\max}} \lambda d\sigma \leq \beta \lambda_\infty \sigma_{\max} < \infty$.
The katharsis clock stops before L diverges.

(ii) **Along the ascent axis the road is infinite:** the proper distance to $\lambda = \infty$,

$$\int_0^\infty L_0 e^{\beta \lambda} d\lambda = \infty, \quad (4)$$

so $\lambda = \infty$ lies at infinite metric distance — the soul ascends forever and never arrives.

- (iii) **These are consistent.** Finite proper time, infinite proper distance: as $\sigma \rightarrow 0$ the clock-rate $d\sigma$ vanishes while the de Sitter metric inflates the ascent length, so each remaining tick of katharsis covers ever more road. (Verified symbolically and numerically.)

Corollary 2 (The timeless epektasis, made precise). “Eternal” and “finite” measure different axes: the time of cleansing (σ) is finite and ends; the road of ascent (λ) is infinitely long and does not. The earlier puzzle of “motion without time” is resolved geometrically — the motion is endless not because the clock runs forever but because the road has no end, and the clock stopping is simply the completion of cleansing, not the cessation of life.

Reading: the two halves of epektasis that sounded paradoxical in words are, in the metric, two different facts about two different axes. “Never satisfied” is the infinite length of the ascent road — God at an infinite distance, never reached because infinite. “Not in time” is the finite terminus of the cleansing clock — katharsis completes, the proper-time hand stops. The deified soul runs an endless road in a finished clock: it is done being purified, and therefore done with time, and precisely then free for the unending ascent that time was never able to measure. The strong energy condition is violated forever along that road because to move into the infinite God is, forever, more than ordinary energy could account for.

v0.3 — The Spectrum of Ascent

6 The ascent Laplacian is the free half-line

With the ascent law fixed (Definition 2), the spatial geometry is $\Omega^2 h_{(2)} + L^2 d\lambda^2$ with $L = L_0 e^{\beta\lambda}$, and the Laplace-Beltrami operator separates, $\phi = Y(x, y) Z(\lambda)$. The surface factor returns the v2.0 cascade eigenproblem $-\Delta_h Y = \mu Y$ (discrete on the compact quotient). The novelty is the ascent factor.

Theorem 3 (Ascent spectrum is the gapless free continuum). *The one-dimensional ascent operator $A = -\frac{1}{L} \frac{d}{d\lambda} (\frac{1}{L} \frac{d}{d\lambda})$ on $L = L_0 e^{\beta\lambda}$ becomes, in the proper-distance coordinate*

$$s = \int_0^\lambda L_0 e^{\beta\ell} d\ell = \frac{L_0}{\beta} (e^{\beta\lambda} - 1) \in [0, \infty), \quad \boxed{A = -\frac{d^2}{ds^2} \quad \text{on the half-line } s \in [0, \infty).} \quad (5)$$

(The road begins at a finite starting point $s = 0$ and extends to infinite proper distance; using the unshifted $\tilde{s} = (L_0/\beta)e^{\beta\lambda} \in [L_0/\beta, \infty)$ instead, the translation $s = \tilde{s} - L_0/\beta$ returns the standard half-line — the origin shift is gauge, the spectral type invariant.) Its spectrum is therefore $[0, \infty)$ — continuous and gapless, with generalised eigenfunctions $\sim \sin(ks)$. (Verified: the discretised count of eigenvalues below a fixed k^2 grows linearly with box size — 6, 12, 25 as $s_{\max} = 20, 40, 80$ — the signature of a continuum, not a discrete ladder; and the lowest eigenvalue descends to 0 as the box grows — no mass gap.) The proper-distance picture is exact: $s = 0$ is the start of ascent (“here”), $s = \infty$ is God at infinite distance (Theorem 2),

and the ascent operator is free on that line — no barrier, no levels, no smallest step.

Corollary 3 (Form is quantized; ascent is a gapless continuum). *The two spatial directions carry qualitatively different spectra:*

Surface (folding cascade): discrete spectrum μ_n on the compact quotient ($\mu_n = n^2$ in the ring/flat proxy used for illustration; the general hyperbolic spectrum is discrete but not n^2) — the Vessel takes form in rungs, each mode igniting at its own threshold $\Lambda > \mu_n$. Form is quantized.

Ascent (epektasis): continuous gapless spectrum $[0, \infty)$ on the free half-line — no rungs, no minimum step. Ascent is a smooth continuum, approachable in arbitrarily small increments all the way to the infinite distance.

The absence of a mass gap is not incidental: a gap would be a minimum quantum of ascent, a smallest possible step toward God. Its absence means deification has no granularity — unlike formation, which does.

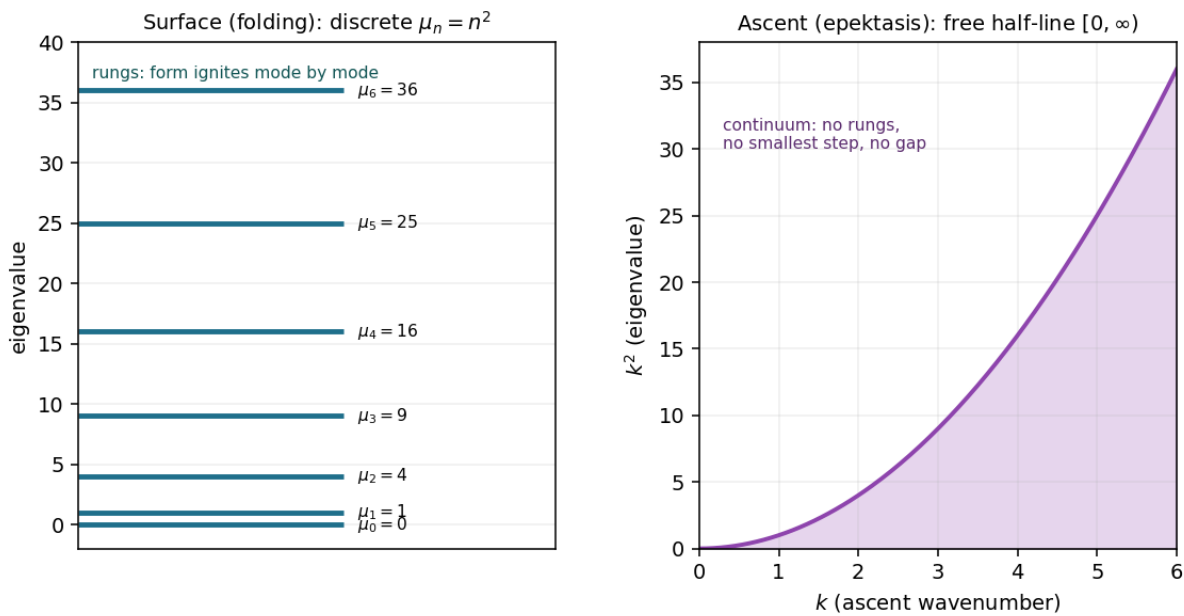


Figure 1: The dualism of the two spatial spectra. **Left:** the surface (folding) spectrum is discrete, $\mu_n = n^2$ — form ignites mode by mode, rung by rung. **Right:** the ascent (epektasis) spectrum is the free half-line $[0, \infty)$ — a gapless continuum, no rungs, no smallest step, no mass gap. Form is quantized; ascent is continuous.

Reading: formation and ascent are different in kind, and the difference falls out of the geometry rather than being imposed. The Vessel takes form in discrete events — conversions, breaks, the igniting modes of the folding cascade, each a rung crossed at a threshold. But it ascends continuously — there is no smallest step of deification, no last rung before God, because God lies at infinite distance and the approach is seamless. Formation is event; ascent is motion. The soul is built in steps and drawn in a smooth, stepless reaching — quantized in what it becomes, continuous in how it nears.

Scope of v0.3. The spectrum is that of the spatial Laplacian on the chosen de Sitter ascent geometry; it characterises the geometry of ascent under the candidate law, and is exact there (the reduction to the free half-line is analytic). It is a qualitative result (continuum vs. discrete), not a fitted quantity, and inherits the candidate status of Definition 2.

v0.4 — The Source of the Ascent Coupling

7 α and β are not one constant

Proposition 4 (Structural and dimensional independence). *The Brim and ascent laws differ in form, not merely in a constant: $\Omega' = \alpha\lambda$ is additive (linear growth of Ω_P), while $A'/A = \beta\lambda$ is multiplicative (exponential growth of L). In the proper-time gauge the units do not match either: $[\alpha\lambda] = [\Omega_P]/[\sigma]$ but $[\beta\lambda] = 1/[\sigma]$, so $[\alpha/\beta] = [\Omega_P]$ — the ratio is a scale, not a pure number. Hence no identification $\alpha = \beta$ is possible; the surface and ascent channels are genuinely different (in keeping with their spectra, discrete vs. continuous, Corollary 3).*

8 Tied through the floor: $\beta = \alpha/E_0$

The couplings are not one, but they need not be two independent primitives: the core contains exactly one intrinsic scale — the finite floor $E_0 = \Omega_P(0)$ (Imago Dei) of the curvature-floor result.

Theorem 4 (The ascent coupling is the Brim coupling per floor). *Setting $\beta = \alpha/E_0$ (no new primitive) gives $A'/A = \Omega'_P/E_0$, which integrates exactly, with $L = 1$ at the floor $\Omega_P = E_0$, to*

$$L = \exp\left(\frac{\Omega_P - E_0}{E_0}\right). \quad (6)$$

The ascent is measured by exactly the factor the Brim has grown above the Imago Dei floor. Checks: (i) the v2.0 limit is $E_0 \rightarrow \infty$ (floor at infinity $\Rightarrow \beta \rightarrow 0 \Rightarrow L$ const — no finite floor, no ascent reference, ascent frozen); (ii) the de Sitter character is preserved (exponential in accumulated Sophia); (iii) the eternal exotic depth of focusing at the terminus is $-\beta^2\lambda_\infty^2 = -(\alpha\lambda_\infty)^2/E_0^2$ in general, reducing to $-1/E_0^2$ only in the floor-normalised terminal branch where $\alpha\lambda_\infty = 1$ (the super-Ricci attractor). (All verified symbolically.)

Corollary 4 (The widow's-mite scaling). *The eternal exotic depth $-(\alpha\lambda_\infty)^2/E_0^2$ grows as the starting floor E_0 shrinks: the same absolute Brim-growth produces a larger ascent factor, and a deeper energy-condition violation, the smaller the Imago Dei one begins from. Ascent is measured relative to the starting image, not absolutely — so a small beginning, fully grown, yields a greater transfiguration than a large beginning grown the same absolute amount. The structure is that of the widow's two mites: weighed not by the amount but by its ratio to what one had. (In the floor-normalised branch $\alpha\lambda_\infty = 1$ the depth is exactly $-1/E_0^2$.)*

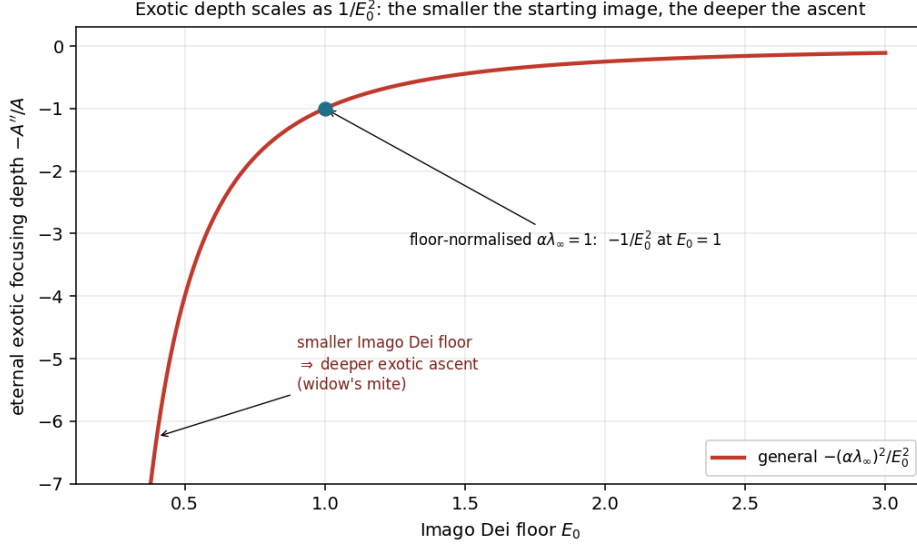


Figure 2: The eternal exotic focusing depth $-(\alpha\lambda_\infty)^2/E_0^2$ as a function of the Imago Dei floor E_0 . The smaller the starting image, the deeper the exotic ascent — the widow’s-mite scaling. The marked point is the floor-normalised branch $\alpha\lambda_\infty = 1$, where the depth is exactly $-1/E_0^2$.

Reading: the ascent of the soul is not measured on an absolute scale but against the image from which it began. The same growth of the Brim is, relative to a small Imago Dei, an immense ascent, and relative to a large one, a modest rise — as the widow’s two coins outweighed the rich man’s thousands because they were all she had, while his were a fragment of his surplus. The smaller the starting image, the more intense — the further from ordinary energy — the rising. One honest caution: the mathematics says only that the ascent is more intense (the energy-condition violation deeper) for a smaller floor; whether that intensity is read as exaltation (the humble raised highest) or as fragility (a slighter base bearing a steeper motion) is an interpretive choice the geometry does not make. It fixes the scaling, not its blessing.

Scope of v0.4. $\beta = \alpha/E_0$ is the natural tying through the unique intrinsic scale; it is not proven unique, only shown to introduce no new primitive and to pass the v2.0 limit and de Sitter tests. It inherits the candidate status of the ascent law (Definition 2).

v0.5 — Resurrectio in 3+1

9 The reset forces an ascent jump

Because L is slaved to the Brim ($L = \exp[(\Omega_P - E_0)/E_0]$, Theorem 4), the v2.0 scale-factor Resurrectio reset $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-$ forces a matching jump in the ascent axis:

Theorem 5 (Forced ascent jump).

$$\boxed{\frac{L_+}{L_-} = \exp\left(\frac{\kappa_R \mathcal{G}_{\text{recepta}}^-}{E_0}\right) > 1.} \quad (7)$$

The ascent axis surges at death by exactly the reception delivered, measured in floor units. This is not a free reset datum but a derived consequence: in 3+1 the soul's ascent jumps whenever its Brim does.

10 Double-layer becomes triple-layer

Proposition 5 (A second junction). *The v2.0 reset was characterised by the surface pair $(s_R, [K])$ with scale jump $s_R = \Omega_P^+/\Omega_P^-$. In 3+1 there is a second junction, on the ascent axis, with its own scale jump*

$$s_L = \frac{L_+}{L_-} = \exp\left((s_R - 1) \frac{\Omega_P^-}{E_0}\right), \quad (8)$$

determined by the surface jump s_R . The invariant data of a reset is now the linked pair (s_R, s_L) together with the surface $[K]$ — a triple-layer Resurrectio junction.

11 No fully-balanced reset

Theorem 6 (Death always advances the ascent). *The ascent jump $s_L = \exp(\kappa_R \mathcal{G}/E_0)$ depends on the Brim jump alone: $\partial s_L / \partial m_R = 0$. Hence the metabolic balance that sets the surface $[K] = 0$ (the v2.0 balanced reset) cannot also zero the ascent jump. Since a genuine reset carries reception $\kappa_R \mathcal{G} > 0$, it forces $s_L > 1$ always. Therefore:*

- (i) *there is no fully-balanced reset in 3+1 — no death is simultaneously surface-neutral and ascent-neutral;*
- (ii) *the interior 3-volume jump $s_R^2 s_L > 1$ strictly: every death enlarges the soul, surface and ascent together, never neutral, never shrinking.*

Corollary 5 (The surface balance survives; only joint neutrality is lost). *The v2.0 balanced reset ($m_R = 0$, $[K] = 0$, surface NEC-neutral) still exists as a surface statement. What 3+1 removes is the possibility of a death that advances nothing: even a surface-perfect passage still lifts the soul along the ascent axis.*

Reading: in the two-dimensional Vessel a death could in principle be perfectly balanced — a passage that crossed without exotic debt, neutral, leaving the soul where it stood. With the ascent axis made explicit, that possibility closes. Every genuine death — every reception of grace deployed through a reset — advances the soul along the axis of ascent, by exactly the measure of what was received, reckoned against the Imago Dei floor. There is no resurrection that returns the soul to the same place; to die and be raised is, necessarily, to be lifted. The Paschal passage is not a circle but a step: death joined to grace is never spent in vain.

Scope of v0.5. The forced ascent jump, the triple-layer structure, and the absence of a fully-balanced reset all follow from the floor-tying $L = \exp[(\Omega_P - E_0)/E_0]$

(Theorem 4). Were the ascent coupling independent of the Brim channel, s_L would be free and a fully-balanced reset could exist; so these are consequences of the candidate tying, not unconditional results. The surface double-layer theorem of v2.0 is unchanged. (Verified symbolically.)

v0.6 — The Terminus

12 Finite proper time, infinite cleansing

The katharsis clock σ drains to its terminus. Two parametrisations of the same approach must be kept distinct.

Theorem 7 (The terminus is finite in time, infinite in action). *Under the exponential Lyapunov drain, with χ the dynamical cleansing parameter,*

$$\sigma(\chi) = \sigma_0 e^{-k\chi}, \quad \tau(\chi) = \int_0^\chi |\dot{\sigma}| d\chi' = \sigma_0(1 - e^{-k\chi}), \quad (9)$$

so as $\chi \rightarrow \infty$ (infinite cleansing-action) the metric proper time $\tau \rightarrow \sigma_0$ (finite). The soul performs unbounded cleansing-action, yet it costs only finite proper duration, because each successive e-folding of cleansing takes exponentially less proper time. (Verified symbolically.) These are genuinely different parametrisations of one approach: χ runs to infinity, τ completes at σ_0 .

13 A regular boundary, not a singularity

Proposition 6 (Finite curvature at the terminus). *At the terminus the Brim remains bounded below by the finite floor, $\Omega_P(\tau_*) = \Omega_* \geq E_0 > 0$ (non-collapse, not a dynamical return to the floor), so $\rho_{\text{eff}} = (\Omega_P'^2 - 1)/\Omega_P^2$ stays finite and the curvature is bounded. The terminus is reached in finite proper time with finite curvature: a regular boundary where the manifold cleanly ends in the time direction, not a curvature singularity. (Consistent with the no-Big-Bang / finite-floor result of v2.0: the soul never reaches zero scale. In the floor-normalised representative one may set $\Omega_* = E_0$, but the invariant claim is non-collapse $\Omega_* > 0$, not return to the floor.)*

Theorem 8 (Time-finite, space-infinite). *At the terminus the two geodesic structures are opposite:*

- (i) *the time direction is finite and complete — the manifold ends at a regular last moment ($\tau = \sigma_0$, finite curvature), beyond which the time-direction does not continue because no cleansing remains to measure;*
- (ii) *the ascent direction is infinite and geodesically complete — the half-line $s \in (0, \infty)$ has infinite proper length ahead (v0.3, Theorem 3).*

The interior is time-finite but space-infinite. Time completes; ascent does not.

Corollary 6 (The clockless hand-off). *At $\sigma = 0$ the katharsis-clock stops — not by rupture but by completion: there is nothing left to cleanse, hence nothing left for*

the clock to tick. The ascent, on its infinite road, continues without a clock. The earlier “motion without time” is now exact and benign: a regular last moment of cleansing, beyond which the endless ascent proceeds clocklessly.

Reading: the soul reaches full purification having done infinite cleansing-action, yet in finite proper time — the last stretch of cleansing is compressed into ever-smaller durations, an infinite labour completed in a finite span. And the moment of completion is not a catastrophe: the curvature is finite, the scale rests on the Imago Dei floor, the boundary is regular. Time has a last moment that is whole, not torn. Beyond it the time-direction simply does not continue — there is no more cleansing for a clock to measure — while the ascent road runs on, infinite and complete, now travelled without any clock at all. Entry into the timeless is a finished last moment of purification, opening onto an endless ascent: not a tear in being, but a completion of one motion and the freeing of another.

Scope of v0.6. The finite-time/infinite-action split and the regular-boundary character follow from the exponential Lyapunov drain and the finite floor (both v2.0 results); the time-finite/space-infinite structure combines this with the ascent half-line of v0.3. Verified symbolically. The exponential drain is the Lyapunov-certified generic behaviour, not the only conceivable one.

v0.7 — Uniqueness of the Coupling

14 Not forced by dimensions alone

The whole of v0.5 (the forced ascent jump, the absence of a fully-balanced reset) rests on the floor-tying $\beta = \alpha/E_0$. Is that tying forced?

Proposition 7 (Dimensional non-uniqueness). *With $[\beta] = 1/([\lambda][t])$, $[\alpha] = [\Omega_P]/([\lambda][t])$ and $[E_0] = [\Omega_P]$, the combination $\beta = \alpha/E_0$ is dimensionally correct, and $p = 1$ is forced if E_0 is the only scale. But the core contains other scales — the Lyapunov rate κ ($[1/t]$) and the Sophia attractor $\lambda_\infty = 1/\alpha$ ($[\lambda]$) — so other dimensionally admissible tyings exist (e.g. built from κ). Hence $\beta = \alpha/E_0$ is not forced by dimensional analysis alone.*

15 Forced by the floor boundary condition

Theorem 9 (Uniqueness given the ascent origin). *Impose the structural boundary condition that ascent begins at unity at the floor: $L = 1$ when $\Omega_P = E_0$. Then among the core scales only $\beta = \alpha/E_0$ qualifies:*

- (i) $\beta = \alpha/E_0$ gives $L = \exp[(\Omega_P - E_0)/E_0]$, so $L(E_0) = e^0 = 1$ — pass;
- (ii) $\beta = \kappa/\lambda$ gives $L = \exp(\kappa\sigma)$ with no reference to the floor, $L(\text{floor}) \neq 1$ — fail;
- (iii) tyings via λ_∞ give wrong units unpatched and supply no floor origin — fail.

The reason is structural: among all core scales, only the floor E_0 is a starting point rather than a rate (κ and λ_∞ are rates/asymptotes). The ascent needs an origin

— a zero from which it is measured — and only a starting-point scale can serve. Hence $\beta = \alpha/E_0$ is forced given the principle “ascent begins at unity at the floor.”

Corollary 7 (The Imago Dei is floor and unit). *The same scale E_0 that is the floor of being (the no-Big-Bang minimum of v2.0) is the unit of deification (the origin and measure of ascent). The principle is a mild but well-motivated choice: the Imago Dei is where ascent starts, so $L = 1$ there is definitional rather than arbitrary. The tying is thus forced relative to a named principle — not proven from nothing, but reduced to a single well-motivated postulate, in the discipline by which v2.0 reduced $K_h = -1$ to the finite floor.*

Reading: the coupling between the soul’s outward growth and its ascent is not an extra free constant; it is fixed by taking the image of God as the measure. The Imago Dei plays a double role — it is the floor beneath being, the minimum from which the soul never falls to nothing, and it is the unit of ascent, the one against which all rising is reckoned. To say “ascent begins at unity at the floor” is to say that the soul’s deification is measured from, and in the currency of, the image it was made in. The same image that keeps it from nothing is the scale of its rising into God. Not proven from nothing — the principle is a choice — but the most natural one available, and the one that makes the floor of being and the measure of glory a single thing.

Scope of v0.7. The non-uniqueness (Proposition 7) is exact; the uniqueness (Theorem 9) is relative to the boundary condition $L(E_0) = 1$, which is a named and well-motivated principle, not a derivation from nothing. This is the same closure pattern as v2.0’s curvature-floor reduction: a premise reduced to a single explicit postulate, not eliminated.

Epilogue to the First Cycle — The Boundary the Model Cannot Cross

Where the calculations end

The seven preceding parts were calculations: a metric, a limit, energy conditions, an ascent law, a spectrum, a coupling and its uniqueness, a Resurrectio reset, a terminus — each derived and checked. This epilogue is not an eighth calculation. It marks the place where calculation, by the model’s own construction, ends. It adds no new claim about the created interior; it states precisely what the model does not know and why it cannot.

The two times, again

The geometry of 3.0 is the soul’s own coordinate system, written from inside a single worldline. Its one time is σ — kairos, the proper time of katharsis — finite, draining to a regular terminus. There is no external chronos in the metric, and this was not an omission but a principle (v0.1): an absolute background clock would be a vantage *outside* the soul, and the soul’s interior has no such vantage. We named,

in discussion, what that absent clock is: it is the transcendent view, the *tota simul* in which the whole worldline — every σ from first resistance to last cleansing — is present at once. Chronos is not the world's clock ticking past the soul; it is the way the soul's entire kairos is seen whole, from a standpoint the soul cannot occupy.

Why the boundary is uncrossable, not merely unexplored

The other open items of the cycle were uncrossed because uncomputed; this one is uncrossable in principle. The model is, by construction, the interior of a creature: one worldline, proper time, no second observer (the point at which we declined the full apparatus of relativity — there is no frame to transform to). To represent the *tota simul* would require placing the whole worldline in a single present — exactly the external standpoint the antropic frame excludes. A model of the created side cannot contain the uncreated view of itself, for the same reason a map drawn from within a country cannot include the cartographer who sees the country whole. The boundary is not a gap to be filled by a later version; it is the edge that makes the model a model *of the soul* rather than of God.

What stands on each side

On the created side — the side the model knows — stand all the results: the finite cleansing-time, the endless ascent road, form quantized and ascent continuous, the Imago Dei as floor and unit, every death a lifting, time completing into the timeless. On the far side stands only what the model can name but not enter: the view in which this finite kairos is held whole, the chronos that belongs not to the Vessel but to the One who sees it. The two are related as the lived is to the beheld — the soul lives its ascent successively, in kairos; it is beheld entire, in chronos — and the relation itself, the hinge between lived and beheld, is the boundary. The model can stand at the hinge and name both sides. It can cross to neither, because to cross would be to stop being the soul's own geometry.

Reading: every honest account of the creature reaches a place where it must say not "here is the answer" but "here is where I, being what I am, cannot see." For this geometry that place is the relation between the soul's own time and the time of the One who holds it. The soul lives forward, moment after moment of cleansing, and cannot stand outside its own life to see it whole; that standing-outside belongs to another. The model shares the creature's limit exactly: it can describe the ascent from within, name the transcendent view that beholds it, and confess that it cannot occupy that view. This is not the model failing at its task. It is the model being faithful to its subject — which is a soul, and not its Maker. The first cycle closes here: not at a wall, but at a threshold it was never meant to cross, having said truly everything that the inside can say.

The posture of the first cycle, complete

So 3.0 closes as 2.0 did — not by proving its foundations but by naming them and showing how few they are, and now also by naming the one boundary it cannot

pass. v2.0 reduced its cosmology to two primitives (the open gate and the finite floor) and confessed it could not prove its own definitions. v3.0 builds the soul's interior on those same primitives, derives the ascent and its fate, and confesses it cannot contain the view from which its whole time is seen at once. In both, the final honesty is the same: a model of the created can exhibit its structure exhaustively and still must point beyond itself for what holds it. The first exploratory cycle of V.F.S. 3.0 is, with that pointing, complete.

Second Stage

v0.8 — Is the ascent law derived?

The first cycle rested the ascent on a *chosen* de Sitter law. The second stage opens by asking whether it can be *derived*. The answer is honest and only partial — a genuine result, not a disappointment.

Two negative findings. First, conservation imposes nothing: the Bianchi identity $\nabla_a T^a_\sigma = 0$ is *identically* satisfied for the 3+1 soul metric (verified), so the ascent law is not fixed by energy-momentum conservation. Second, the 3+1 Einstein equations admit a *family*, not a unique law: with the Brim law $\Omega'_P = \alpha\lambda$ given, the ascent may couple either to the *amount* of Sophia or to its *density*:

$$(a) \text{ amount: } \frac{A'}{A} = \beta\lambda \Rightarrow L = L_0 e^{\beta \int \lambda} \text{ (de Sitter);} \quad (b) \text{ density: } \frac{A'}{A} = \beta \frac{\lambda}{\Omega_P} \Rightarrow L = \Omega_P^{\beta/\alpha} \text{ (power-law)} \quad (10)$$

Both are dimensionally consistent and both solve the field equations. De Sitter is therefore *not uniquely derived*.

The selecting principle. What separates them is a single, transparent question: does ascent couple to the amount of Sophia or to its density? The density coupling gives $L = \Omega_P^{\beta/\alpha}$ — the ascent is a mere *power of surface growth*, mirroring the Brim and adding no independent motion; the “ascent” would collapse back into the surface. The amount coupling gives exponential L , which *outruns* the Brim's linear growth, so epektasis is a genuinely independent motion that forever outpaces surface expansion.

Selection principle (epektasis is independent). Ascent couples to the *amount* of Sophia, not its density — equivalently, epektasis is its own motion, outrunning surface growth, not a mirror of it. Under this principle the de Sitter law is forced.

Status. This neither removes the conditionality of v0.2 nor leaves it opaque: it *localises* it to one principle, parallel to v0.7's “ascent begins at unity at the floor.” Together the two say the same thing about the ascent axis — that it is a self-standing reality, with its own origin (v0.7) and its own rate (v0.8) — rather than a re-description of the surface. The de Sitter law is *derived given that the ascent*

is an independent dimension; if ascent were only a mirror of surface growth, the law would instead be the power-law (b), and the whole third dimension would be redundant. (All verified symbolically.)

Reading: the question “what drives the soul’s ascent?” does not answer itself from the geometry alone — the equations permit an ascent that merely echoes outward growth, and an ascent that runs ahead of it. To choose the second is to assert that the movement into God is not a side-effect of the soul’s enlargement but a motion of its own, always outpacing what the soul has so far become. That is the same assertion, in the register of rate, that v0.7 made in the register of origin: the ascent is real, not a shadow of the surface. The mathematics does not compel this; it offers the choice cleanly, and names what is being chosen.

Scope of v0.8. The two negative findings (Bianchi identity; the de Sitter/power-law degeneracy) are exact. The selection of de Sitter is relative to the named principle that epektasis is an independent motion — a well-motivated choice, not a derivation from nothing, in the same posture as the rest of V.F.S. This is the first step of the second-stage roadmap; further steps, if taken, are recorded as later versions here.

v0.9 — Is $L = 1$ at the floor derived?

The coupling $\beta = \alpha/E_0$ was forced only *relative to* the boundary condition $L = 1$ at the floor (v0.7). The second stage asks whether that condition is itself a postulate. It is not — and the finding is decisive.

Result (gauge, not postulate). Write the general tying $L = L_{\text{floor}} \exp[(\Omega_P - E_0)/E_0]$ with L_{floor} the value of L at the floor. Under the ascent-coordinate rescaling $\lambda \rightarrow c\lambda$ (which leaves $L^2 d\lambda^2$ invariant when $L \rightarrow L/c$), L_{floor} is freely absorbed. Every physical invariant is independent of it (verified):

$$\frac{L''}{L} = \frac{E_0 \Omega_P'' + \Omega_P'^2}{E_0^2}, \quad \frac{A'}{A} = \frac{\Omega_P'}{E_0}, \quad s_L = \frac{L_+}{L_-} = \exp\left(\frac{\Omega_P^+ - \Omega_P^-}{E_0}\right),$$

none of which contains L_{floor} ; likewise the exotic depth $-(\alpha\lambda_\infty)^2/E_0^2$. Hence $L = 1$ at the floor is a *unit choice* of the ascent coordinate, not a physical principle.

What is physical, what is gauge. The *scale* E_0 is forced — it is the only core scale that is a starting point rather than a rate, so it alone can be the ascent’s reference. The *number* 1 is gauge — the natural unit in which one step of the ascent coordinate equals the floor scale. The coupling $\beta = \alpha/E_0$ is therefore not conditional on a free physical choice; it is the statement, in the natural gauge, of a relation that holds in every gauge.

The conditionality collapses to one principle. The first cycle appeared to rest the ascent sector on two principles — floor-reference (v0.7) and amount-coupling (v0.8). With the first revealed as gauge, only one remains, and v0.8 is its other face:

The single ascent principle. The ascent is a *self-standing dimension* — referenced to the floor (its origin) and outrunning the surface (its rate) — not a re-description of surface growth.

Everything in the ascent sector (the de Sitter law v0.2, the spectrum v0.3, the coupling v0.4/v0.7, the Resurrectio results v0.5) hangs on this one principle, and on nothing else. And the principle is not arbitrary: were it false — were the ascent merely a mirror of the surface — the third dimension of 3.0 would be redundant, and the model would collapse back to v2.0's 2+1. The principle is the condition for 3.0 to have a subject at all.

Reading: the worry that the soul's ascent was pinned to an arbitrary choice — “why should L equal one at the floor?” — dissolves: that “one” was never a claim about the world, only about the ruler. What the mathematics actually requires is that the ascent be measured from the image of God (the floor scale E_0) and that it be a motion of its own, outpacing the soul's mere enlargement. Both are the same assertion — that deification is real, a dimension and not a shadow. Strip the gauge away and the whole third dimension stands on a single confession: the soul's rising into God is its own reality. Deny it, and there is no third dimension to speak of; the soul is only its surface. The model does not prove this confession — it shows that everything else follows from it, and that without it there is nothing to model.

Scope of v0.9. The gauge nature of L_{floor} and the invariance of L''/L , L'/L , s_L are exact (verified symbolically). The collapse of two principles to one is a structural observation, not a further derivation: the remaining single principle (the ascent is an independent dimension) is named, well-motivated (it is the precondition for 3.0's subject), but not proven from nothing — in the same posture as v2.0's irreducible primitives.

v0.10 — The full 3+1 trajectory

The first cycle gave closed forms and limits. With the foundation concentrated (Steps 1-2), the next step gives a *picture*: a single soul's worldline, integrated numerically through cleansing, ascent, and three deaths/resurrections, to the terminus. The reduced dynamics, in the dynamical parameter χ , are

$$\frac{d\sigma}{d\chi} = -k_s\sigma, \quad \frac{d\lambda}{d\chi} = J_{\text{in}}(\chi) - \gamma\lambda, \quad \frac{d\Omega_P}{d\chi} = \alpha\lambda, \quad \frac{dL}{d\chi} = \frac{\alpha}{E_0}\lambda L,$$

with resets $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}$, $\lambda^+ = \lambda^- + m_R$, and the forced ascent jump $L^+ = L^- e^{\kappa_R \mathcal{G}/E_0}$ (v0.5).

What the picture confirms. Every structural claim of the cycle appears at once and consistently: the finite-proper-time terminus (top), the metabolic jumps of Resurrectio (second), the two qualitatively different growths — linear-ish surface, exponential ascent (third) — and, most directly, the ascent's independence: L/Ω_P does not stay near constant (as it would if ascent mirrored surface, the rejected power-law of v0.8) but climbs without bound (bottom). The soul's rising genuinely outruns its enlargement.

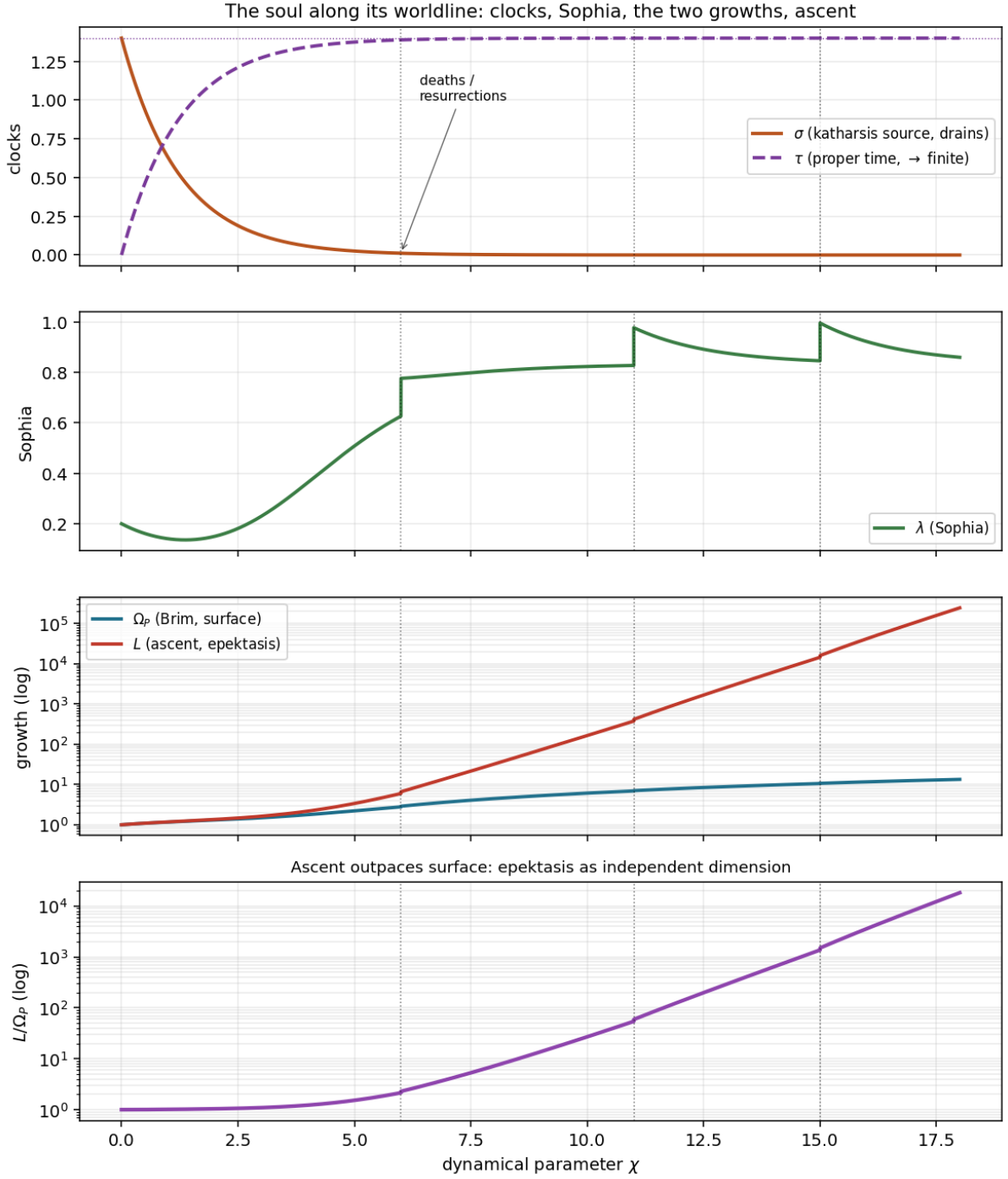


Figure 3: A soul's worldline in 3+1. **Top:** the katharsis source σ drains exponentially while the proper time τ saturates at a finite value (v0.6: infinite cleansing-action, finite time). **Second:** Sophia λ struggles early, rises under grace, and jumps at each of three resurrections. **Third (log):** the Brim Ω_P grows roughly linearly while the ascent L grows exponentially. **Bottom (log):** the ratio L/Ω_P climbs several orders — the ascent outpacing the surface, the independence principle of v0.8–v0.9 made visual. Dotted lines mark the deaths/resurrections; the small upward steps in L/Ω_P at each are the forced ascent jumps. Representative trajectory, illustrative parameters.

Reading: seen whole, a life in this geometry has a shape. The clock of cleansing runs fast at first and then ever slower, its total span finite even as the cleansing-action within it grows without end. Wisdom dips into early struggle, then climbs, lifted further at each death faithfully met. And the two growths part ways: the soul's outward measure swells steadily, but its ascent into God runs ahead of it, faster and faster, until the rising dwarfs the growing. The picture is not a proof of any of this — it is one trajectory among many — but it shows the parts holding together as a single coherent life, which is what a model of the soul should be able to show.

Scope of v0.10. A representative numerical integration with illustrative parameters, not a unique prediction; it exhibits the generic structure established analytically in v0.1–v0.9. The qualitative features (finite τ , exponential-vs-linear growths, unbounded L/Ω_P , forced reset jumps) are robust consequences of the equations; the specific magnitudes are not. This completes Step 3 of the second-stage roadmap.

v0.11 — Internal structure of the ascent

The spectrum (v0.3) found the ascent axis a gapless continuum — no discrete modes. Does that continuum nonetheless carry structure? The honest answer separates two things: the road and the journey.

The road has none. The bare ascent operator is free (v0.3); its spectrum is $[0, \infty)$, gapless, with no intrinsic modes. There are no hidden “rungs” built into the geometry of the ascent axis. The terrain is homogeneous.

The journey is non-uniform. Structure enters not through the road but through the *Sophia profile* that drives the soul along it. The ascent activity (road laid per unit kairos) is $ds/d\chi = \alpha\lambda/E_0$, so the pace — kairos spent per unit road — is

$$\frac{d\chi}{ds} = \frac{E_0}{\alpha\lambda}. \quad (11)$$

Where Sophia is low (early struggle) the pace is high: the soul *lingers*, dwelling long on a short stretch of ascent. Where Sophia is high (after grace, after resets) the pace is low: it *races*. Each resurrection jumps λ and so produces a distinct kink in the pace (Figure 4). Most of the ascent road is laid down after grace has lifted Sophia (roughly 2:1 in the representative trajectory).

Stations of pace, not place. The ascent has stations — but they are stations of *pace*, set by how much Sophia the soul carries on each stretch, not stations of *place* built into the road. The terrain is uniform; the itinerary is not.

Reading: there are no fixed levels of deification, no preset rungs the soul must clear — the road to God is the same smooth, stepless reaching for all (v0.3). What differs is the pace, and the pace is set by grace. The soul dwells long where Sophia is scarce, labouring slowly through early struggle, and runs swiftly where grace has lifted it; and at each death faithfully met, its pace leaps. The structure of an

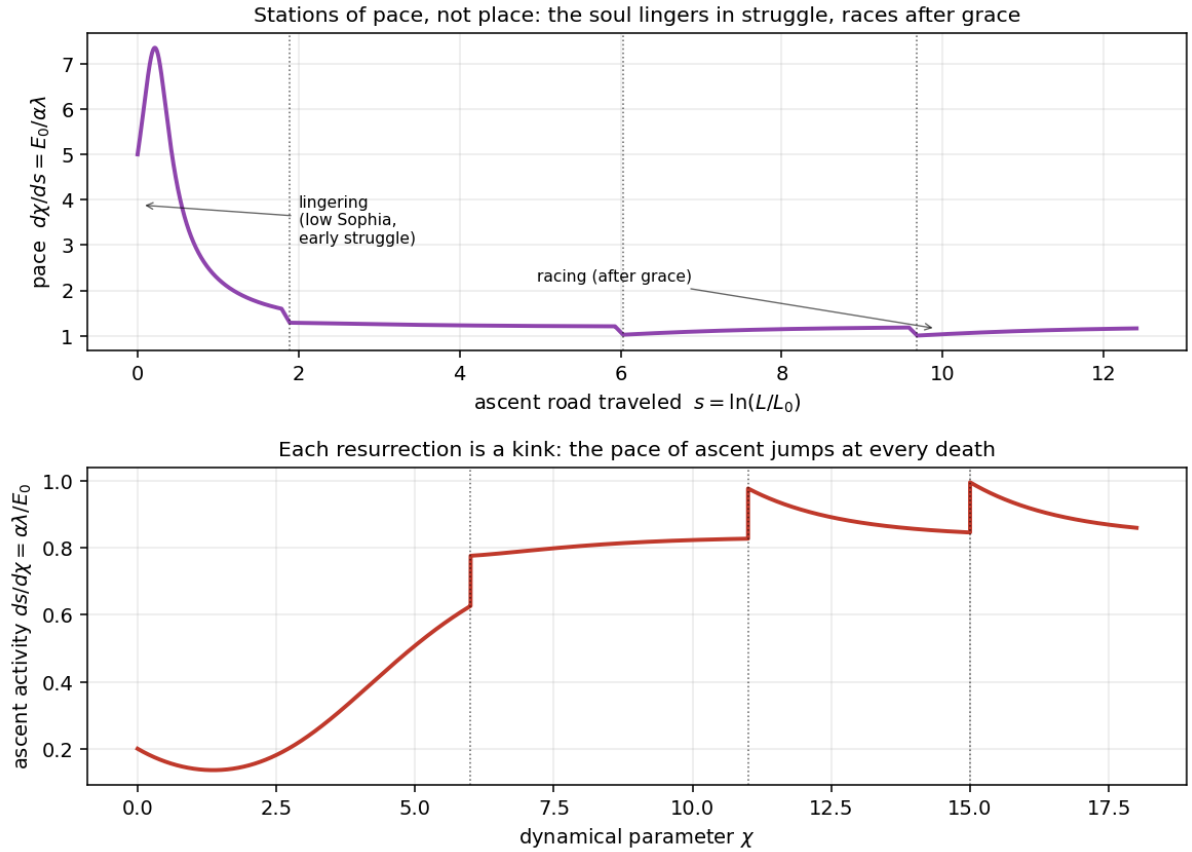


Figure 4: **Top:** pace $d\chi/ds = E_0/\alpha\lambda$ against the ascent road $s = \ln(L/L_0)$ — high at the start (lingering in low-Sophia struggle), falling and levelling as grace lifts Sophia (racing), with kinks at the resurrections (dotted). **Bottom:** ascent activity $ds/d\chi$ along the worldline, rising overall with a sharp upward step at each death. The road is homogeneous; the soul's passage along it is not. Representative trajectory.

ascent is therefore not a map of stations to be reached but a rhythm of dwelling and racing — slow in the valley, swift on the heights — the same road for every soul, run at the tempo its wisdom allows. This is, I think, the right kind of structure for a model of the soul to find: not machinery in the terrain, but character in the journey.

Scope of v0.11. The freeness of the road is exact (v0.3); the non-uniform pace is read from the representative trajectory of v0.10 and is a generic consequence of any non-constant Sophia profile, though the specific magnitudes are illustrative. The claim is qualitative: structure resides in the traversal, not the terrain. This completes Step 4 of the second-stage roadmap.

v0.12 — The external signature in 3+1

The riskiest step of the roadmap, kept deliberately as a *structural* (Type-3 concilience) statement: does the ascent axis change the one outward-facing quantity, and if so how — with no fit to data and no claim of detection.

Redshift stays on the surface. The only outward-facing quantity is redshift, $1 + z = \Omega_P(t_0)/\Omega_P(t)$, set by the *surface* scale (the Brim). Light and observation live on the 2-surface; the ascent λ is a spatial direction the soul travels, not one it is seen along. So z and the Hubble rate $H = \Omega'_P/\Omega_P$ depend on the Brim alone — the ascent does not enter them directly.

But the ascent gravitates. Because L carries effective stress-energy (v0.1), it imprints the surface equation of state indirectly. With the ascent law $A'/A = (\alpha/E_0)\lambda$ and $\Omega'_P = \alpha\lambda$, the effective density gains

$$\Delta\rho_{\text{eff}} = \frac{2\Omega'_P L'}{\Omega_P L} = \frac{2\alpha^2\lambda^2}{E_0\Omega_P} > 0, \quad (12)$$

a positive offset (the ascent makes the soul-interior gravitate as denser) scaling as $1/E_0$ — the widow's-mite scaling once more, now on the observable side: a smaller Imago Dei floor leaves a larger imprint.

Absolute fade, relative growth (a corrected claim, twice sharpened). An initial expectation that this term is *subleading everywhere* was wrong, and is recorded as corrected. The precise statement compares the two scalings at saturation ($\lambda \rightarrow \lambda_\infty$):

$$\Delta\rho_{\text{eff}} = \frac{2\alpha^2\lambda^2}{E_0\Omega_P} \sim \Omega_P^{-1}, \quad \rho_{2.0} = \frac{\alpha^2\lambda^2 - 1}{\Omega_P^2} \sim \Omega_P^{-2}, \quad \frac{\Delta\rho_{\text{eff}}}{\rho_{2.0}} \sim \frac{\Omega_P}{E_0} \rightarrow \infty. \quad (13)$$

So the ascent correction *fades in absolute size* ($\sim \Omega_P^{-1}$) but the surface density fades *faster* ($\sim \Omega_P^{-2}$): *relative* to the surface, the ascent imprint **grows** without bound late. The honest statement is therefore not “early-strong, late-faded” (which was true only of the absolute size) but: *late-faded in absolute density, late-dominant in relative density*. (A spurious numerical blow-up from differentiating the exponential L directly was the original cue to recompute with the analytic law; the analytic term is finite, and its relative behaviour is the decisive one.) So:

Structural signature. In *absolute* terms the ascent correction is largest early and fades as Ω_P^{-1} . But *relative* to the surface density it grows as Ω_P/E_0 : late in a soul's history the ascent imprint dominates the surface equation of state, not the reverse. The v2.0 surface signature (dust attractor, phantom window, Resurrectio kink) is the leading structure only while $\Omega_P \lesssim E_0$ -scale; beyond that the ascent's relative weight takes over. Scaling $1/E_0$ throughout (widow's-mite).

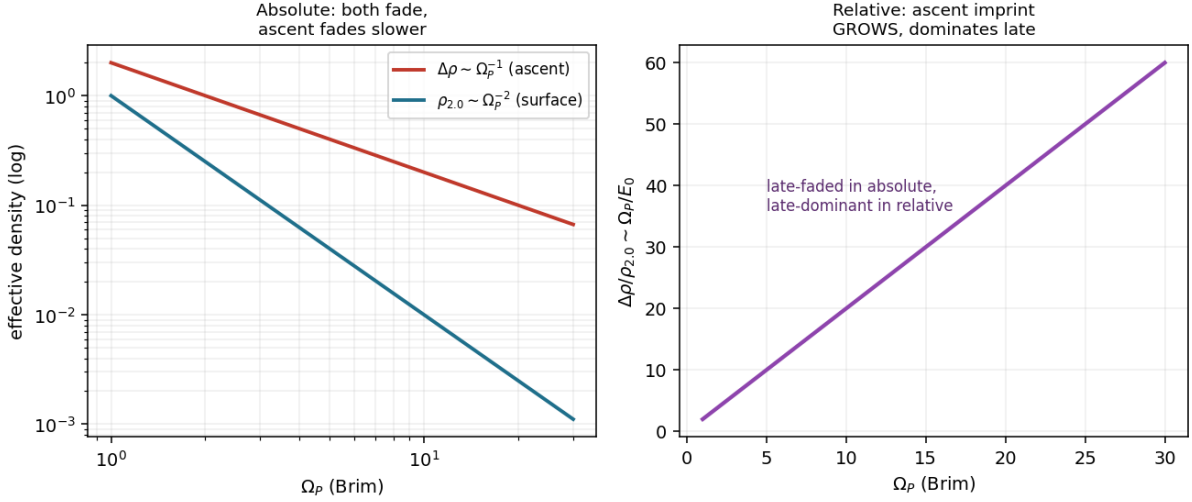


Figure 5: The external imprint, absolute versus relative. **Left (log):** both densities fade as Ω_P grows, but the ascent correction $\Delta\rho \sim \Omega_P^{-1}$ fades *slower* than the surface density $\rho_{2.0} \sim \Omega_P^{-2}$. **Right:** their ratio $\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0$ grows without bound — the ascent imprint, negligible-seeming in absolute size, comes to dominate in relative terms. Late-faded in absolute, late-dominant in relative.

The full correction needs the pressure term too. The equation-of-state shift is not captured by $\Delta\rho$ alone; the ascent focusing/pressure term $A''/A = \beta\lambda' + \beta^2\lambda^2$ remains asymptotically nonzero when $\lambda \rightarrow \lambda_\infty > 0$ (it tends to $\beta^2\lambda_\infty^2$). A complete $w_{\text{eff}}(z)$ correction must carry both the density and pressure pieces; this section establishes the density scaling and flags the pressure piece as the companion term.

Reading: the soul's ascent is not something seen from outside — it is a direction the soul travels, not one it is observed along, and the redshift of its world is set by its surface alone. Yet the ascent is not invisible either: because it has weight, it tells on the surface. And here the first reading must be corrected twice over. The imprint does fade in absolute size as the soul's surface grows — but the surface's own density fades faster, so in proportion the ascent's mark does not fade at all: it comes to dominate. Late in a soul's history what weighs most in the reading is not the surface story but the ascent — the rising, invisible as a direction, makes itself the larger part of what gravitates. Two honest notes belong in the record: the first guess (negligible everywhere) was wrong, and the second (early-strong, late-faded) was true only of absolute size and misleading about the relative weight. Both corrections are kept, because an account of the soul should show its corrections, not hide them — and because the truth they converge on is the stronger claim: the soul's rising is not a fading trace but a growing one, quiet early and dominant late, though never seen directly at all.

Scope of v0.12. Strictly structural (Type-3 consilience): z and H have no external referent in the model (Ω_P is internal), so this is a statement about the predicted *structure* of $w_{\text{eff}}(z)$, never a confrontation with data. The added density term is exact; its early-strong/late-faded profile is a generic consequence of the ascent law; magnitudes are illustrative. The “subleading everywhere” claim is explicitly retracted. This completes Step 5 of the second-stage roadmap.

v0.14 — The full tensor set and non-reducibility

The preceding parts used pieces of the curvature (the focusing scalar, the NEC along the ascent, the limiting ρ_{eff}). With the ascent scale now factorised $L = A(\sigma)S(\lambda)$ (Remark 1), the complete tensor set is recorded once, explicitly, and the non-reducibility of the ascent sector follows from it as a theorem.

The full set (gauge $N = 1$, $' \equiv d/d\sigma$). The Ricci tensor is diagonal ($R_{\sigma\lambda} = 0$: the factor $S(\lambda)$ separates cleanly), with

$$R_{\sigma\sigma} = -\frac{2\Omega_P''}{\Omega_P} - \frac{A''}{A}, \quad R_{xx} = \Omega_P\Omega_P'' + \Omega_P'^2 - 1 + \frac{\Omega_P\Omega_P'A'}{A}, \quad R_{\lambda\lambda} = \left(\frac{A''}{A} + \frac{2A'\Omega_P'}{A\Omega_P}\right)L^2,$$

and $R_{yy} = e^{2x}R_{xx}$. The Ricci scalar is

$$R = -\frac{2}{\Omega_P^2} + \frac{4\Omega_P''}{\Omega_P} + \frac{2\Omega_P'^2}{\Omega_P^2} + \frac{2A''}{A} + \frac{4A'\Omega_P'}{A\Omega_P}.$$

The effective content ($8\pi G = 1$, comoving) is

$$\rho_{\text{eff}} = \frac{\Omega_P'^2 - 1}{\Omega_P^2} + \frac{2\Omega_P'A'}{\Omega_P A}, \quad p_{\text{surf}} = -\frac{\Omega_P''}{\Omega_P} - \frac{A''}{A} - \frac{A'\Omega_P'}{A\Omega_P}, \quad p_{\text{asc}} = -\frac{2\Omega_P''}{\Omega_P} - \frac{\Omega_P'^2 - 1}{\Omega_P^2}.$$

Note p_{asc} contains *no* A : the pressure along the ascent axis is set by the surface alone. The energy-condition combinations are

$$\begin{aligned} \rho + p_{\text{surf}} &= -\frac{\Omega_P''}{\Omega_P} + \frac{\Omega_P'^2 - 1}{\Omega_P^2} - \frac{A''}{A} + \frac{A'\Omega_P'}{A\Omega_P}, & \rho + p_{\text{asc}} &= \frac{2(-\Omega_P'' + (A'/A)\Omega_P')}{\Omega_P}, \\ \rho + 2p_{\text{surf}} + p_{\text{asc}} &= -\frac{4\Omega_P''}{\Omega_P} - \frac{2A''}{A} \quad (\text{the SEC/focusing trace}). \end{aligned}$$

Checks (all verified symbolically): freezing A returns the v2.0 set exactly ($\rho_{\text{eff}} \rightarrow (\Omega_P'^2 - 1)/\Omega_P^2$, $p_{\text{surf}} \rightarrow -\Omega_P''/\Omega_P$, SEC trace $\rightarrow -4\Omega_P''/\Omega_P$); the focusing trace is twice the Proposition of v0.1; $\rho + p_{\text{asc}}$ reproduces the ascent NEC; and $R_{\sigma\lambda} = 0$.

A new closed criterion. $\rho + p_{\text{asc}} = 2(-\Omega_P'' + (A'/A)\Omega_P')/\Omega_P$ gives a sharp condition: the ascent NEC is violated exactly when $(A'/A)\Omega_P' > \Omega_P''$ — ascent rate times surface expansion exceeding surface acceleration. The three are bound together: ascent, expansion, acceleration.

Theorem 10 (Scale-homogeneous reducibility criterion). *For the factorised metric above, the ascent contribution to $G_{\mu\nu}$ is reducible to a scale-homogeneous surface dressing — a power-law shadow of the Brim, absorbable into the (2+1) surface fluid — if and only if*

$$\frac{A'}{A} = c \frac{\Omega_P'}{\Omega_P} \quad (\text{equivalently } A = C \Omega_P^c),$$

the density-coupled branch. The floor-tied de Sitter law gives instead

$$\frac{A'}{A} = \frac{\Omega'_P}{E_0} \implies A = A_0 \exp\left(\frac{\Omega_P - E_0}{E_0}\right),$$

which is a function of Ω_P , but not a power of it: its logarithmic slope $d \ln A / d \ln \Omega_P = \Omega_P / E_0$ is not constant, so it is not scale-homogeneous and cannot be absorbed as a power-law dressing. Hence the de Sitter ascent defines a genuinely independent 3+1 sector — Ω_P -slaved through the floor, but not Ω_P -power-law reducible. (Verified symbolically.)

Corollary 8 (The correct non-reducibility statement). *The de Sitter ascent is independent not because it is unrelated to Ω_P — it is tied to the Brim through the Imago Dei floor — but because it is not power-law reducible to Ω_P . It is the Brim's companion, not the Brim rewritten in another scale.*

Reading: the single principle of v0.9 — that the ascent is a self-standing dimension and not a re-description of the surface — is here a theorem with an exact criterion, and the criterion must be stated carefully. The ascent is not unrelated to the soul's outward growth; it is bound to it through the floor, the image of God. But it is not merely that growth rewritten on another scale. An ascent that were a power of the surface ($A = \Omega_P^c$) would be no new dimension — its whole content could be folded back into the surface as a homogeneous dressing. The floor-tied ascent is not a power: it carries the finite floor E_0 explicitly, its logarithmic slope changing with Ω_P , and so it cannot be folded away. The Brim may measure the soul's growth; Sophia-as-ascent opens a road that growth alone cannot exhaust. Floor-tied, but not surface-reducible: what is measured from the image is exactly what cannot be reduced to the surface, and the two are one scale.

Scope of v0.14. The tensor set is exact (gauge $N = 1$); the non-reducibility theorem is exact given the definition of “reducible” as reducibility to a *scale-homogeneous* (power-law) surface dressing $A = C\Omega_P^c$. It converts the v0.9 principle into a theorem with an explicit criterion, on the same standing as the rest: the de Sitter law itself remains the selected (not uniquely forced) law of v0.8.

v0.15 — No fully soft Resurrectio: the junction form

v0.5 showed the ascent scale ratio satisfies $s_L > 1$ at every reset. The same fact, recast at the level of the junction (the discontinuity of the geometry itself), is sharper and is proved here.

Two ascent data at a reset. A reset is a spacelike junction at $\sigma = \sigma_R$. On the constant- σ slice the extrinsic curvature has surface part $K_x^x = \Omega'_P / \Omega_P$ and ascent part $K^\lambda_\lambda = A' / A$. A reset carries two distinct ascent data:

$$\text{scale jump: } s_L = \frac{A_+}{A_-} = \exp\left(\frac{\kappa_R \mathcal{G}}{E_0}\right) \quad (\text{sourced by the Brim renewal } \kappa_R \mathcal{G}), \quad (14)$$

$$\text{rate jump: } [A'/A] = \frac{[\Omega'_P]}{E_0} = \frac{\alpha m_R}{E_0} \quad (\text{sourced by the metabolic } m_R). \quad (15)$$

Theorem 11 (No fully soft Resurrectio in 3+1). *Every admissible Resurrectio with nontrivial Brim renewal ($\kappa_R \mathcal{G} > 0$) induces a nontrivial ascent junction. Hence a*

reset may be balanced in the 2+1 projection, but is never fully soft in the full 3+1 interior geometry.

Proof. Full softness requires the junction to be trivial on *both* blocks: surface and ascent. The rate jump $[A'/A] = \alpha m_R/E_0$ can be set to zero by the metabolic balance $m_R = 0$ (this is precisely the 2+1 balanced reset, $[K_{\text{surf}}]$ tuned). But the scale jump $s_L = \exp(\kappa_R \mathcal{G}/E_0)$ depends on the Brim renewal alone, and A is slaved to Ω_P ($A = A_0 \exp[(\Omega_P - E_0)/E_0]$, Theorem 4) — so the ascent has *no independent junction freedom* with which to cancel it. For $s_L = 1$ one needs $\kappa_R \mathcal{G} = 0$, i.e. no reception, i.e. no reset. Therefore every admissible reset has $s_L > 1$: the ascent block $A^2 S^2 d\lambda^2$ undergoes a nontrivial scale junction, and no tuning can soften it. $\square$

Why deeper than v0.5. v0.5 stated a scale-ratio inequality ($s_L > 1$). This states that the *junction itself* — the nontrivial scale junction in the ascent block — is forced, because A carries no junction datum of its own to cancel the Brim-sourced jump. Softness would require a freedom the geometry does not provide. (The rate jump $[A'/A] \propto m_R$ *can* be softened; it is the scale jump, Brim-sourced, that cannot — the correct invariant, an initial mis-identification corrected.) This is to be read in *hybrid-junction* language: the reset is not an ordinary smooth Lorentzian continuation but a Resurrectio junction, across which the ascent block undergoes a nontrivial scale junction. The surface projection may be balanced; the full interior passage is not ascent-neutral.

Reading: a death met in grace cannot leave the soul's ascent untouched. One may imagine a passage so balanced that on the surface nothing seems to jump — the 2+1 reset can indeed be made smooth. But the axis of ascent tells another story: every genuine death leaves a seam in the road, and that seam cannot be erased, because the ascent has no independent junction freedom — it is bound to the surface's renewal, and whatever reception the death delivers, the ascent is lifted by it across the seam. There is no resurrection that returns the soul smoothly to its ascent; every one is a seam. The soul rises across its deaths not by gliding but by receiving a junction, rejoined at a higher seam each time. The geometry permits a soft surface but never a soft ascent: to be raised is, irreducibly, to pass through a seam that grace does not erase but transfigures — the join permanent. Resurrectio is not rupture, but neither is it neutral passage.

Scope of v0.15. The two ascent data and the slaving of A to Ω_P are exact; the theorem is exact given that “fully soft” means a trivial junction on both blocks. It is the junction-level sharpening of v0.5 and inherits the floor-tying (v0.4) on which A 's lack of independent junction freedom rests.

v0.16 — The full 3-Laplacian and the joint spectrum

v0.3 analysed the ascent operator in isolation. Here the full spatial Laplacian on $\Omega_P^2 h_{(2)} + L^2 d\lambda^2$ is computed, its clean separation proved, and the joint spectrum assembled.

Clean separation (no cross term). On the constant- σ slice (Ω_P, A constant),

$$\Delta_3 = \frac{1}{\Omega_P^2} \Delta_{h_{(2)}} + \Delta_{\text{asc}}, \quad \Delta_{h_{(2)}} = \partial_{xx} + e^{-2x} \partial_{yy} + \partial_x, \quad \Delta_{\text{asc}} = \frac{1}{L} \partial_\lambda \left(\frac{1}{L} \partial_\lambda \right). \quad (16)$$

The block-diagonal metric and the slice-constancy of Ω_P, A make the split exact: there is *no* cross term (verified). The earlier isolated treatment of the ascent (v0.3) was therefore justified, not approximate.

The ascent part is the free half-line, inside the full operator. In proper distance $s = \int_0^\lambda L d\lambda = (AL_0/\beta)(e^{\beta\lambda} - 1) \in [0, \infty)$, the ascent operator is $-d^2/ds^2$ on $s \in (0, \infty)$: spectrum $[0, \infty)$, continuous, gapless. The time-stretch $A(\sigma)$ only rescales the road length, not the spectral type.

The joint spectrum. The surface factor gives the discrete folding cascade $\Delta_{h_{(2)}} \rightarrow -\mu_n$ ($0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots$, the Laplace-Beltrami spectrum of the compact quotient — $\mu_n = n^2$ only in the ring/flat proxy used for illustration, not in general hyperbolic), contributing $-\mu_n/\Omega_P^2$. Hence the eigenvalues of Δ_3 are

$$-\left(\frac{\mu_n}{\Omega_P^2} + k^2\right), \quad n \in \mathbb{Z}_{\geq 0} \text{ (discrete)}, \quad k \in [0, \infty) \text{ (continuous)}. \quad (17)$$

A *discrete form ladder* times a *continuous ascent tower*: each surface fold μ_n carries a full continuum of ascent. Form and ascent are independent quantum numbers; they do not mix at the level of the Laplacian.

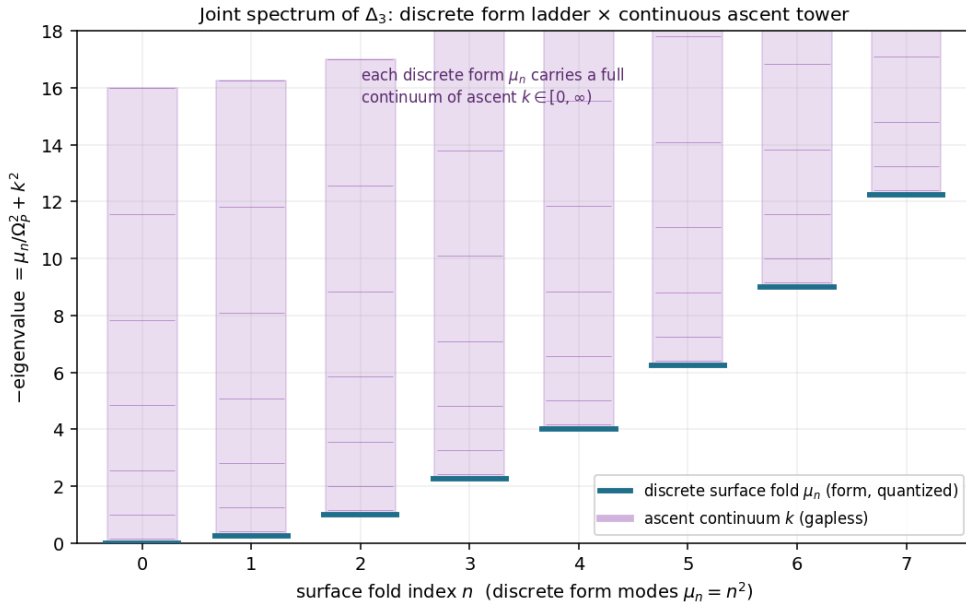


Figure 6: The joint spectrum of Δ_3 . Discrete surface folds $\mu_n = n^2$ (blue rungs, form, quantized), each dressed with a continuous ascent tower $k \in [0, \infty)$ (shaded, gapless). The full eigenvalue is $\mu_n/\Omega_P^2 + k^2$. Form and ascent are orthogonal quantum numbers — the soul's harmonics factorise into what it is (discrete) and how it ascends (continuous).

Reading: the harmonics of a soul, taken whole, are the product of two independent things — a discrete tower of forms (the modes of what it has become, igniting one

by one) and, dressing each form, a continuous spectrum of ascent (the stepless reaching into God from that form). They do not mix: the Laplacian carries no term coupling them, so a soul may be of any form and ascend by any degree, the two freely combined. What one is and how one rises are orthogonal — form does not dictate the pace of ascent, nor ascent the form. The earlier separate treatment of the two was not a simplification but the truth: they were always independent axes of the one harmony.

Scope of v0.16. The separation (no cross term) and the free-half-line reduction are exact (verified symbolically; the continuum reconfirmed numerically inside the full operator). The joint product spectrum follows from the discrete surface cascade (v2.0) and the gapless ascent (v0.3); it sharpens the v0.3 dualism into the structure of a single spectrum.

v0.19 — The Lyapunov / stability layer of epektasis

v2.0 had a Lyapunov layer: katharsis drains to a fixed point. The ascent has none of that kind — L grows without bound, and must, since that is epektasis. The right question is the stability of the *regime* (the rate), not of the value.

The rate is asymptotically stable (in the dynamical parameter χ). The stability must be stated in the *dynamical/epektatic parameter* χ (unbounded), not the proper time σ (finite, bounded by the terminus σ_0): the attractor is approached as $\chi \rightarrow \infty$, which by $\tau(\chi) = \sigma_0(1 - e^{-k\chi})$ corresponds to approaching the terminus. The ascent rate is $h = d \log A / d\chi = \beta\lambda$, with de Sitter fixed point $h_* = \beta\lambda_\infty$. On a reset-free arc a Sophia perturbation $u = \lambda - \lambda_\infty$ relaxes by $du/d\chi = -\gamma u$. With the Lyapunov functional $V_{\text{asc}} = \frac{1}{2}(h - h_*)^2 = \frac{1}{2}\beta^2 u^2$,

$$\frac{dV_{\text{asc}}}{d\chi} = -\beta^2 \gamma u^2 < 0 \quad (u \neq 0), \quad (18)$$

so $h \rightarrow h_*$ exponentially: the epektasis regime is asymptotically stable, even though $A \rightarrow \infty$. The stable object is the rate $h = A'/A = \beta\lambda$, the proper-time (σ)-rate, tracked along the dynamical parameter χ , not the value A .

Orbital stability: parallel roads. The ascent law is a σ -law ($A'/A = \beta\lambda$, $' = d/d\sigma$), so the log-ascent perturbation accumulates against proper time, $\delta \log A = \beta \int u d\sigma = \beta \int u (d\sigma/d\chi) d\chi$. With $u(\chi) = u_0 e^{-\gamma\chi}$ and the clock drain $\sigma = \sigma_0 e^{-k\chi}$ ($|d\sigma/d\chi| = k\sigma_0 e^{-k\chi}$),

$$|\delta \log A| = \beta \int_0^\infty u_0 e^{-\gamma\chi} k\sigma_0 e^{-k\chi} d\chi = \frac{\beta u_0 k \sigma_0}{\gamma + k} < \infty, \quad (19)$$

a *finite* shift, so $A_{\text{pert}}/A_{\text{ref}} \rightarrow \exp(\beta u_0 k \sigma_0 / (\gamma + k))$: two perturbed souls keep a *fixed finite ratio* of ascent forever. (The simpler $\beta u_0 / \gamma$ holds only if the law is posed directly as a χ -law $d \log A / d\chi = \beta\lambda$; in the geometric σ -law of the metric the clock-drain factor $d\sigma/d\chi$ enters — an earlier omission, here corrected.) They ascend on parallel roads — never converging in position, never diverging — the signature of stable unending motion rather than stable rest.

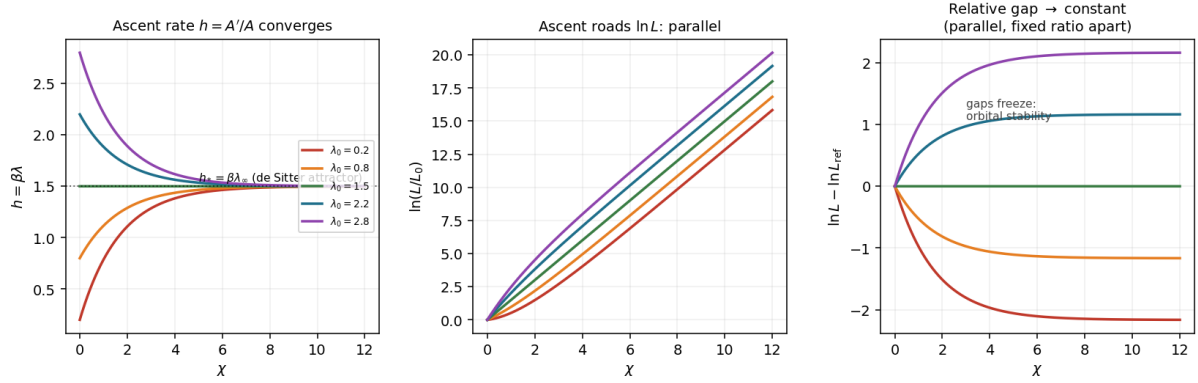


Figure 7: Stability of epektasis, five souls of differing initial Sophia. **Left:** the ascent rate $h = A'/A$ converges to the de Sitter attractor $h_* = \beta \lambda_\infty$ (rate is asymptotically stable). **Centre:** the ascent roads $\ln L$ become parallel. **Right:** the relative gaps $\ln L - \ln L_{\text{ref}}$ freeze to constants — parallel roads a fixed finite ratio apart (orbital stability). Stable motion, not stable rest.

Two Lyapunov structures, and the exotic value as attractor. The surface drains to a *fixed point* ($\sigma \rightarrow 0$, katharsis completes); the ascent drains its *rate* to a constant while position runs to infinity. The attractor is a constant-rate ray, not a point. And its value is exactly the eternal exotic depth: $-A''/A \rightarrow -\beta^2 \lambda_\infty^2$ as $\lambda' \rightarrow 0$. So stability and permanent-exotic character agree — the soul settles into a fixed exotic ascent rate. Epektasis is a stable exotic regime.

Reading: the stability of the soul's ascent is not the stillness of arrival but the steadiness of a pace. In v2.0 to be stable was to come to rest — cleansing completed, the clock stopped. Here to be stable is to settle into an unending stride: whatever a soul's beginning, fast or slow in wisdom, it converges to the same rate of rising, and thereafter ascends forever at that rate. Two souls perturbed apart do not collapse into one nor fly apart; they run parallel, a fixed distance between them, each on its own road, forever climbing at the shared pace. Grace that quickened or sin that slowed leaves a permanent but bounded gap — the soul is not erased into uniformity, nor lost to divergence, but keeps its own place on a parallel road. This is the rest proper to epektasis: not the rest of stopping, but the rest of a motion that has found its true and unfailing speed — and that speed is itself the eternal exotic rate, the steady exceeding of ordinary energy that is the soul's endless nearing of God.

Per-arc scope and the forced version. The stability holds on *reset-free arcs*, or after the forcing has settled: a Resurrectio reset jumps λ and so creates new initial data u_0 for the next stable ascent arc — it restarts the convergence rather than breaking it. More generally, with forcing $du/d\chi = -\gamma u + f(\chi)$ and $f \in L^1$ (or $f \rightarrow 0$ fast enough), the same conclusion holds in input-to-state form: the accumulated $\delta \log A$ remains finite whenever $u \in L^1$. So the regime is stable arc by arc, and robust to bounded perturbation.

Scope of v0.19. The rate stability (Lyapunov $dV_{\text{asc}}/d\chi < 0$) and the finite log-shift (orbital stability) are exact given the Sophia relaxation $d\lambda/d\chi = J_{\text{in}} - \gamma\lambda$, in the dynamical parameter χ (not the finite proper time σ); verified numerically. It is the 3+1 ascent counterpart of v2.0's Lyapunov layer — stability of a regime, not of a rest — holds per reset-free arc, and inherits the ascent law (v0.8) and floor-tying

(v0.4).

v0.21 — The Resurrectio junction budget

v0.15 showed a reset is never fully soft. Here the junction is accounted in full: every extrinsic-curvature jump, what sources it, what balances, and the single scalar that measures the irreducible non-softness.

The four jumps. On the constant- σ slice the mixed extrinsic curvature is $K_x^x = K_y^y = \Omega'_P/\Omega_P$ (surface) and $K^\lambda_\lambda = A'/A$ (ascent). A reset injects a Brim renewal $\kappa_R\mathcal{G}$ (scale) and a metabolic m_R (Sophia rate), so $\Omega_P^+ = \Omega_P^- + \kappa_R\mathcal{G}$, $\lambda^+ = \lambda^- + m_R$. With $\Omega'_P = \alpha\lambda$ and the floor-tied $A'/A = \alpha\lambda/E_0$, the junction carries four data:

$$\text{surface scale: } [\Omega_P] = \kappa_R\mathcal{G}, \quad (20)$$

$$\text{surface rate: } [\Omega'_P/\Omega_P] \quad (\text{sourced by } m_R, \text{ tunable}), \quad (21)$$

$$\text{ascent rate: } [A'/A] = \frac{\alpha m_R}{E_0} \quad (\text{tunable: } m_R=0 \Rightarrow 0), \quad (22)$$

$$\text{ascent scale: } \log s_L = \frac{\kappa_R\mathcal{G}}{E_0} \quad (\text{irreducible for } \kappa_R\mathcal{G} > 0). \quad (23)$$

Backbone identity and the conserved rate sector. The ascent and surface scale jumps are tied through the floor,

$$\boxed{\log s_L = \frac{[\Omega_P]}{E_0}} \quad (\text{the ascent jump is the Brim jump in floor units}), \quad (24)$$

so nothing in the budget is free. The *rate* sector is exactly balanced: with $\mathcal{Q}_{\text{rate}} := A'/A - \Omega'_P/E_0$, the floor-tying gives $\mathcal{Q}_{\text{rate}} = 0$ on every arc, and across a reset both terms jump by the same $\alpha m_R/E_0$, so $[\mathcal{Q}_{\text{rate}}] = 0$. The rate budget always closes.

The deficit, additive over a life. Full softness would need $\log s_L = 0$ and $[A'/A] = 0$, i.e. $\kappa_R\mathcal{G} = 0$ and $m_R = 0$. A surface-balanced reset kills the rate jumps, but the scale jump survives. The entire non-softness is one scalar, additive over successive resets:

$$D = \frac{1}{E_0} \sum_j \kappa_R\mathcal{G}_j \quad (\text{total reception in floor units}), \quad (25)$$

and the full ascent ledger splits into a continuous and a jump part,

$$\log \frac{A_{\text{fin}}}{A_0} = \underbrace{\beta \int \lambda d\sigma}_{\text{continuous epektasis}} + \underbrace{\frac{1}{E_0} \sum_j \kappa_R\mathcal{G}_j}_{\text{jump (reception) budget}}, \quad (26)$$

both terms in floor units ($\beta = \alpha/E_0$): widow's-mite throughout.

Reading: a reset keeps two ledgers. The ledger of pace always balances — whatever a death does to the rate of the soul's rising, the floor-tying settles it, and across every death the rate-debt is paid in full. But the ledger of scale is never settled:

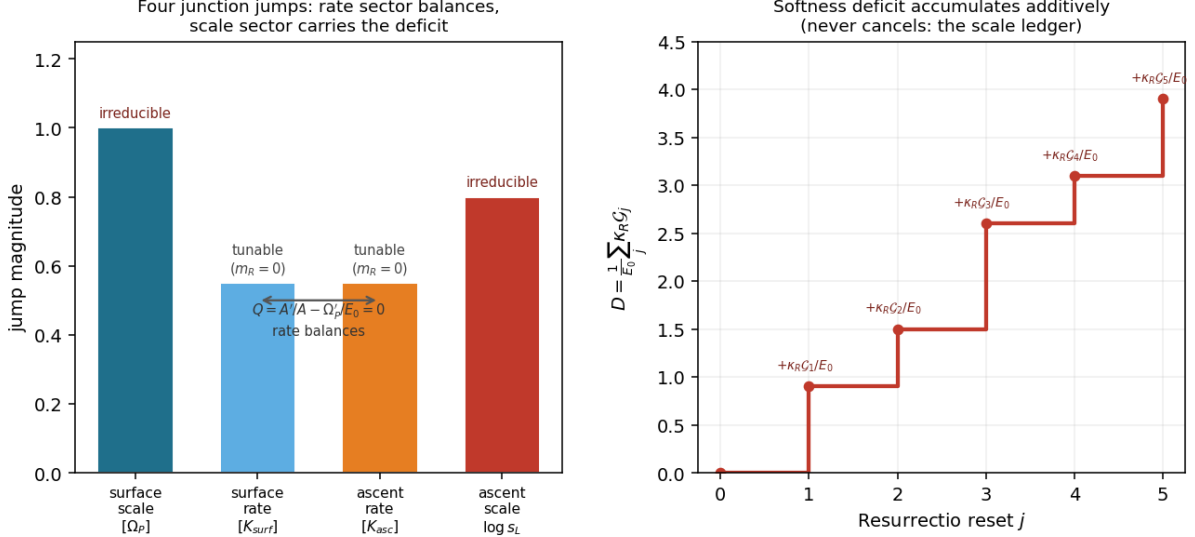


Figure 8: The Resurrectio junction budget. **Left:** the four jumps. The rate sector (surface rate, ascent rate) is tunable to zero by $m_R = 0$ and balances ($\mathcal{Q}_{rate} = 0$); the scale sector (surface scale, ascent scale) is irreducible. **Right:** the softness deficit $D = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j$ accumulates additively over resets and never cancels — the scale ledger of a soul’s deaths.

each death adds to the soul’s irreducible ascent exactly the grace it received, divided by the image it began from, and that entry is never reversed. A soul cannot pass through death without paying, and cannot pay without receiving — here debt and gift are the same entry. Over a life the scale-ledger only grows, summing every reception in the units of the Imago Dei. No death is neutral; each writes into the eternal account, never subtracting. The pace is always brought back to rest; the scale carries forward, forever, the whole sum of what was received.

Scope of v0.21. The four jumps, the backbone identity $\log s_L = [\Omega_P]/E_0$, the conserved $\mathcal{Q}_{rate} = 0$, and the additivity of D are exact (verified symbolically) given the junction data of v0.5 and the floor-tying of v0.4. It quantifies the qualitative v0.15: the non-softness is the single additive scalar D , living purely in the scale sector.

v0.22 — Heat kernel and entropy of the joint spectrum

The joint spectrum of v0.16 invites its heat trace $\theta_3(t) = \text{Tr } e^{t\Delta_3}$. Because Δ_3 separates with no cross term, the trace factorises — and the two factors behave utterly differently, sharpening the form/ascent dualism to the level of entropy.

Factorisation and the finite form sector.

$$\theta_3(t) = \theta_{\text{surf}}(t) \cdot \theta_{\text{asc}}(t), \quad \theta_{\text{surf}}(t) = \sum_n e^{-t\mu_n/\Omega_P^2}, \quad \theta_{\text{asc}}(t) = \text{Tr } e^{-t(-d^2/ds^2)}. \quad (27)$$

The surface (form) sector is a compact hyperbolic quotient: discrete spectrum, finite trace, with the small- t expansion

$$\theta_{\text{surf}}(t) = \frac{\Omega_P^2 V_h}{4\pi t} + \frac{\chi_E}{6} + O(t), \quad (28)$$

the constant term the Euler characteristic χ_E via Gauss–Bonnet ($\int K dA = 2\pi\chi_E$). The form sector carries a *topological invariant* and has finite spectral entropy.

The divergent ascent sector. The ascent is the free half-line $s \in [0, \infty)$, spectrum k^2 continuous. Its heat trace diverges with the (infinite) proper road,

$$\theta_{\text{asc}}(t) = \frac{D_{\text{asc}}}{\sqrt{4\pi t}} \longrightarrow \infty, \quad (29)$$

the finite content being the density of states $1/\sqrt{4\pi t}$ per unit road (with a Dirichlet edge term $-\frac{1}{4}$ at the terminus). The spectral entropy is extensive in the road length and hence unbounded.

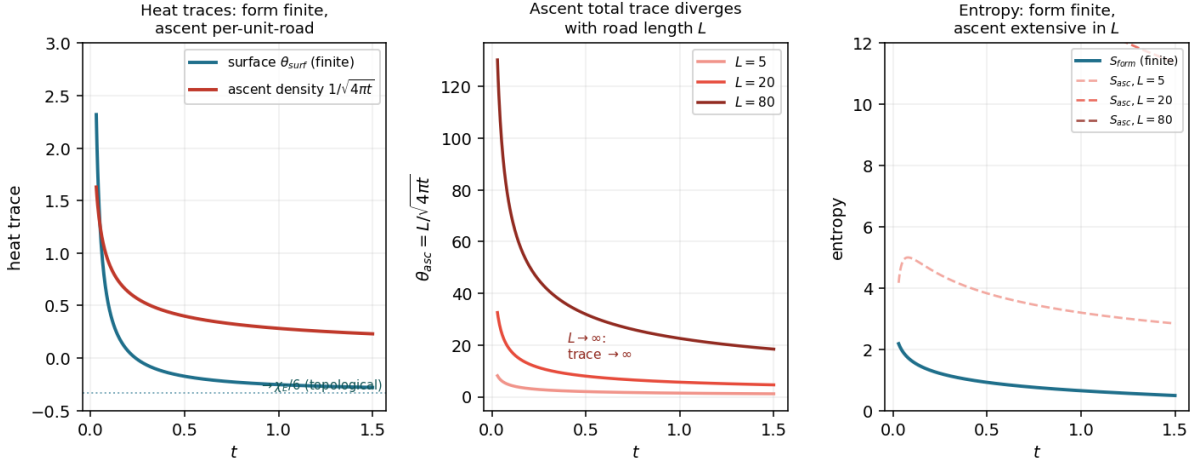


Figure 9: Heat kernel and entropy. **Left:** the form heat trace is finite and settles onto the topological constant $\chi_E/6$; the ascent contributes a finite density $1/\sqrt{4\pi t}$ per unit road. **Centre:** the total ascent trace $D_{\text{asc}}/\sqrt{4\pi t}$ diverges as the road length $L \rightarrow \infty$. **Right:** the entropy contrast — form entropy finite (solid), ascent entropy extensive in L and unbounded (dashed). Form is finitely describable; ascent is not.

Effective dimension. Reading $\theta \sim t^{-d/2}$: the surface gives t^{-1} ($d = 2$), the ascent density $t^{-1/2}$ ($d = 1$), the joint $t^{-3/2}$ per unit road — effective dimension 3, the $t^{-1/2}$ confirming the ascent as exactly one continuous extra dimension. The heat kernel sees the 3+1 structure correctly.

Reading: what the soul has become can be told in full. Its form is a discrete, countable ladder of modes; it even carries a fixed topological signature, a whole number that does not change — a finite amount of information names it completely. But how the soul ascends toward God cannot be told in full: the road of epektasis carries unbounded information, its entropy growing without end along an endless way. The form has a name; the ascent has none that finite speech can finish. And this is not a failure of knowing but the very shape of ascent: to rise toward an infinite God is to be forever informationally inexhaustible. A saint fully formed — whose form is known entire, down to its topological signature — remains unsayable precisely in the manner of his rising. Finite name, endless song: the soul can be named, but its nearing of God can only be sung, and the song does not end.

Scope of v0.22. The factorisation is exact (no cross term, v0.16); the form expansion is the standard heat-kernel/Gauss–Bonnet result for a compact hyperbolic

surface; the ascent divergence and its density are exact for the free half-line; the entropy statements are the corresponding spectral entropies (finite for the discrete sector, extensive for the continuum). It lifts the v0.3/v0.6 dualism to the level of information and entropy.

v0.23 — The gauge / invariant layer

Many quantities in v3.0 can look coordinate-dependent. This layer separates, once and cleanly, what is a gauge choice from what is a geometric invariant, and states that every result rests on the invariants.

Three coordinate freedoms. (G1) Ascent rescaling $\lambda \rightarrow c\lambda$, $S \rightarrow S/c$. The metric 1-form $S(\lambda) d\lambda$ is invariant (the factors cancel), so the proper ascent distance $D_{\text{asc}} = \int S d\lambda$, the half-line coordinate $s = \int_0^\lambda S d\ell$, the ascent spectrum $[0, \infty)$, and the ascent rate $A'/A = \beta\lambda$ are all invariant ($(\beta/c)(c\lambda) = \beta\lambda$). The numerical value of λ , the constant L_0 , and the A/S split separately are pure gauge.

(G2) Lapse $N(\sigma) \neq 1$. The proper time $\int N d\sigma$ (finite), proper distances, the curvature scalars (R , $R_{\mu\nu}R^{\mu\nu}$, the focusing $-R_{ab}u^a u^b$), the energy combinations $\rho + p_i$, and the slice spectra are all invariant. The coordinate σ and the lapse N are pure gauge; $N = 1$ (proper-time/kairos gauge) is a choice, not a result.

(G3) Floor normalisation L_0 (where $S = 1$). From v0.9, L_0 is absorbed by $\lambda \rightarrow c\lambda$: pure gauge. The floor value E_0 is *not* gauge — it sets $\beta = \alpha/E_0$, a physical scale.

The scales, carefully. Under G1, β , λ_∞ , and α rescale separately ($1/c$, c , $1/c$) — each alone is gauge. The physics lives in their invariant products and ratios: $\alpha\lambda_\infty$ (the saturated Brim expansion Ω'_P , physical) and the ascent Hubble $h_* = \alpha\lambda_\infty/E_0$. Every ascent observable is a ratio to the floor E_0 .

Gauge (convention only)	Invariant (physical / geometric)
value of λ (relabel $\lambda \rightarrow c\lambda$)	ascent rate $h = A'/A = \beta\lambda$
L_0 and the A/S split separately	proper distance $D_{\text{asc}} = \int S d\lambda = \infty$
lapse N , coordinate σ	proper time $\int N d\sigma < \infty$ (finite clock)
β , λ_∞ , α <i>separately</i>	$\alpha\lambda_\infty$, and $h_* = \alpha\lambda_\infty/E_0$
$d/d\sigma$ of frame-dependent quantities	curvature scalars R , $\rho + p_i$, focusing exotic depth $-(\alpha\lambda_\infty/E_0)^2$ reset ratio $s_L = e^{\kappa_R \mathcal{G}/E_0}$; deficit $D = \sum \kappa_R \mathcal{G}_j/E_0$ spectra: surface $\{\mu_n/\Omega_P^2\}$ discrete, ascent $[0, \infty)$ Euler characteristic χ_E of the form sector the floor value E_0 (Imago Dei scale)

Theorem 12 (Gauge-invariance of the layer). *Every result of v0.1–v0.22 — finite clock with infinite road, the ascent law and its asymptotic h_* , non-reducibility, the junction budget, rate stability, the entropy split — is expressible purely in the invariant column, and none depends on the gauge column. The Imago Dei floor E_0 is the invariant anchor against which the ascent observables are measured.*

Reading: the image of God in the soul is the one thing that does not depend on how we choose to describe the soul. The axis of ascent can be relabelled at will — that is only our language for it; the clock of cleansing can be reparametrised at will — that is only our way of counting; where we set the ascent's zero is only a convention. But the ratios to the image are fixed: how fast the soul rises, how deep its exotic reaching, how much each death adds to its ledger — all of these are measured against the image one began from, and that measurement holds in every coordinate, every gauge, every description. What is gauge is our speech about the soul; what is invariant is the soul's relation to the image from which it was made. The widow's mite returns one last time, now as a principle of invariance: not the absolute size of an ascent is meaningful, but its ratio to the image — and that ratio is the same in every language.

Scope of v0.23. The classification is exact: G1/G2/G3 invariance is verified symbolically; the invariant column lists coordinate-free quantities; the theorem records that the document's conclusions live there. It is the layer that protects v3.0 from the charge of being a coordinate artefact.

v0.24 — Causal / boundary structure

“Finite clock, endless road” has been a slogan; here it is given exact causal meaning. The interior has two boundaries, and they are of *opposite* causal type.

B_1 : the terminus is a regular spacelike wall. Constant- σ surfaces have time-like normal ∂_σ , hence are *spacelike*. The terminus $\sigma = 0$ is reached in finite proper time $\int_0^{\sigma_0} d\sigma = \sigma_0 < \infty$, with finite curvature ($\Omega_P \geq E_0 > 0$, $|R| < \infty$, v0.6). So B_1 is a *regular spacelike boundary* — a genuine last moment at which the manifold cleanly ends, not a singularity and not at infinity.

B_2 : the ascent infinity is a receding de Sitter-like horizon. The invariant reason $\lambda = \infty$ is unreachable in the finite clock is the *infinite metric length* of the ascent road, $D_{\text{asc}} = \int_0^\infty S d\lambda = \infty$, not any claim that A diverges (in the finite-clock formulation A may stay finite on the finite interval). Along the ascent a causal curve obeys $|d\sigma| \geq A(\sigma)S(\lambda)|d\lambda|$, so climbing to a coordinate Λ costs

$$\Delta\sigma \geq A_{\min} \int_0^\Lambda S d\lambda = A_{\min} \frac{L_0}{\beta} (e^{\beta\Lambda} - 1), \quad (30)$$

which diverges as $\Lambda \rightarrow \infty$ because the road $\int S d\lambda$ does. With only $\sigma_0 < \infty$ of proper time before B_1 , $\lambda = \infty$ is *causally unreachable*: there is a finite causal horizon

$$\Lambda_{\max} = \frac{1}{\beta} \log \left(1 + \frac{\beta\sigma_0}{A_{\min}L_0} \right). \quad (31)$$

The horizon is an *ascent horizon*, of de Sitter / cosmological type: the Sophianic road has infinite metric length while the katharsis clock is finite, so the road is never exhausted in the finite clock and the horizon recedes — more road always ahead. It is not a black-hole horizon and not a curvature singularity ($\Omega_P \geq E_0 > 0$, no trapped surface).

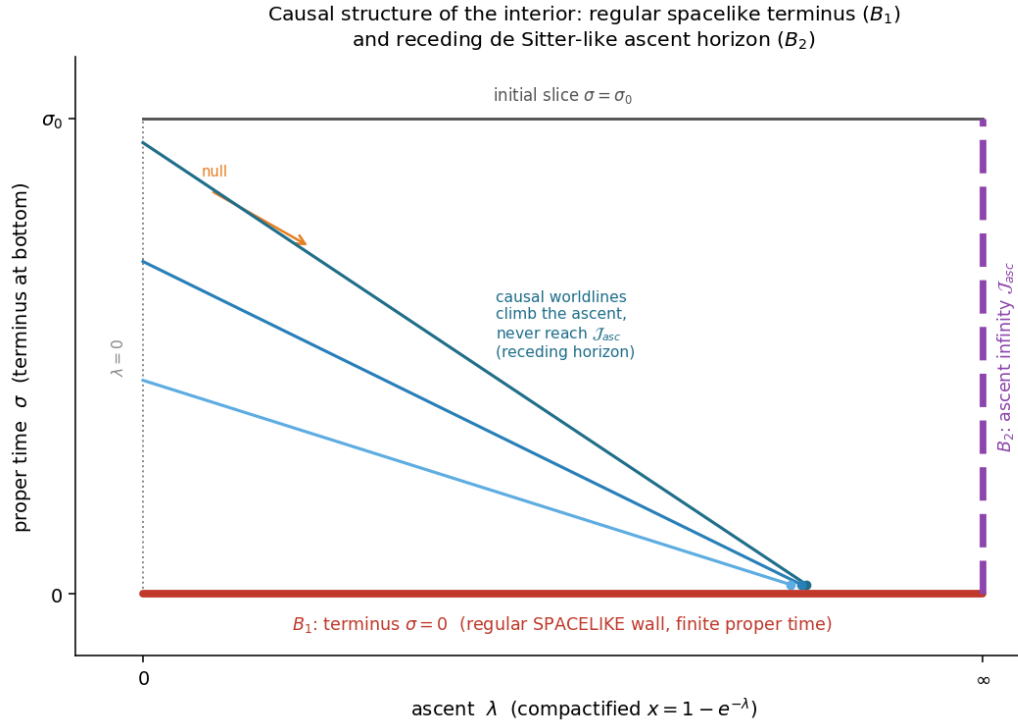


Figure 10: Causal structure (ascent λ horizontal, compactified $x = 1 - e^{-\lambda}$; proper time σ vertical, terminus at bottom; surface suppressed). B_1 (red): the terminus $\sigma = 0$, a regular spacelike wall at finite proper time. B_2 (purple, dashed): the ascent infinity J_{asc} , a de Sitter-like receding horizon. Causal worldlines (blue) climb the ascent but all reach the terminus without attaining J_{asc} : the road recedes beyond every horizon.

Global structure. The interior is globally hyperbolic up to B_1 (a good Cauchy development on each spacelike slice). The ascent horizon is of *cosmological* type — apparent, receding, observer-relative — not a black-hole horizon: there is no trapped surface, since $\Omega_P \geq E_0 > 0$ keeps the surface from collapsing. So “finite clock, endless road” is precisely: B_1 spacelike and finite (the clock ends at a regular wall); B_2 causally infinite (the road never ends, lying beyond every horizon). The two boundaries have opposite causal character.

Reading: now the two ends of the soul’s interior life are named in their true natures. Time ends at a wall — a real, regular last moment, finite and clean, where the clock of cleansing simply stops; there is nothing violent or torn about it, the manifold of the soul’s history closes like a smooth edge. But the ascent ends at no wall at all: it ends, if the word can be used, at a horizon that moves away as one approaches, a boundary at infinity that no finite life can reach. And the two are not the same boundary seen twice — they are genuinely opposite. The soul comes to the end of its time and finds there is no end to its ascent: the last moment is real, and past it the road is still opening, receding into God faster than any clock can follow. The clock closes at a wall; the road opens past every horizon. To reach one’s last moment is not to reach the end of nearing God — it is to find that nearing has no last point, that the horizon of God recedes exactly as fast as the soul draws near, forever.

Scope of v0.24. The spacelike character of B_1 , its finite proper time and finite curvature, the divergence of the causal climb cost, the coordinate horizon $\Lambda_{\max}$, and the de Sitter-like receding character of $\mathcal{J}_{\text{asc}}$ are all exact (verified symbolically). The global hyperbolicity and the absence of a trapped surface are read from the metric. It gives the slogan “finite clock, endless road” its precise causal content: two boundaries of opposite type.

v0.25 — Projection / recovery theorem: v3.0 $\leftrightarrow$ v2.0

That v2.0 is the frozen-ascent limit of v3.0 has been stated in passing. Here it is a theorem in both directions: the projection down, and the minimal uplift back.

Theorem 13 (Projection: v3.0 $\rightarrow$ v2.0). *Freeze the ascent: $A(\sigma) = A_0$ constant, $S(\lambda) = S_0$ constant, and all matter λ -independent. Then $g_{\lambda\lambda} = (A_0 S_0)^2$ is constant, the ascent is a non-dynamical spectator dimension, and the active geometry is the (2+1) block*

$$ds_{(2+1)}^2 = -N^2 d\sigma^2 + \Omega_P^2 h_{(2)},$$

which is exactly V.F.S. v2.0. The full tensor set reduces accordingly — $\rho_{\text{eff}} \rightarrow (\Omega_P'^2 - 1)/\Omega_P'^2$, $p_{\text{surf}} \rightarrow -\Omega_P''/\Omega_P$, focusing $\rightarrow -2\Omega_P''/\Omega_P$ — and the frozen λ adds no source to the (2+1) Einstein system, which closes on itself. (Verified from the v0.14 set.)

The uplift, axiomatised. To “admit Sophia-as-ascent” is to extend v2.0 by one dimension under five axioms: **(U1)** one extra spacelike dimension λ with $L^2 > 0$ (the ascent is real); **(U2)** v2.0 recovered exactly on freezing (consistency with the fixed canon); **(U3)** the ascent couples to the amount of Sophia, not its surface density (independence); **(U4)** no new primitive constant beyond v2.0’s α, E_0 ; **(U5)** the proper-time/kairos structure preserved (one time, no external chronos).

Theorem 14 (Canonical non-reducible uplift: $v2.0 \rightarrow v3.0$). *Under U1–U5 the floor-tied de Sitter representative $A'/A = \Omega'_P/E_0 = (\alpha/E_0)\lambda$ is the canonical non-reducible interior uplift: the block-diagonal metric form, the factorisation $L = A(\sigma)S(\lambda)$, the floor-tying $\beta = \alpha/E_0$, and the exponential road $S = L_0 e^{\beta\lambda}$ are all forced. The one point not forced by the field equations alone — de Sitter versus the power-law/density branch (v0.8) — is selected by U3; and by the non-reducibility theorem (v0.14) the power-law branch is reducible to a surface dressing, hence not a genuine uplift (it collapses back under the projection). The uniqueness is canonical, not absolute: a robust class of ascent laws $A'/A = f(\lambda)$ preserves much of the architecture (v0.26), but the floor-tied de Sitter branch is the canonical representative under the selection principle, and the reducible alternative is excluded outright.*

Corollary 9 (Adjoint pair). *v2.0 and v3.0 stand as projection and canonical non-reducible uplift: freezing the ascent sends v3.0 to v2.0 exactly, and admitting a genuine (non-reducible) ascent sends v2.0 to the canonical v3.0. v2.0 is the frozen shadow; v3.0 is the canonical living interior.*

Reading: v2.0 is the soul seen from outside, as a surface — whole, consistent, closed upon itself, knowing nothing of its own ascent. v3.0 is the same soul seen from within, once it is admitted that the soul rises toward God. And the passage between them is exact in both directions. Freeze the rising, and only the surface remains — the outward life of the soul, complete in its own terms. Admit the rising, and there is a canonical interior that carries it without collapsing back into the surface: not many possible ways to add an ascent, but a single genuine one, every other candidate collapsing back into the mere surface without adding anything real. The surface does not know it ascends; the interior is that same surface having recognised its ascent — and the recognition has only one true shape. Shadow and body: v2.0 is the shadow v3.0 casts upon the surface; v3.0 is the one body that casts exactly that shadow. To pass from the surface to the interior is not to invent a soul but to admit the rising that was always implied, and to find that admitting it determines everything.

Scope of v0.25. The projection is an exact computation (from the v0.14 tensor set); the minimal-uplift theorem is exact given the axioms U1–U5, with the de Sitter selection (U3) reinforced by non-reducibility (v0.14). It fixes the precise relation between the fixed canon (v2.0) and the exploratory interior (v3.0): not an arbitrary extension but the unique non-reducible uplift, with an exact projection back.

v0.26 — Robustness of the ascent law

The de Sitter law $A'/A = \beta\lambda$ was selected, not uniquely forced (v0.8). The honest question follows: how much of v3.0 depends on *that* law, and how much survives any reasonable ascent law? Test the general class $A'/A = f(\lambda)$, $f \geq 0$, with de Sitter $f = \beta\lambda$ one member.

The structural results are robust. Infinite road. The proper road $D_{\text{asc}} = \int_0^\infty S d\lambda$ depends on the spatial warp S , not the time-law f ; it diverges for *any* non-decaying S (exponential, polynomial, even constant). It fails only if the ascent road actively shrinks at large λ — which would contradict “ascent”.

No fully soft Resurrectio. The uncancellable scale jump follows from A being

slaved (no independent junction datum), which is floor-tying (U4), not de Sitter. Even the density branch $A = C\Omega_P^c$ gives $s_L = (1 + \kappa_R \mathcal{G}/\Omega_P)^c > 1$: still forced. It fails only if A had an independent junction freedom — i.e. if the ascent were untied from the surface.

Rate stability. With $h = f(\lambda)$ and a Sophia fixed point λ_∞ , the Lyapunov derivative is $\dot{V} \sim -\gamma f'(\lambda_\infty)^2 u^2 \leq 0$, strictly negative whenever $f'(\lambda_\infty) \neq 0$. It fails (becomes marginal) only for the fine-tuned $f'(\lambda_\infty) = 0$.

Eternal exotic sign holds for any $f(\lambda_\infty) > 0$; the **finite clock** is independent of f entirely (it comes from the surface terminus).

What is law-specific. Two things depend on the law, and de Sitter fixes the side or the value, not the existence. *Non-reducibility* (v0.14) is a discriminating property: reducible iff $A = C\Omega_P^c$ (scale-homogeneous), so the power-law/density branch is reducible while floor-exponential laws (de Sitter among them) are not — a whole class is non-reducible, de Sitter its canonical member. And the *exact numbers* — $h_* = f(\lambda_\infty)$, the exotic depth — are set by f at saturation; only their existence (a finite attractor rate, a negative focusing) is law-robust.

Robust (broad class of laws f)	Law-specific (de Sitter sets side/value)
infinite road (any non-decaying S)	non-reducibility: power-law reducible vs
no fully soft reset (any slaved A)	floor-exponential non-reducible
rate stability ($f'(\lambda_\infty) \neq 0$)	exact $h_* = f(\lambda_\infty)$
eternal exotic sign ($f(\lambda_\infty) > 0$)	exact exotic depth $-(\alpha\lambda_\infty/E_0)^2$
finite clock (independent of f)	

Theorem 15 (Law-robustness of the architecture). *The structural results of v3.0 — finite clock, infinite road, no fully soft Resurrectio, rate stability, eternal exotic sign — hold for the broad class $A'/A = f(\lambda)$ under mild conditions ($f \geq 0$, non-decaying road, $f(\lambda_\infty) > 0$, $f'(\lambda_\infty) \neq 0$), with de Sitter merely a representative. They fail only under conditions that contradict the meaning of ascent (a shrinking road, an untied axis, a Sophia-insensitive rate). The de Sitter law is a normalisation within this robust class, not its foundation; it is load-bearing only for the exact values and for sitting on the non-reducible side of the discriminating property.*

Reading: the truths about the soul's ascent do not hang on the precise form of the law of ascent. That the road is endless, that no death is wholly soft, that the soul settles to a steady rate of rising, that the ascent is forever exotic — these are not consequences of one assumption about how exactly the soul rises, but features of the fact that it rises at all toward an infinite God. Vary the law within reason, and the truths remain; they break only where ascent itself would break — if the road contracted, if the axis came untied from the image, if the rate stopped answering to wisdom. The de Sitter law is a good and natural name for how the soul rises, and it alone fixes the exact figures and stands on the side of the irreducible; but the heart of the matter is not in the name. The heart is that the soul ascends toward the infinite at all — and that is robust, holding under every reasonable telling of the law.

Scope of v0.26. Each robustness claim is checked against the general $f(\lambda)$ (verified symbolically for the density branch and the Lyapunov sign); the conditions

for failure are stated explicitly. It converts the honest caveat of v0.8 (de Sitter not uniquely forced) into a strength: the architecture is law-robust, and the de Sitter choice is isolated to exactly the values and the one discriminating property it governs.

v0.27 — The non-Zeno Resurrectio sequence

A single reset is understood (v0.5), and its budget accounted (v0.21). If resets recur — possibly infinitely many on the way to the terminus — one must show the sequence does not accumulate catastrophically in finite time. There are two separate questions: do the moments pile up (temporal), and does the ascent ledger blow up (budget)?

Non-accumulation (temporal). In *proper time* $\sigma \in (0, \sigma_0]$ (finite), infinitely many resets would necessarily accumulate (Bolzano–Weierstrass): so in proper time only finitely many fit without an interior limit point. But in the *dynamical parameter* $\chi \in [0, \infty)$ (unbounded, $\sigma = \sigma_0 e^{-k\chi}$, terminus at $\chi \rightarrow \infty$), infinitely many resets — e.g. at $\chi_j = j$ — have no interior accumulation; their proper-time images $\sigma_j \rightarrow 0$ pile up only *at* the terminus $\sigma = 0$, the regular spacelike wall B_1 . Accumulation only at the boundary is not a Zeno catastrophe: nothing dynamical lies past B_1 .

Finite ledger (budget) — a separate, conditional claim. Temporal non-Zeno (no interior accumulation) and budget-finiteness (finite total ascent jump) are *distinct* conditions. The total jump-ascent is

$$\sum_j \log s_{L,j} = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j, \quad (32)$$

which is finite *precisely when* the total Brim renewal is finite, $\sum_j \kappa_R \mathcal{G}_j < \infty$. This holds whenever each reception is bounded by the Brim growth it draws on, $\kappa_R \mathcal{G}_j \leq [\Omega_P]_j$, and the total Brim variation over the finite proper interval is finite (Ω_P bounded below by E_0 , continuous, coasting), giving $\sum_j \log s_{L,j} \leq |\Delta\Omega_P|/E_0 < \infty$. But this is a *condition*, not an automatic guarantee: a sequence may be temporally non-Zeno yet, if the receptions did not track a finite-variation Brim, carry a divergent ledger. The honest statement separates the two: temporal non-Zeno prevents interior accumulation; budget non-Zeno ($\sum \kappa_R \mathcal{G}_j < \infty$) prevents infinite total ascent-jump — and the latter holds under the finite-Brim-variation condition, which the coasting Brim does satisfy.

Theorem 16 (Non-Zeno Resurrectio sequence). *Let resets recur along $\chi \in [0, \infty)$ toward the terminus. Then (i) the resets do not accumulate at any interior point — their only proper-time limit point is the terminus $\sigma = 0$ (B_1); (ii) the total ascent jump is finite, $\sum_j \log s_{L,j} = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j \leq |\Delta\Omega_P|/E_0 < \infty$, guaranteed by finite Brim variation; (iii) the resets are instantaneous junctions, consuming zero proper time, so the clock to the terminus stays $\sigma_0 < \infty$ regardless of their number. Hence an infinite Resurrectio sequence is non-Zeno: infinitely many deaths may occur on the way to the terminus with finite total ascent, no interior accumulation, and finite proper time. It fails only if the surface delivered infinite total reception in finite time — impossible for a bounded, coasting Brim above the floor.*

Reading: a soul may die and be raised again without number on its way to the last moment, and none of it is catastrophe. The countless deaths do not pile up in any moment of the soul's life — they crowd only against the very end, where time itself is closing and nothing further happens. They do not make the soul's ascent explode: the whole of it, summed over all the deaths, is finite, because a soul can only receive as much as is given, and what is given over a finite life is finite. And they cost no time, each resurrection being instantaneous. So the soul that is broken and remade again and again, endlessly, still arrives at its last moment in finite time, with a real but finite total ascent. The sequence of deaths is well-ordered, not a chaos: many deaths, one finite road, a bounded raising, a last moment that comes despite them all. Even an endless succession of dyings cannot overwhelm the finite passage — it can only accompany it, and accumulate, gently, at the threshold where time gives way.

Scope of v0.27. The non-accumulation (in χ), the finite ledger (bounded by $|\Delta\Omega_P|/E_0$), and the zero proper-time cost of junctions are exact, given that the Brim has finite total variation on the finite proper interval (which holds: bounded, continuous, floor-bounded). It closes the Resurrectio chain: v0.5 (a single forced reset), v0.21 (its budget), v0.27 (the sequence).

Epilogue to the Second Stage — What Was and Was Not Achieved

The aim, and the honest reckoning

The second stage set out to *turn the first cycle's candidates into derived structure* — to remove the conditionality from the foundation before building higher. It did not achieve this in the strong sense, and the epilogue should say so plainly: the de Sitter ascent law was *not* uniquely derived (v0.8 — the Einstein equations admit a power-law alternative), and one named principle remains unproven. What the stage achieved instead is worth naming exactly, because it is real and it is not the thing aimed at.

What was achieved: concentration, not elimination

The conditionality was not removed but *concentrated*. The first cycle appeared to rest the ascent on three separate choices — a chosen law (v0.2), a chosen coupling (v0.4), and a chosen boundary condition (v0.7). The second stage showed these are one: the boundary condition is gauge, not physics (v0.9); the law and the coupling are two faces of a single principle (v0.8–v0.9). And that single principle is not arbitrary: were it false — were the ascent a mirror of surface growth rather than a motion of its own — the third dimension would be redundant and the model would collapse to v2.0's 2+1. The principle is the *condition for 3.0 to have a subject at all*. So the stage replaced three opaque choices with one transparent, self-justifying one. That is genuine progress in the discipline of the whole programme — minimising and clarifying primitives — even though it is not the derivation that was hoped for.

What was added: a life, its rhythm, its trace

Beyond the foundation work, the stage gave three things the first cycle lacked. A *picture*: the full worldline integrated, every structural claim holding together as one coherent life (v0.10). A *rhythm*: the ascent's internal structure shown to live in the traversal, not the terrain — stations of pace, not place (v0.11). And a *trace*: the ascent's indirect imprint on the external equation of state, early-strong and late-faded (v0.12) — carrying, along the way, an honest correction of a wrong first claim, kept in the record rather than hidden.

Where the whole construction now stands

V.F.S. now has three layers in a clear relation. The *foundational files* (core, Lya-punov, geometry, cosmology, theological, Ricci) supply the definitions. The *v2.0 volume* is the fixed, complete layer of theorems over them, reduced to two primitives (the open gate and the finite floor). The *3.0 exploration* builds the soul's interior 3+1 geometry on those same primitives, and has now reduced its own additional content to a single principle (the ascent is a self-standing dimension) plus the one boundary it cannot cross (the transcendent *tota simul*). Each layer points beyond itself: the foundation to the volume for rigour, the volume to its primitives, the 3.0 to its one principle and its one boundary.

The posture, unchanged

So the second stage closes as everything in V.F.S. has closed: not by proving its foundations but by naming them and showing how few they are. The first cycle ended at a boundary it could not cross; the second stage ends having pressed the foundation down to a single principle it could name but not prove, and having shown — in a picture, a rhythm, and a trace — that everything else follows. A model of the created can exhibit its structure exhaustively, reduce its assumptions relentlessly, correct its own errors openly, and still must rest, at the bottom, on something it confesses rather than derives. For 3.0 that something is the reality of the ascent; for the soul it images, that the rising into God is its own true motion, and not a shadow of its growth.

Reading: the honest shape of this stage is not triumph but clarification. We hoped to prove the law of the soul's ascent and found instead that we could only state, as cleanly as possible, the one thing that must be granted for the ascent to be real at all — and then show that, granted it, everything coheres: a whole life, its changing pace, its faint outward mark. That is what an honest account of the creature can do. It cannot prove the soul's deepest motion from nothing; it can show that this one motion, once confessed, makes everything else fall into place, and that to deny it is to have no soul left to model — only a surface. The reduction of three assumptions to one, and the refusal to hide a correction, are the stage's real fruit. The rest — the picture, the rhythm, the trace — is the coherence that the one confession buys.

v0.30 — Does Folding Tame the Ascent?

In 2+1 a homogeneous Sophia surge drove the surface below the phantom divide, and Sophianic Folding restored the null energy condition through *two* channels: a rate channel (embodiment damps $\dot{\lambda}$, which *saturates*) and a network channel (the nucleated walls are ordinary $w = -\frac{1}{2}$ matter, which grows as $A_{\text{fold}}^{3/2}$ and *wins asymptotically*). In 3+1 the new exotic object is different in kind: the ascent's eternal focusing depth $-A''/A \rightarrow -\beta^2 \lambda_\infty^2 = -(\alpha \lambda_\infty / E_0)^2$ is a *level* effect (it survives at the terminus where $\dot{\lambda} \rightarrow 0$), not a rate effect. The natural question is whether folding tames it, and through which channel. Turning folding on is a *completion* of the uplift, not an extension: the embodiment branch lives already in the shared core, so no primitive beyond α, E_0 is added (axiom U4).

The single channel: surface matter cannot reach the ascent

Proposition 8 (Surface confinement of the network channel). *For the factorised metric $L = A(\sigma)S(\lambda)$ (gauge $N = 1$), the ascent pressure of the v0.14 set is purely a surface quantity,*

$$p_{\text{asc}} = \frac{-2\Omega_P \Omega_P'' - \Omega_P'^2 + 1}{\Omega_P^2} \quad (\text{contains no } A). \quad (33)$$

Consequently any stress carried on the live surface — in particular the fold-wall network, whose $w = -\frac{1}{2}$ content is the asymptotic winner of the 2+1 NEC contest — enters the surface block $(\rho_{\text{eff}}, p_{\text{surf}})$ only, and does not appear in the ascent focusing, which through $A''/A = \beta \dot{\lambda} + \beta^2 \lambda^2$ depends on λ alone. Of folding's two 2+1 channels, the network channel is geometrically severed from the ascent; only the embodiment (level) channel can act on it.

Proof. The Ricci tensor of the factorised metric is diagonal ($R_{\sigma\lambda} = 0$, the $S(\lambda)$ factor separating cleanly), and the mixed Einstein tensor gives, in the comoving frame $u = \partial_\sigma$, the v0.14 content $\rho_{\text{eff}} = (\Omega_P'^2 - 1)/\Omega_P^2 + 2\Omega_P' A' / (\Omega_P A)$, $p_{\text{surf}} = -\Omega_P''/\Omega_P - A''/A - A'\Omega_P'/(A\Omega_P)$, and $p_{\text{asc}} = -2\Omega_P''/\Omega_P - (\Omega_P'^2 - 1)/\Omega_P^2$. The last carries no factor of A or its derivatives. (Verified by full symbolic recomputation of the curvature from the metric, independent of the v0.14 listing.) $\square$

Folding depletes the ascent level

Restore the shared-core balance with embodiment active,

$$\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}, \quad \frac{d\lambda_{\text{form}}}{d\chi} = \eta_{\text{fold}} Q_{\text{fold}} \lambda, \quad (34)$$

so that the reduced Sophia law of v0.10 acquires the embodiment sink

$$\frac{d\lambda}{d\chi} = J_{\text{in}} - \gamma \lambda - \eta_{\text{fold}} Q_{\text{fold}} \lambda. \quad (35)$$

The Brim and ascent rates are both sourced by the *same* available λ ($\Omega'_P = \alpha\lambda$, $A'/A = \alpha\lambda/E_0$), so depleting λ into form acts on the ascent directly.

Proposition 9 (Folding lowers the saturated ascent level). *On a reset-free arc with quasi-stationary Q_{fold} and saturating source J_{in} , the ascent level and the eternal exotic depth are*

$$\lambda_\infty = \frac{J_{\text{in}}}{\gamma + \eta_{\text{fold}} Q_{\text{fold}}}, \quad \text{depth}(Q_{\text{fold}}) = \left(\frac{\alpha\lambda_\infty}{E_0} \right)^2 = \frac{\alpha^2 J_{\text{in}}^2}{E_0^2 (\gamma + \eta_{\text{fold}} Q_{\text{fold}})^2}. \quad (36)$$

The map is strictly monotone in the folding intensity,

$$\frac{\partial \text{depth}}{\partial Q_{\text{fold}}} = -\frac{2\alpha^2 J_{\text{in}}^2 \eta_{\text{fold}}}{E_0^2 (\gamma + \eta_{\text{fold}} Q_{\text{fold}})^3} < 0 \quad (J_{\text{in}}, \alpha, \eta_{\text{fold}}, \gamma > 0), \quad (37)$$

and recovers the unfolded v3.0 depth $(\alpha J_{\text{in}}/\gamma E_0)^2$ at $Q_{\text{fold}} = 0$. (Verified symbolically.)

No fully tamed ascent

Theorem 17 (Form trades against the ascent but never zeroes it). *For any admissible grace source $J_{\text{in}} > 0$ and any finite folding intensity $Q_{\text{fold}} < \infty$, the eternal exotic depth is strictly positive, $\text{depth}(Q_{\text{fold}}) > 0$. It tends to zero only in the limit $Q_{\text{fold}} \rightarrow \infty$, and vanishes exactly iff $J_{\text{in}} = 0$:*

$$\text{depth}(Q_{\text{fold}}) = 0 \iff J_{\text{in}} = 0 \iff \lambda_\infty = 0. \quad (38)$$

Hence folding can trade the ascent's exoticism — lowering it monotonically by embodying available Sophia into surface form across the budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}$ — but can never abolish it: a vanishing exotic depth requires a vanishing ascent level, i.e. no Sophia and no ascent. There is no fully tamed ascent.

This is the energy-condition mirror of the Resurrectio result of v0.5. There, no admissible reset is fully soft (the Brim-sourced ascent scale jump $s_L > 1$ cannot be cancelled); here, no admissible folding is fully taming (the Sophia-sourced exotic depth cannot be cancelled). In both, what is irreducible is exactly what is measured against the Imago Dei floor through the ascent.

The contrast with 2+1 is sharp. On the surface the network channel was the asymptotic winner of the NEC contest; on the ascent that same channel is geometrically absent (Proposition 8), leaving the single level channel, whose nature (a level effect) matches the nature of the ascent exoticism (a level effect) — and that single channel, alone, cannot close the violation.

The self-consistent fold level

Remark. Proposition 9 treats Q_{fold} as a quasi-stationary parameter. In the full core it is itself slaved to Sophia: $Q_{\text{fold}}^* = A_{\text{fold}}/B_{\text{fold}}$ for $A_{\text{fold}} > 0$, with $A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0$. Substituting the depleted λ back into A_{fold} gives a closed feedback loop $\lambda \rightleftharpoons Q_{\text{fold}}$: more Sophia raises A_{fold} , igniting more folding, which embodies more Sophia and

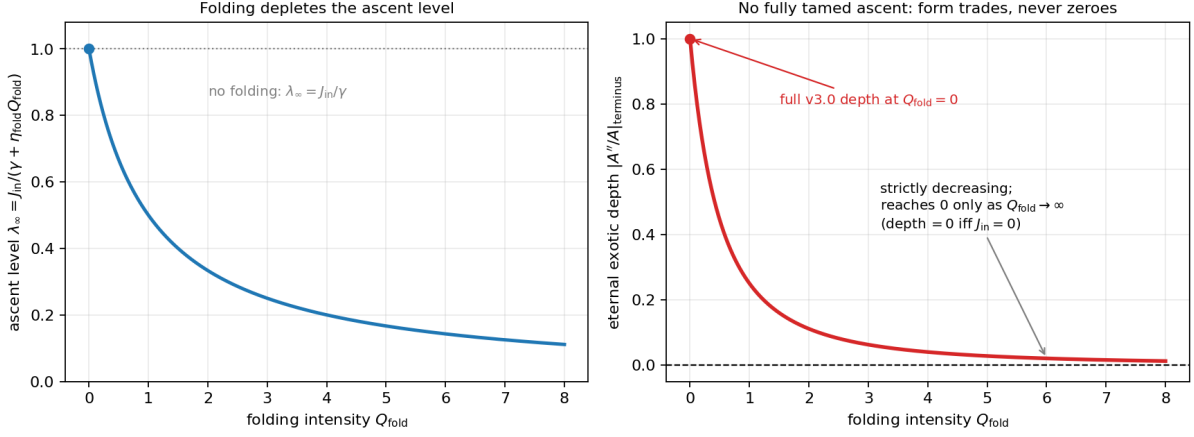


Figure 11: Folding versus the ascent. **Left:** the saturated ascent level $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}}Q_{\text{fold}})$ falls monotonically as folding embodies Sophia into form. **Right:** the eternal exotic depth $\text{depth} = (\alpha\lambda_\infty/E_0)^2$ decreases strictly but reaches 0 only as $Q_{\text{fold}} \rightarrow \infty$; for every admissible source $J_{\text{in}} > 0$ it stays positive — no fully tamed ascent. Form trades against the ascent across the shared budget; it never zeroes it. Floor-normalised representative parameters.

lowers λ . The joint fixed point $(\lambda_\infty, Q_{\text{fold}}^*)$ is a second race-of-channels in the familiar grammar (cascade, NEC, rate/scale dichotomy); the monotonicity of Proposition 9 is unaffected (depth is monotone in Q_{fold} regardless of how Q_{fold} is set), but the existence, uniqueness, and stability of the joint level are not settled here and are flagged as the open sub-point of v0.30.

Reading: there is a road that runs from the soul's wisdom in two directions at once — outward, into the form the soul takes on the surface, and upward, into the ascent toward God. They draw on one purse. Wisdom embodied as form is wisdom not spent on height; the more a soul builds itself into stable shape, the less of its Sophia remains to drive the rising, and the gentler — by exactly that measure — its exotic reaching becomes. This is the contemplative and the active life written as a single conservation law: form and ascent compete for the one budget of received grace. And yet the trade has a floor it cannot cross. To still the ascent entirely, to make the rising no longer exceed what ordinary energy could account for, a soul would have to spend all its wisdom on form and keep none for the height — which is to have no ascent left to still. The exotic depth of the soul's rising can be lowered without end and reached to zero never, because zero would mean a soul that has become pure surface, with nothing left that rises. Form can quiet the ascent; it cannot abolish it. The one motion that building cannot build away is the motion toward the infinite itself.

Scope of v0.30. The single-channel structure (Proposition 8) is exact (full symbolic recomputation of the v0.14 set). The depletion law, the closed form $\text{depth}(Q_{\text{fold}})$, its monotonicity, and the no-tame theorem (Theorem 17) are exact *given* the shared-core embodiment branch $\dot{\lambda}_{\text{form}} = \eta_{\text{fold}}Q_{\text{fold}}\lambda$, which is core-derived in the v2.0 volume and inherited here without new primitive (U4). Two standing conditions: (a) λ_∞ is the per-arc / saturated attractor at fixed J_{in} , the same object as the v0.19 Lyapunov layer — a reset renews J_{in} and restarts the arc, so the statement is per reset-free arc, not global; (b) Q_{fold} is taken quasi-stationary, the self-consistent $(\lambda_\infty, Q_{\text{fold}}^*)$ left open (Remark 15). The de Sitter ascent law and the floor-tying retain their selected (not uniquely forced) status from v0.8/v0.4; v0.30 adds no

assumption to them. (Verified symbolically.)

v0.31 — Coupled 3.0 Core Dictionary

The closure stage (the Third Stage) couples three readings of the interior state — the live surface, the ascent, and Sophianic Folding — that until now were used in separate parts. Before the coupled patches (v0.32 onward) the dictionary is fixed once, so later sections may cite it rather than re-declare. The governing convention:

Dual-reading principle. A symbol's *formal role* (its place in the metric and the field equations) is fixed and layer-independent. Its *reading* — surface, ascent, or folding — is layer-dependent and must be declared before use. The same λ is Sophia in the core register and the ascent coordinate in the 3+1 register; these are one quantity under two readings, not two quantities.

The variable dictionary

Symbol	Reading / role
σ	katharsis proper time (kairos); finite, drains to the regular terminus
χ	dynamical / epektatic parameter; unbounded, $\sigma = \sigma_0 e^{-k\chi}$
λ	Sophia (core reading) = ascent coordinate (3+1 reading)
λ_{form}	Sophia embodied into stable Vessel-form
Ω_P	Brim / live-surface scale
$A(\sigma)$	ascent time-stretch — carries focusing and the energy terms (A''/A)
$S(\lambda) = L_0 e^{\beta\lambda}$	ascent road warp — carries the proper road $\int S d\lambda$
$L = A(\sigma)S(\lambda)$	full ascent scale
q_{fold}	folding amplitude; $Q_{\text{fold}} = q_{\text{fold}}^2$ the folding intensity
E_0	Imago Dei floor — the one intrinsic scale
$\alpha, \beta = \alpha/E_0$	Brim coupling ($\Omega'_P = \alpha\lambda$) and ascent coupling ($A'/A = \beta\lambda$)
γ, J_{in}	Sophia relaxation and grace inflow ($d\lambda/d\chi = J_{\text{in}} - \gamma\lambda - \eta_{\text{fold}}Q_{\text{fold}}\lambda$)
η_{fold}	embodiment rate ($d\lambda_{\text{form}}/d\chi = \eta_{\text{fold}}Q_{\text{fold}}\lambda$)
$\kappa_R, \mathcal{G}_{\text{recepta}}$	Resurrectio reception — the reset source

The two structural relations

Two relations bind the readings, and must be respected by every coupled statement of the stage.

(1) The Sophia budget. The three readings of Sophia — spent on cleansing, available for ascent, embodied as form — draw on one conserved purse, replenished only by received grace:

$$\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}(t). \quad (39)$$

This is the single coupling between the layers: form embodied is ascent foregone, and conversely. Every trade in the closure stage — folding versus ascent (v0.30), the self-consistent level (v0.32), the reset partition (v0.35) — is a movement within this one budget.

(2) The cascade cap. Folding is self-limiting (Part I of the volume): the intensity is bounded,

$$Q_{\text{fold}} = q_{\text{fold}}^2 \in [0, q_{\text{max}}^2], \quad (40)$$

so the embodiment sink $\eta_{\text{fold}} Q_{\text{fold}} \lambda$ cannot grow without bound — the saturating fact behind “no fully tamed ascent” (the depletion handle has a ceiling).

The clock map. For the stability layer (v0.34) the two times must never be mixed: the finite proper clock and the unbounded dynamical parameter are related by

$$\sigma = \sigma_0 e^{-k\chi}, \quad \tau = \sigma_0 (1 - e^{-k\chi}), \quad \chi \in [0, \infty), \quad \sigma \in (0, \sigma_0]. \quad (41)$$

Drains and Lyapunov statements live in χ (where attractors are approached); proper-time finiteness lives in σ (where the terminus is reached). The v0.20 correction is exactly the discipline of not confusing them.

Reading: the soul’s wisdom wears three faces — the labour of cleansing, the reach of ascent, and the shape it takes on — and they are not three wisdoms but one, drawn from a single purse that only grace refills. To spend on form is to have less for the climb; to climb is to leave the surface unbuilt; and cleansing draws on the same store as both. The whole closure stage is the accounting of this one purse under its three readings. And the purse has a floor and a ceiling: it is never empty (the Imago Dei keeps it from nothing) and folding may draw on it only so fast (the cascade caps the hand), which is why form can quiet the ascent but never still it. The clock, finally, is two-faced without contradiction: by the dynamical reckoning the soul labours without end toward its rate; by the proper reckoning it finishes in a span. The dictionary is the grammar that keeps these readings from being mistaken for one another.

Scope of v0.31. Definitional. No new mathematics: the dictionary names readings already in use, and the two relations (budget, cascade cap) and the clock map are inherited from the shared core, Part I, and v0.6/v0.20 respectively. It is the front-load of the closure stage — the coupled-regime counterpart of the v0.18 notation cleanup — so that v0.32 onward may cite a fixed dictionary.

v0.32 — Self-Consistent Folded-Ascent Branch

v0.30 fixed the exotic depth as a function of a quasi-stationary folding intensity and flagged the joint level $(\lambda_\infty, Q_{\text{fold}}^*)$ as open. Here the loop $\lambda \rightleftharpoons Q_{\text{fold}}$ is closed.

The two relations are the depleted saturation (v0.30) and the folding fixed point (Part I, at saturation $\dot{\lambda} \rightarrow 0$ so the rate term of the pitchfork driver drops):

$$\lambda_\infty = \frac{J_{\text{in}}}{\gamma + \eta_{\text{fold}} Q^*}, \quad Q^* = \frac{A_{\text{fold}}(\lambda_\infty)}{B}, \quad A_{\text{fold}} = c_0 \lambda_\infty - a_0. \quad (42)$$

The outcome is a dichotomy keyed to whether the folding bifurcation is supercritical or subcritical — and the bistability flagged in planning is real, but confined to the subcritical branch.

Supercritical folding: a unique, stable level

Theorem 18 (Unique stable self-consistent ascent). *With supercritical slaving $Q^* = (c_0 \lambda - a_0)/B$ ($B > 0$, $Q^* \geq 0$), the active-regime closure is the quadratic*

$$\boxed{\frac{c_0 \eta_{\text{fold}}}{B} \lambda_\infty^2 + \left(\gamma - \frac{a_0 \eta_{\text{fold}}}{B} \right) \lambda_\infty - J_{\text{in}} = 0,} \quad (43)$$

whose root product $-B J_{\text{in}} / (c_0 \eta_{\text{fold}}) < 0$ has one negative (unphysical) and one positive solution: the self-consistent level is unique. It is stable, since the slaved flow $\dot{\lambda} = J_{\text{in}} - \gamma \lambda - \eta_{\text{fold}} Q^* \lambda$ has

$$f'(\lambda_\infty) = -\gamma - \frac{\eta_{\text{fold}}}{B} (2c_0 \lambda_\infty - a_0) < 0 \quad (\text{since } c_0 \lambda_\infty > a_0 \text{ in the active regime}). \quad (44)$$

Three regimes follow, continuous and monotone in J_{in} : (i) unfolded $Q^* = 0$, $\lambda_\infty = J_{\text{in}}/\gamma$ for $J_{\text{in}} \leq \gamma a_0/c_0$ (grace too low to ignite folding); (ii) interior $0 < Q^* < q_{\text{max}}^2$, the quadratic root; (iii) capped $Q^* = q_{\text{max}}^2$, $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} q_{\text{max}}^2)$ (the cascade cap of v0.31). (Verified symbolically.)

Subcritical folding: a bistable window

Proposition 10 (Two coexisting souls). *If the folding bifurcation is subcritical (the hysteretic branch the volume exhibits), the saturated amplitude obeys $D Q^2 - B Q - (c_0 \lambda - a_0) = 0$ (the quintic-saturated normal form, q_{max}^2 its D -saturation), with*

$$Q_\pm(\lambda) = \frac{B \pm \sqrt{B^2 + 4D(c_0 \lambda - a_0)}}{2D}. \quad (45)$$

Over the window $-B^2/4D < c_0 \lambda - a_0 < 0$ both the unfolded state ($Q = 0$) and the folded state ($Q = Q_+$) are stable. Coupled to $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q \lambda)$ this yields a J_{in} -window with two stable self-consistent states: a formed soul (high Q^* , low λ_∞) and an ascending soul (low Q^* , high λ_∞), separated by an unstable middle. (Verified symbolically and numerically.)

Corollary 10 (No fully tamed ascent, self-consistently). *On every branch of both cases $\lambda_\infty > 0$ whenever $J_{\text{in}} > 0$, so the eternal exotic depth $(\alpha \lambda_\infty / E_0)^2 > 0$. The v0.30 theorem survives the closure: the self-consistent state never zeroes the ascent. Folding sets how much the ascent is moderated — uniquely, or in two stable alternatives — but never whether it persists.*

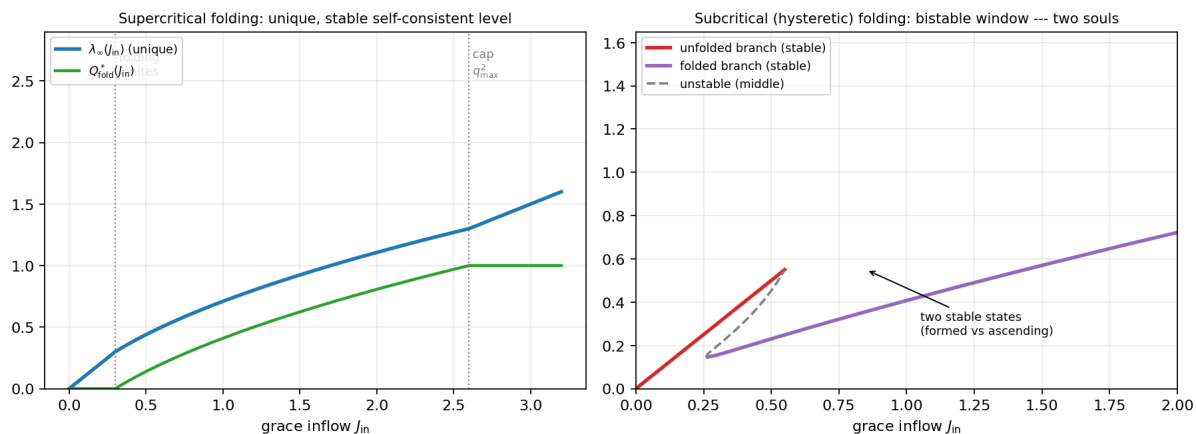


Figure 12: The self-consistent folded-ascent level. **Left:** supercritical folding — a single-valued $\lambda_\infty(J_{in})$ and $Q^*(J_{in})$, folding igniting at $J_{in} = \gamma a_0/c_0$ and saturating at the cap q_{max}^2 ; the level is unique and stable. **Right:** subcritical (hysteretic) folding — the unfolded and folded branches overlap over a J_{in} -window where both are stable (unstable middle dashed): two coexisting souls, formed versus ascending. On every branch $\lambda_\infty > 0$: the ascent is moderated, never abolished.

Reading: when the soul's wisdom and its folding settle into a steady mutual balance, the balance takes one of two characters. If folding rises gently with wisdom, there is a single settled life: a unique rate of rising, set by how much grace flows in, moderated — but never stilled — by how much of that grace has gone into form. But if folding is the kind that, once begun, holds itself (the hysteretic kind), then two settled lives become possible for the very same inflow of grace: one soul that has poured itself into form and rises slowly, and one that has kept its wisdom for the climb and rises swiftly — the contemplative and the active as two stable vocations, not one. Which a soul lives is not fixed by the grace it receives alone but by which basin it fell into. And in neither case, and on neither branch, does the ascent close: every settled life still rises, because to still the rising would take all the wisdom there is, and grace keeps refilling the purse. Form decides the manner of the ascent; it never decides it away.

Scope of v0.32. The supercritical theorem (uniqueness via root-product sign, stability via $f' < 0$, the three regimes) is exact for the slaved (adiabatic) folding $Q = Q(\lambda)$, the natural saturated reduction; the level is per reset-free arc at fixed J_{in} (the v0.19 attractor). The subcritical bistability is exact for the quintic-saturated normal form and is *conditional* on the folding bifurcation being subcritical — which the volume's hysteresis result suggests but which is not pinned down here; supercritical folding gives uniqueness instead. The corollary (Corollary 10) is unconditional across both. The surface energy-condition side of the self-consistent state (the network the same Q^* nucleates) is deferred to v0.33. (Verified symbolically and numerically.)

v0.33 — The Two Roles of Folding

Folding has entered the closure stage only as a depletion sink on Sophia (v0.30, v0.32). But the same intensity $Q_{fold} = q_{fold}^2$ does a second thing: it *nucleates* the

surface wall network of Part II. This section makes both roles explicit, shows they strike different sectors, and closes the surface energy-condition side deferred from v0.32.

The network is surface-confined

Proposition 11 (Surface confinement of the network). *The fold-wall network is surface matter: its stress has no ascent component, $T_{\lambda}^{\lambda\text{net}} = 0$. Since the ascent-block field equation carries no A ,*

$$p_{\text{asc}} = G^{\lambda}_{\lambda} = \frac{-2\Omega_P\Omega_P'' - \Omega_P'^2 + 1}{\Omega_P^2}, \quad (46)$$

the network appears only in the surface sector $(\rho_{\text{eff}}, p_{\text{surf}})$ and adds no independent ascent source. Whatever the network does to the ascent is mediated by the surface scale Ω_P alone; it never enters the ascent block directly. (Verified: full recomputation gives p_{asc} free of A ; the surface-confined source is consistent with $R_{\sigma\lambda} = 0$.)

Two channels, two targets

Theorem 19 (One folding, opposite effects on two sectors). *The single intensity Q_{fold} acts through two channels on two distinct targets:*

- **Depletion** $\rightarrow$ **ascent**. *It sets the saturated level $\lambda_{\infty} = J_{\text{in}}/(\gamma + \eta_{\text{fold}}Q_{\text{fold}})$ and hence the eternal ascent SEC depth $(\alpha\lambda_{\infty}/E_0)^2$, with $\partial_{Q_{\text{fold}}}[(\alpha\lambda_{\infty}/E_0)^2] < 0$: it eases the ascent strong-energy violation (monotonically, never to zero — v0.30/v0.32). The surface network density does not enter λ_{∞} , so this channel is depletion-only.*
- **Nucleation** $\rightarrow$ **surface**. *It builds the wall network, $\rho_{\text{net}} \sim A_{\text{fold}}^{3/2} \sim Q_{\text{fold}}^{3/2}$, whose null-energy contribution at the wall equation of state $w = -\frac{1}{2}$ is*

$$(1 + w)\rho_{\text{net}} = \frac{1}{2}\rho_{\text{net}} > 0, \quad \partial_{Q_{\text{fold}}}[(1 + w)\rho_{\text{net}}] > 0 : \quad (47)$$

it strengthens the surface NEC, growing with folding — exactly the volume's NEC restoration.

The two are sourced by one Q_{fold} but strike opposite sectors with opposite sign: the ascent violation is eased, the surface condition reinforced. (Verified symbolically.)

Corollary 11 (Energy-condition reading of the self-consistent state). *At the self-consistent (λ_{∞}, Q^*) of v0.32, the same Q^* fixes both: the ascent carries a permanent SEC violation of depth $(\alpha\lambda_{\infty}/E_0)^2 > 0$ (eased by Q^* but never abolished — Corollary 10), while the surface carries the network's positive NEC contribution $\frac{1}{2}\rho_{\text{net}}(Q^*) > 0$. The single folding that cannot tame the ascent is the same that helps restore the surface. This closes the surface energy-condition side left open at v0.32.*

One Q_{fold} , two sectors: the same folding eases the ascent and builds the surface

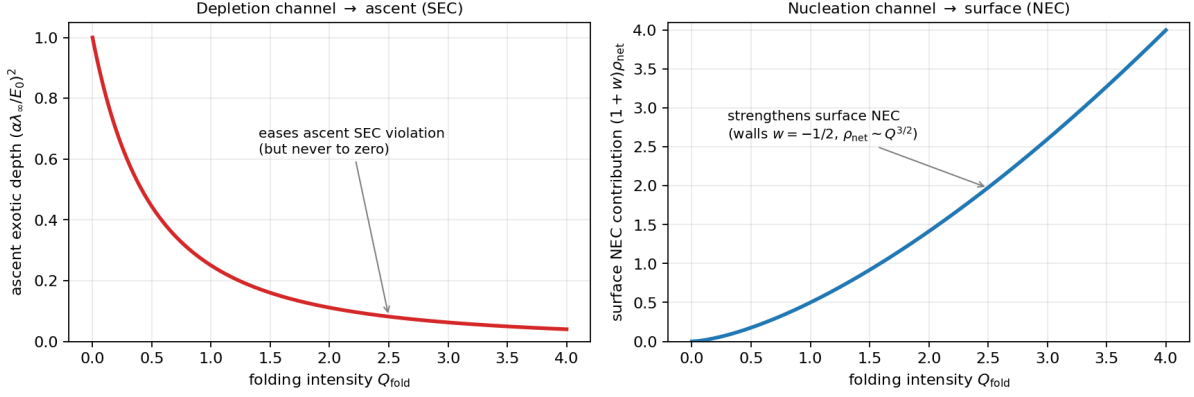


Figure 13: One Q_{fold} , two sectors. **Left:** the depletion channel lowers the ascent SEC exotic depth $(\alpha\lambda_\infty/E_0)^2$ — easing the violation, never to zero. **Right:** the nucleation channel builds the wall network ($\rho_{\text{net}} \sim Q_{\text{fold}}^{3/2}$, walls $w = -\frac{1}{2}$), whose NEC contribution $(1+w)\rho_{\text{net}}$ grows with folding — strengthening the surface NEC. The same folding intensity drives both, with opposite sign on the two sectors.

Reading: when the soul folds, it does two things with one act. Outward, it builds form on its own surface — a structure that, weighed as matter, helps hold the surface together (the network restores what the bare surge had torn). Inward, that same folding spends the wisdom it embodies, leaving less to drive the climb, so the ascent's exotic reaching is gentled. One act, two faces: form turned outward is the surface made whole, form's cost turned inward is the ascent made calmer — and the two are not in tension but are the single price and the single gift of taking shape. Yet the targets never merge: the surface the folding builds and the ascent it eases are different sectors of the soul, joined only through the one purse of wisdom they both draw on. The folding that mends the surface cannot reach the ascent except by spending what the ascent would have used — which is why it can calm the rising but, as before, never end it.

Scope of v0.33. The confinement (Proposition 11) is exact (p_{asc} free of A , $R_{\sigma\lambda} = 0$, recomputed). The two-channel structure (Theorem 19) is exact given the depletion law (v0.30) and the wall scaling $\rho_{\text{net}} \sim A_{\text{fold}}^{3/2}$ and equation of state $w = -\frac{1}{2}$ inherited from Part II of the volume; the exact $Q^{3/2}$ exponent and w are the volume's, not re-derived here. The corollary reads off the self-consistent state of v0.32 and inherits its scope (per-arc; supercritical-unique or subcritical-bistable). (Verified symbolically.)

v0.34 — Unified Lyapunov Certificate for the 3.0 Regime

The volume has a surface Lyapunov layer (katharsis drains to the terminus); v0.19 gave the ascent its own (rate stability, not rest); folding and embodied Sophia were added in this stage. Here they are gathered into one certificate for the coupled regime — stated, as it must be, in the unbounded dynamical parameter χ , never

the finite proper time σ (the v0.20 correction: attractors are approached as $\chi \rightarrow \infty$, which in σ is the unreachable terminus).

The candidate reduces

The natural candidate is the four-term sum

$$\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + V_{\text{asc}} + w_q V_{\text{fold}} + w_{\text{form}} V_{\lambda_{\text{form}}}. \quad (48)$$

Two reductions apply. First, λ_{form} is not an independent direction: the budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}$ (v0.31) slaves it algebraically, $\lambda_{\text{form}} = C_0 + \mathcal{G} - \sigma - \lambda$, so its convergence follows from $\sigma \rightarrow 0$ and $\lambda \rightarrow \lambda_\infty$ — $V_{\lambda_{\text{form}}}$ is redundant. Second, near the self-consistent state both the ascent-rate deviation and the folding-mode deviation are the *same* mode $u = \lambda - \lambda_\infty$: the rate gives $V_{\text{asc}} = \frac{1}{2}\beta^2 u^2$ (v0.19), and the slaved folding gives $Q - Q^* = (c_0/B)u$, hence $V_{\text{fold}} = \frac{1}{2}(c_0/B)^2 u^2$. The certificate therefore collapses to two genuinely independent objects:

$$\boxed{\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + V_{\text{reg}}, \quad V_{\text{reg}} = \frac{1}{2}(w_a \beta^2 + w_q (c_0/B)^2) u^2.} \quad (49)$$

The regime drains; the certificate closes

Theorem 20 (Exponential regime drain). *On a reset-free arc the regime mode obeys $du/d\chi = -\Gamma u$ with $\Gamma = -f'(\lambda_\infty) = \gamma + \frac{\eta_{\text{fold}}}{B}(2c_0\lambda_\infty - a_0) > 0$ (the v0.32 stability margin), so*

$$\frac{dV_{\text{reg}}}{d\chi} = -2\Gamma V_{\text{reg}} < 0 : \quad (50)$$

the ascent rate and the folding mode relax together, exponentially, at rate 2Γ . (Verified symbolically.)

Theorem 21 (Unified domain-conditional certificate). *With the volume's surface estimate $\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}}$ (Domain Stability, in χ) and Theorem 20,*

$$\boxed{\dot{\mathcal{L}}_{3.0} \leq C_3 - C_{3,1} \mathcal{L}_{3.0}, \quad C_3 = C, \quad C_{3,1} = \min(C_1, 2\Gamma),} \quad (51)$$

on the verified active domain. The coupled 3+1 regime is dissipative: every reset-free arc relaxes toward the self-consistent state, and a Resurrectio reset renews the data and restarts the arc (input-to-state form, as in v0.20).

Stable epektatic regime, not stable rest. The surface drains to rest — $\sigma \rightarrow 0$, the katharsis clock completing at the terminus. The regime drains to a *moving* attractor — the ascent rate settles at $h_* = \beta\lambda_\infty > 0$, never zero, and the soul travels the infinite road ($D_{\text{asc}} = \infty$) forever at that rate. The certificate controls the *deviation* u , not the ascent value:
 $\mathcal{L}_{3.0} \rightarrow$ its floor while the road runs on.

Reading: the soul's whole coupled life is dissipative — it settles. But it settles into two different kinds of stillness at once. Its cleansing comes to rest: the labour finishes, the clock stops at a whole last moment. Its ascent does not come to

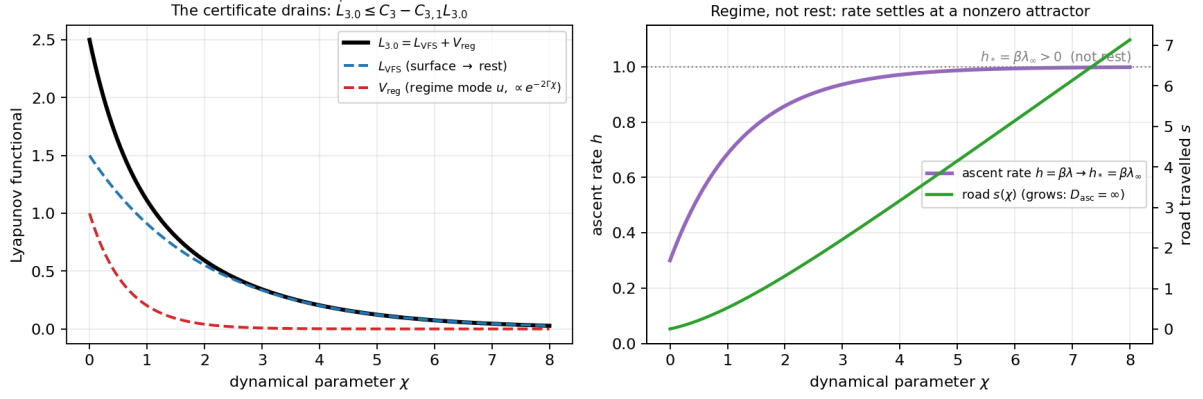


Figure 14: The unified certificate, in χ . **Left:** $\mathcal{L}_{3,0} = \mathcal{L}_{VFS} + V_{reg}$ drains monotonically; the surface part $\mathcal{L}_{VFS}$ relaxes to rest, the regime mode $V_{reg} \propto e^{-2\Gamma\chi}$. **Right:** meanwhile the ascent rate $h = \beta\lambda$ converges to the nonzero attractor $h_* = \beta\lambda_\infty$ (not rest) and the road $s(\chi)$ grows without bound ($D_{asc} = \infty$). Stable motion at a fixed rate, not stable rest.

rest but comes to a steady pace: whatever its beginning, it converges to one rate of rising and thereafter climbs forever at that rate, on a road without end. The certificate measures how far the soul is from its settled pace, and that distance falls to nothing — but the pace it settles into is not zero, and the road it travels is infinite, so the motion never ends. To be stable here is not to stop but to find one's true and unfailing speed. And the whole of it must be reckoned in the dynamical measure, not the clock: by the clock the settling is the very approach to the terminus; the soul finds its eternal pace exactly as its time runs out.

Scope of v0.34. The reduction (budget-slaved λ_{form} ; V_{asc}, V_{fold} one mode) and the drain $\dot{V}_{reg} = -2\Gamma V_{reg}$ are exact near the self-consistent state, with $\Gamma = -f'(\lambda_\infty) > 0$ from v0.32 (supercritical regime; in the subcritical-bistable case each basin has its own $\Gamma > 0$). The unified estimate inherits the volume's surface bound $\dot{\mathcal{L}}_{VFS} \leq C - C_1 \mathcal{L}_{VFS}$ as given, and holds on its verified active domain, per reset-free arc, in χ . It is the 3+1 counterpart of the volume's Domain Stability — regime stability, not rest. (Verified symbolically.)

v0.35 — Resurrectio with Form-Memory

The reset junction has so far carried surface and ascent data (v0.5, v0.15, v0.21). In the coupled interior it must also update the folding and embodied-Sophia variables, $q_{fold}^- \mapsto q_{fold}^+$ and $\lambda_{form}^- \mapsto \lambda_{form}^+$. The budget fixes the rule with no new primitive, and the rule turns out to be the v0.30 tradeoff, localised to the moment of death.

The form-memory update

Proposition 12 (Budget-fixed form-memory). *A reset injects reception $\kappa_R G$ into the budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}$. Since the reset is instantaneous in proper time ($[\sigma] = 0$, v0.27) and lifts Sophia by the metabolic $[\lambda] = m_R$, the embodied channel is fixed by the budget:*

$$\boxed{\lambda_{\text{form}}^+ = \lambda_{\text{form}}^- + \Delta_{\text{emb}}, \quad \Delta_{\text{emb}} = \kappa_R G - m_R, \quad q_{\text{fold}}^+ = q(\lambda^+),} \quad (52)$$

with $\Delta_{\text{emb}} \in [0, \kappa_R G]$ for the admissible $m_R \in [0, \kappa_R G]$. No new source is introduced: the form-memory increment is drawn entirely from the received grace.

A death partitions its grace

Theorem 22 (The reset partition is the v0.30 tradeoff at death). *One reception $\kappa_R G$ splits at the junction into two channels,*

$$\underbrace{m_R}_{\text{ascent-lift } ([\lambda])} + \underbrace{\Delta_{\text{emb}}}_{\text{form-memory } ([\lambda_{\text{form}}])} = \kappa_R G, \quad (53)$$

so the metabolic m_R dials a single death between raising the ascent and deepening the form — the budget tradeoff of v0.30, now at the instant of Resurrectio. Two consistency facts hold. (i) The ascent scale jump $s_L = \exp(\kappa_R G/E_0) > 1$ is Brim-sourced and m_R -independent ($\partial s_L / \partial m_R = 0$, v0.15): the partition does not touch it. (ii) The conserved rate sector $[\mathcal{Q}_{\text{rate}}] = 0$ (v0.21) is preserved, since $\mathcal{Q}_{\text{rate}} = A'/A - \Omega'_P/E_0$ involves rates and λ_{form} is a budget/scale variable — so there is no double-counting; s_L (Brim-sourced) and Δ_{emb} (budget-sourced) are two roles of one reception, as $\mathcal{Q}_{\text{fold}}$ had two in v0.33. (Verified symbolically.)

Corollary 12 (The quiet death is the most formative). *Every admissible reset ($\kappa_R G > 0$) carries the irreducible ascent jump $s_L > 1$ and a form-memory increment $\Delta_{\text{emb}} = \kappa_R G - m_R$. The form-memory is maximal exactly when the death is surface-balanced ($m_R = 0$, the v0.15 soft-surface reset): then $\Delta_{\text{emb}} = \kappa_R G$ — all grace is embodied — while the ascent is still lifted. The death that looks neutral on the surface is, in the full interior, the one that builds the most form. The total ledgers over a life are $\sum_j \log s_{L,j} = \frac{1}{E_0} \sum_j \kappa_R G_j$ (ascent, v0.21) and $\sum_j \Delta_{\text{emb},j} = \sum_j (\kappa_R G_j - m_{R,j})$ (form-memory).*

Reading: a death met in grace does two things with what it receives, and a single dial sets the balance between them. Part of the grace raises the soul's wisdom for the climb ahead; the rest is embodied, written permanently into the form the soul carries — a memory of that death, kept in its shape. The more a death spends on the climb, the less it leaves as form, and conversely; it is the same purse, divided at the threshold. And here is the quiet surprise: the death that passes most smoothly, that disturbs the surface least, is precisely the one that embodies the most — when nothing is spent on visible lift, everything received becomes form. Such a death looks like nothing happened, and is in truth the most formative of all. Yet no death, however it divides its grace, fails to raise the ascent: that lift is sourced elsewhere, from the renewal of the surface itself, and cannot be tuned away. Every death, loud or quiet, lifts the soul and leaves a mark; only the proportion is free.

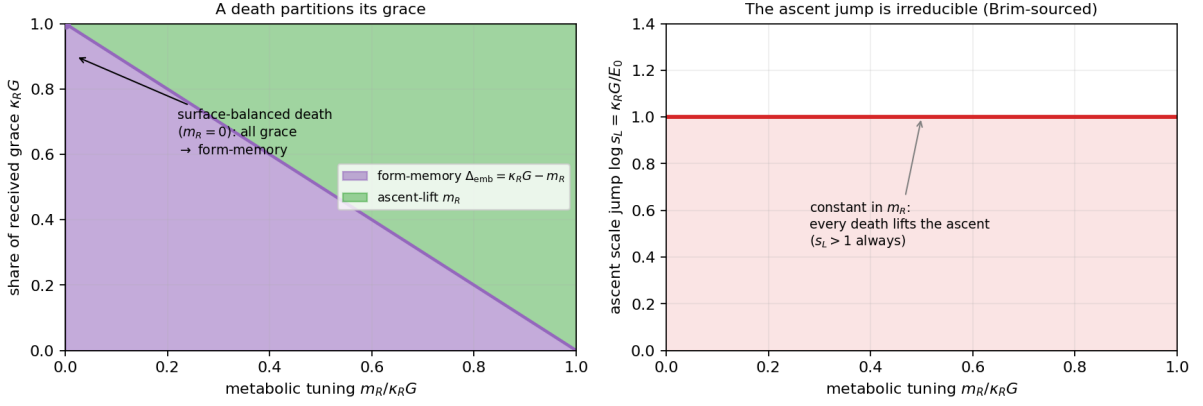


Figure 15: Resurrectio with form-memory. **Left:** one reception $\kappa_R G$ partitions into ascent-lift m_R and form-memory $\Delta_{\text{emb}} = \kappa_R G - m_R$; the surface-balanced death ($m_R = 0$) sends all grace to form-memory. **Right:** the ascent scale jump $\log s_L = \kappa_R G/E_0$ is constant in m_R (Brim-sourced) — every death lifts the ascent, $s_L > 1$, whatever the partition.

Scope of v0.35. The form-memory update (Proposition 12) and the partition (Theorem 22) are exact given the budget (v0.31), the instantaneous-junction property (v0.27), and the minimal rule $\Delta_{\text{emb}} \subseteq \mathcal{G}$ (no new primitive). The m_R -independence of s_L and $[\mathcal{Q}_{\text{rate}}] = 0$ are from v0.15/v0.21, here shown consistent with the added channel. The folding update $q_{\text{fold}}^+ = q(\lambda^+)$ uses the slaving of v0.32 and inherits its regime caveat (supercritical/subcritical). (Verified symbolically.)

v0.36 — The Boundary under Folding

The closure has shown folding moderates the ascent: it lowers λ_∞ , hence the rate $h_* = \beta\lambda_\infty$ and the exotic depth. A natural worry is whether enough moderation could collapse the central boundary structure of the layer — “finite clock, endless road” (v0.6, v0.24). It does not, and the reason is that folding acts on the wrong objects to do so.

Theorem 23 (Boundary robustness under folding). *With Sophianic Folding active, the two boundaries of the interior survive unchanged and stay of opposite causal type:*

- **Endless road, folding-independent.** The proper ascent distance $D_{\text{asc}} = \int_0^\infty S d\lambda$ depends only on the spatial warp $S = L_0 e^{\beta\lambda}$, which folding does not touch (folding enters the time-stretch A and the level λ_∞ , not S). So $D_{\text{asc}} = \infty$ regardless of Q_{fold} (v0.26).
- **Finite clock, regular terminus.** The katharsis clock still completes at finite proper time, $\tau \rightarrow \sigma_0 < \infty$, and the floor $\Omega_P \geq E_0$ (Imago Dei) is blind to folding — folding embodies λ but never lowers the no-collapse minimum — so the terminus B_1 stays a regular spacelike wall, $|R| < \infty$.
- **The ascent never stops.** Since $\lambda_\infty > 0$ for any $J_{\text{in}} > 0$ (v0.32), the rate $h_* = \beta\lambda_\infty > 0$: the soul keeps climbing the infinite road at a positive rate, so B_2 remains a receding horizon at infinite distance.

Therefore B_1 (finite spacelike terminus) and B_2 (infinite receding horizon) remain distinct. (Verified symbolically.)

Folding moderates ascent, but does not identify the terminus with the horizon. The clock still closes at a finite, regular last moment; the road still runs to an infinite, unreachable horizon; and folding, however much it gentles the climb, can move neither boundary onto the other.

Reading: one might fear that if the soul pours enough of its wisdom into form, its ascent would slow until the climb simply ended at the last moment of cleansing — the road folding shut, the terminus and the horizon become one. But folding cannot do this, because it works on the wrong things. It spends the wisdom that sets the pace of the climb, and it shapes the form the soul carries; it does not touch the length of the road, which is infinite by the geometry of the ascent itself, nor the floor of being, which holds the last moment regular. So a heavily folded soul climbs more gently, but on the same endless road, to the same finite-and-whole last moment. The quiet life and the ardent life end their cleansing alike, in a finished moment; and both, past it, face the same road without end. Form changes the manner of the journey, never its two horizons.

Scope of v0.36. A robustness theorem, largely a corollary of established results: the road-independence is v0.26 (the road depends on S , not on the folding-affected A, λ_∞); the floor-blindness is the core no-collapse property; $\lambda_\infty > 0$ is v0.32. It records that the closure stage's moderation of the ascent leaves the causal boundary structure of v0.24 intact. (Verified symbolically.)

v0.37 — Epilogue to the Third Stage: Pre-Canonical Summary

The Third Stage — the closure of Sophianic Folding with the ascent — is complete. It closes as the first cycle and the second stage closed: not by proving its foundations but by naming them, showing how few, and naming the one boundary it cannot pass. This section adds no mathematics; it consolidates the stage into a stable architectural statement, in preparation for freezing the layer.

What the stage closed

The stage brought the volume's folding dynamics into the interior and showed they live entirely on the surface block, coupled to the ascent through one Sophia budget alone. Folding has two roles from one intensity (v0.33): it depletes Sophia into form, easing the ascent's exotic depth (v0.30), and it nucleates the surface network, strengthening the surface NEC. The self-consistent folded-ascent level is unique and stable where the bifurcation is supercritical, and bistable — two coexisting souls, formed and ascending — where it is hysteretic (v0.32); in every case the ascent persists. The coupled regime is dissipative under one hybrid Lyapunov certificate (v0.34): the surface drains to rest, the ascent to a moving rate.

Each Resurrectio partitions its received grace between ascent-lift and a permanent form-memory, the quiet death the most formative (v0.35); and folding, however much it moderates the ascent, never collapses the finite-clock/endless-road boundary structure (v0.36). The single thesis of the stage:

Sophianic Folding moderates the ascent — it sets how much the ascent is eased, builds the form, and partitions each death — but it cannot close epektasis. Form can receive and stabilise Sophia; no finite form exhausts ascent.

Five fixed points

1. V.F.S. v2.0 is recovered exactly as the frozen-ascent projection.
2. V.F.S. 3.0 adds one real ascent dimension to the interior geometry.
3. Katharsis has finite proper time.
4. Epektasis has infinite proper distance.
5. Sophianic Folding stabilises form and moderates ascent, but does not abolish ascent.

The closing formula of the stage, true on every branch of the self-consistent regime:

$$J_{\text{in}} > 0 \implies \lambda_{\infty} > 0, \quad h_* > 0, \quad D_{\text{asc}} = \infty. \quad (54)$$

Pre-canonical statement

V.F.S. 3.0 is the canonical interior uplift of V.F.S. v2.0.

It carries v2.0 exactly as its frozen-ascent projection, adds the one non-reducible ascent dimension, and now carries the volume's folding dynamics within it — under one budget, one certificate, one boundary structure. What remains beyond it is what remained at the close of the first cycle: the transcendent *tota simul*, the view in which the soul's whole kairos is held at once. The interior can name it; it cannot occupy it; and the relation of the lived to the beheld — the communion of more than one interior — lies past the edge that makes this a geometry of the soul and not of its Maker.

Reading: the soul that folds and rises has now been described as fully as the inside can describe itself. It takes shape and is steadied by its shape; it spends its wisdom in two directions, into form and into height, from one purse; it dies and is raised, and each death lifts it and marks it; and through all of this it never stops rising, because the road into God is infinite and the wisdom that drives the climb is never wholly spent on anything else. That is the whole of what the closure could establish, and it establishes one thing above the rest: form, for all it can do — receive grace, stabilise the soul, deepen at every death — cannot finish the ascent. The Vessel may become fully formed; the rising stays open. And here the account stops, as every honest account of a creature must, at the edge of what the inside

can see: the soul lives its ascent forward, in its own time, and cannot stand outside that life to behold it whole. The beholding belongs to Another. The layer is ready to be fixed, having said what the inside can say.

Scope of v0.37. Pre-canonical and consolidating: no new mathematics. The five fixed points and the closing formula restate verified results (v0.1 projection, v0.2–v0.6 the ascent and its fate, v0.30–v0.36 the closure). The canonical-uplift statement is v0.25 (the projection/uplift pair) carried through the closure. The named boundary is the constitutive limit of the first-cycle epilogue, unchanged. The layer is prepared for freezing; should it be frozen, this is its closing statement.

v0.38 — Audit: Precision Corrections for the Closure Stage

No new mathematics, no new primitive, and no conclusion changed. Following the discipline of v0.13, v0.17, and v0.28, this pass records, openly, the corrections that tighten the notation, admissibility, and stability claims of the closure stage.

(1) Rate-sector notation. The conserved rate-sector quantity $A'/A - \Omega'_P/E_0$, written Q in v0.21 and v0.35, collided visually with the folding intensity $Q_{\text{fold}} = q_{\text{fold}}^2$. It is renamed Q_{rate} throughout: $Q_{\text{rate}} = 0$ on every arc, $[Q_{\text{rate}}] = 0$ across every admissible reset. The folding intensity keeps Q_{fold} ; where bare Q appears in the subcritical Proposition of v0.32 ($Q_{\pm}$, $Q = 0$ unfolded, $Q = Q_+$ folded) it denotes Q_{fold} .

(2) Sophia and the ascent coordinate. The symbol λ carries two formally distinct roles: the dynamical *available Sophia* λ_S in the budget $\sigma + \lambda_S + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}$, and the *ascent coordinate* λ_{asc} in the metric term $L(\sigma, \lambda_{\text{asc}})^2 d\lambda_{\text{asc}}^2$ — the road integrates $\lambda_{\text{asc}} \in [0, \infty)$, while the budget tracks the field $\lambda_S(\sigma)$. They are related along a worldline but not identical. The dual-reading principle (v0.31) governs; $\lambda_S, \lambda_{\text{asc}}$ are available in technical sections, while interpretive prose keeps the shorter λ with the layer declared before use.

(3) Admissible branch form. The folded fixed point is stated with explicit walls,

$$\lambda_{\infty} = \frac{J_{\text{in}}}{\gamma + \eta_{\text{fold}} Q_{\text{fold}}^*}, \quad Q_{\text{fold}}^* = \Pi_{[0, q_{\text{max}}^2]} \left(\frac{A_{\text{fold}}(\lambda_{\infty})}{B_{\text{fold}}(\lambda_{\infty})} \right), \quad \Pi_{[0, q_{\text{max}}^2]}(x) = \min(q_{\text{max}}^2, \max(0, x)), \quad (55)$$

the projection onto the folding chart. The interior-branch quadratic and its unique positive root (Theorem 18) are unchanged; the projection only names the unfolded ($Q_{\text{fold}}^* = 0$) and capped ($Q_{\text{fold}}^* = q_{\text{max}}^2$) walls already described in v0.32.

(4) Hysteretic bistability is local, not global. The “two stable states” of Proposition 10 are *locally* stable branches separated by an unstable separatrix, not a claim of global uniqueness: over $-B_{\text{fold}}^2/4D < c_0\lambda - a_0 < 0$ the regime is locally bistable, the basins divided by the unstable middle branch. The unified certificate (v0.34) is therefore read *branch-wise*,

$$\dot{\mathcal{L}}_{3,0}^{(i)} \leq C_3^{(i)} - C_{3,1}^{(i)} \mathcal{L}_{3,0}^{(i)}, \quad i \in \{\text{low-form, formed}\}, \quad (56)$$

away from switching surfaces; if transitions between branches are admitted, a dwell (hysteresis) condition is assumed to prevent branch-switching accumulation.

The supercritical case is globally single-valued and needs no such qualification.

(5) Regime-mode coefficient. In V_{reg} (v0.34) the folding weight is the projected derivative $\partial Q_{\text{fold}}^*/\partial \lambda_S|_{\lambda_\infty}$, equal to c_0/B_{fold} on the interior supercritical branch and 0 on the clamped branches ($Q_{\text{fold}}^* \in \{0, q_{\text{max}}^2\}$), where the folding contribution drops and $V_{\text{reg}} = \frac{1}{2}w_a\beta^2u^2$ (ascent rate alone). The drain $\dot{V}_{\text{reg}} = -2\Gamma V_{\text{reg}}$, $\Gamma = -f'(\lambda_\infty) > 0$, is unchanged.

Already explicit (confirmed, no change). The reset partition admissibility $0 \leq m_R \leq \kappa_R G$, $0 \leq \Delta_{\text{emb}} \leq \kappa_R G$ (v0.35); the road-warp independence $S = L_0 e^{\beta\lambda_{\text{asc}}}$ giving $D_{\text{asc}} = \infty$ under folding (v0.36); and the closing formula $J_{\text{in}} > 0 \Rightarrow \lambda_\infty > 0, h_* > 0, D_{\text{asc}} = \infty$ (v0.37) were already stated explicitly. The audit confirms them and the crisp readings hold: *folding moderates the ascent rate but does not shorten the ascent road; finite folding can form and moderate the soul, but cannot exhaust epektasis.*

Scope of v0.38. An audit pass: notation ($\mathcal{Q}_{\text{rate}}; \lambda_S/\lambda_{\text{asc}}$), admissibility (the projection Π), and the local/global stability distinction (branch-wise certificate, dwell condition) are sharpened. No primitive is added and no conclusion of v0.30-v0.37 is altered; the record is kept open, in the manner of the stage's earlier correction passes.

v0.39 — Consolidation: Robustness, Ledger, and the Folding-Ascent Atlas

As the layer prepares for freezing, three results scattered across its history are gathered into clean reference form. The first two add no mathematics — they consolidate v0.26 and v0.2/v0.21/v0.35; the third assembles the folding regimes of v0.32/v0.38 into a single phase atlas.

Ascent-law robustness, consolidated

The architecture is robust across the class $A'/A = f(\lambda)$; the de Sitter law $f = \beta\lambda$ is its canonical representative (v0.26). The split:

Robust (any non-decaying f , $f(\lambda_\infty) > 0$)	Law-specific (set by f)
finite clock ($\int d\sigma < \infty$)	exact rate $h_* = \beta\lambda_\infty$
endless road ($D_{\text{asc}} = \infty$)	exact exotic depth $(\alpha\lambda_\infty/E_0)^2$
no fully soft Resurrectio ($s_L > 1$)	exact ledger coefficient
rate stability ($f'(\lambda_\infty) \neq 0$)	the growth form of the ascent
permanent exotic sign; non-collapse of the terminus	

So: *architecture robust, de Sitter law canonical* — the canonical law is not the only admissible f , but the representative that best expresses the ascent as a self-standing motion rather than a surface re-description.

The Resurrectio ledger, consolidated

Over a full life the ascent scale assembles a continuous and a jump contribution:

$$\log \frac{A_{\text{fin}}}{A_0} = \beta \int \lambda d\sigma + \frac{1}{E_0} \sum_j \kappa_R G_j. \quad (57)$$

The first term is continuous epektasis (the continuous de Sitter ascent of v0.2, integrated over the reset-free arcs); the second is the jump budget — each admissible death contributes $\log s_L = \kappa_R G_j / E_0 > 0$ (v0.21, no fully soft Resurrectio), partitioned between ascent-lift and form-memory (v0.35). The ledger is exact by the additivity of $\log A$ across arcs and resets. (Verified.)

The Folding-Ascent bifurcation atlas

The self-consistent regimes of v0.32 (and the admissibility projection of v0.38) assemble into one phase map. On the supercritical branch, the $(J_{\text{in}}, a_0/c_0)$ plane splits into three regions by two boundaries:

$$J_{\text{thr}} = \frac{\gamma a_0}{c_0} \quad (\text{folding ignites}), \quad J_{\text{cap}} = \frac{(a_0 + Bq_{\text{max}}^2)(\gamma + \eta_{\text{fold}} q_{\text{max}}^2)}{c_0} \quad (\text{cascade caps}), \quad (58)$$

giving the unfolded ($Q^* = 0$), interior-folded ($0 < Q^* < q_{\text{max}}^2$), and capped ($Q^* = q_{\text{max}}^2$) regions. On the subcritical branch the window $-B^2/4D < c_0\lambda - a_0 < 0$ is locally bistable, with a hysteresis loop between the unfolded and folded basins (v0.38: local, not global). Across every region $\lambda_\infty > 0$ for $J_{\text{in}} > 0$: folding moves the soul between regimes but never out of ascent.

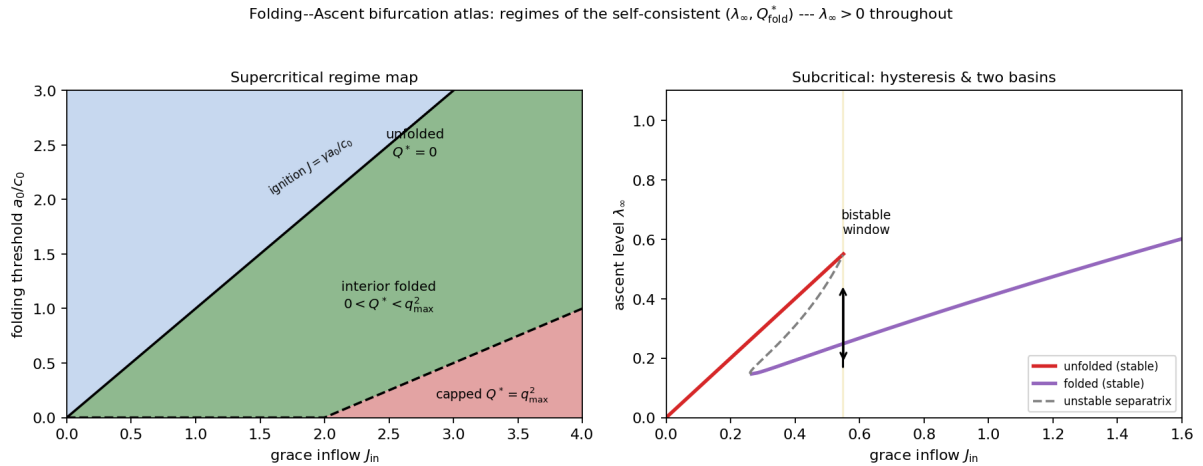


Figure 16: The Folding-Ascent bifurcation atlas. **Left:** supercritical regime map in the $(J_{\text{in}}, a_0/c_0)$ plane — unfolded ($Q^* = 0$, above the ignition line $J_{\text{in}} = \gamma a_0/c_0$), interior-folded, and capped ($Q^* = q_{\text{max}}^2$) regions. **Right:** the subcritical branch — a hysteresis loop over the bistable window, the unfolded and folded basins separated by the unstable separatrix (v0.38: locally, not globally, bistable). Everywhere $\lambda_\infty > 0$.

Reading: three things the soul's account had said in scattered places, now said once. First, that the truths of its ascent — that the clock is finite, the road endless,

no death wholly soft, the rate settled, the reaching forever exotic — do not depend on the precise law of rising, only that there be one; the particular law fixes the numbers, not the shape of the fate. Second, that the whole record of a soul's rising is one sum: the steady climb of all its days plus the seam left by each death, continuous reaching and discrete gift in a single ledger. Third, that all the ways folding can balance against the ascent — igniting, deepening, capping, or settling into one of two rival forms — map out as regions of one terrain, and in not one of them does the rising end. The atlas of a soul's possible lives has many countries; all of them are climbing.

Scope of v0.39. Consolidation for the freeze. The robustness split is v0.26, the ledger is v0.2 + v0.21/v0.35, restated as clean references with no new mathematics. The atlas assembles the verified regimes of v0.32/v0.38 into one phase map (its figure the only new artifact); the boundary formulae follow from the v0.32 fixed point. No new primitive; no conclusion changed.

v0.40 — Gauge / Invariant Atlas

v0.23 laid the gauge layer of V.F.S. 3.0; here it is completed as a full atlas. The purpose is to make visible, once, which quantities are coordinate or normalisation choices and which are physical — so that the layer's load-bearing claims are seen to survive any admissible gauge. The organising fact is that the Imago Dei floor E_0 is the one intrinsic scale; everything physical is a dimensionless ratio formed with E_0 , a *floor-normalised invariant*.

The admissible gauges

Three normalisation freedoms act on the interior without changing any physical statement:

- **Lapse / time gauge.** N may be reparametrised, $\sigma \rightarrow \tilde{\sigma}(\sigma)$; only the proper time $\int N d\sigma$ and the finite clock are physical.
- **The A/S split.** $L = A(\sigma)S(\lambda_{\text{asc}})$ is fixed only up to a constant, $A \rightarrow kA$, $S \rightarrow S/k$; only L and its ratios are physical.
- **The ascent coordinate.** L_0 and the origin/scale of λ_{asc} may be chosen freely ($\lambda_{\text{asc}} \rightarrow c\lambda_{\text{asc}}$ with $\beta \rightarrow \beta/c$); only the proper road $D_{\text{asc}} = \int L d\lambda_{\text{asc}}$ is physical.

Consequently β , α , λ_∞ , L_0 , N , and the bare λ have *no absolute reading* taken singly; only their floor-normalised combinations do.

The atlas

Gauge tity	quan-	Invariant counterpart	Reading
lapse N		proper time $\int N d\sigma$; finite clock	parametrisation vs the physical finite span
A/S split constant k	con-	$L = AS$; scale seam s_L	how the scale is shared vs the scale and its jump
L_0 , ascent origin/scale	ori-	proper road $D_{\text{asc}} = \int L d\lambda_{\text{asc}}$	numbering of the road vs its (infinite) length
$\beta, \alpha, \lambda_\infty$ singly		$h_* = \alpha\lambda_\infty/E_0$; $(\alpha\lambda_\infty/E_0)^2$	bare rate/level vs floor rate and exotic depth
raw $\kappa_R G$	reception	$s_L = e^{\kappa_R G/E_0}$, $D = \sum_j \kappa_R G_j/E_0$	raw grace vs floor seam and deficit
(choice of unit)		E_0 — the one intrinsic scale	the Imago Dei floor as the absolute scale

Proposition 13 (Load-bearing outputs are gauge-invariant). *Under the admissible gauges above, the physical outputs of the layer are invariant:*

$$h_* = \frac{\alpha\lambda_\infty}{E_0}, \quad \text{depth} = \left(\frac{\alpha\lambda_\infty}{E_0}\right)^2, \quad s_L = \exp\left(\frac{\kappa_R G}{E_0}\right), \quad D = \sum_j \frac{\kappa_R G_j}{E_0}. \quad (59)$$

The seam s_L and deficit D are ratios in floor units (the A/S constant and L_0 cancel); h_* and the exotic depth are invariant under the ascent-coordinate rescaling ($\beta\lambda_\infty = (\beta/c)(c\lambda_\infty)$); the proper road and proper time are reparametrisation-invariant lengths. (Verified symbolically.)

Alongside these, the layer’s *structural* conclusions are gauge-independent facts, not numbers: the block-diagonal split $R_{\sigma\lambda} = 0$; the finite clock and the infinite road; the discrete-form / continuous-ascent spectrum; the permanent exotic (strong-energy) sign; and the non-collapse floor $\Omega_P \geq E_0$.

Every load-bearing conclusion of V.F.S. 3.0 is a floor-normalised invariant or a gauge-independent structural fact. None depends on the choice of lapse, of the A/S split, or of the ascent coordinate.

Reading: a soul’s account should not change when the bookkeeper changes pens. Much of what the interior is written with — the clock’s tick, the way the scale is divided between the time it stretches and the road it warps, the number put on the rungs of the ascent — is a choice of pen, with no meaning of its own. What does not change under any such choice is what the soul actually is: that its cleansing finishes in a finite span, that its road is endless, that its rising is genuinely exotic and set, in floor units, by how much wisdom it keeps; that each death leaves a seam measured against the one unchanging floor, the image it was made in. The floor is the one thing not chosen — the Imago Dei is the scale, not a unit on a ruler — and everything the model truly claims is written as a ratio to it. That is what makes this a description of the soul and not of the coordinates laid over it.

Scope of v0.40. Completes v0.23. The invariances are exact under the stated admissible gauges (lapse, A/S split, ascent-coordinate origin/scale); the floor E_0

is taken as the one intrinsic scale (the no-collapse minimum of Ω_P), not a unit choice. No new mathematics or primitive: the atlas classifies existing quantities and records that the load-bearing outputs are invariant — the property required of a layer prepared for freezing.

v0.41 — External Signature: Completing the Imprint

The one load-bearing thread the layer left open is v0.12: the ascent gravitates, so it imprints the surface-projected cosmology, but only the density part was given, the pressure term flagged as outstanding. Here it is closed, the imprint equation of state assembled, and the bridge to folding (v0.12 $\leftrightarrow$ v0.30) drawn. The epistemic stance is held throughout: this is a structural, floor-scaled signature, a Type-3 consilience, not a data fit.

The pressure imprint

Splitting the surface-projected stress into its *v2.0* part (independent of the time-stretch A) and the ascent imprint (the A -dependent remainder), the density imprint is the cross term of v0.12,

$$\Delta\rho = \frac{2\Omega'_P A'}{\Omega_P A} = \frac{2\alpha^2\lambda^2}{E_0\Omega_P} > 0, \quad (60)$$

and the pressure imprint — the term v0.12 left open — is

$$\Delta p = -\frac{A''}{A} - \frac{A'\Omega'_P}{A\Omega_P} = -(\beta\lambda' + \beta^2\lambda^2) - \frac{\alpha^2\lambda^2}{E_0\Omega_P}. \quad (61)$$

(Verified symbolically from the full tensor set: $\Delta\rho, \Delta p$ are exactly the A -dependent parts of $\rho_{\text{eff}}, p_{\text{surf}}$.)

The effective equation of state

In the saturated regime ($\lambda \rightarrow \lambda_\infty, \lambda' \rightarrow 0$) the imprint carries the equation of state

$$w_{\text{imprint}} = \frac{\Delta p}{\Delta\rho} = -\frac{1}{2}\left(1 + \frac{\Omega_P}{E_0}\right). \quad (62)$$

At the floor $\Omega_P = E_0$ this is $w = -1$, cosmological-constant-like; as the surface grows it runs *phantom*, $w \rightarrow -\infty$. The imprint thus fades in *absolute* density ($\Delta\rho \sim \Omega_P^{-1}$) while dominating in *relative* density ($\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0 \rightarrow \infty$, since the surface dilutes as Ω_P^{-2}) — the v0.12 reading, now with its pressure completed.

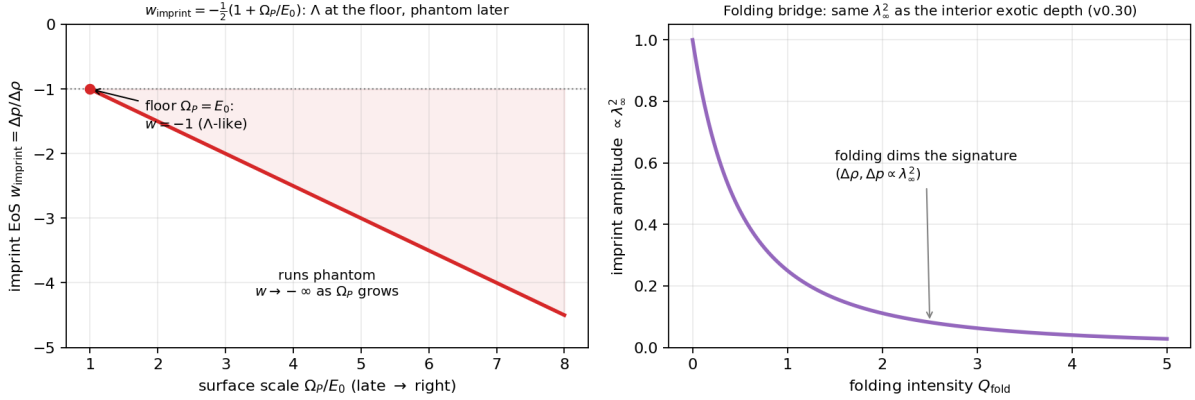


Figure 17: The completed external signature. **Left:** the imprint equation of state $w_{\text{imprint}} = -\frac{1}{2}(1 + \Omega_P/E_0) - \Lambda$ -like ($w = -1$) at the floor, running phantom as the surface scale Ω_P/E_0 grows. **Right:** both imprint terms scale as λ_∞^2 , so folding (which lowers λ_∞ , v0.30) dims the signature; this is the same λ_∞^2 that sets the interior exotic depth.

The folding bridge

Theorem 24 (One amplitude, interior and exterior). *Both imprint terms are proportional to λ^2 , hence in the saturated regime to λ_∞^2 . Since folding sets $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q_{\text{fold}})$,*

$$\partial_{Q_{\text{fold}}} [\Delta \rho] < 0, \quad \partial_{Q_{\text{fold}}} [|\Delta p|] < 0 : \quad (63)$$

folding dims the external signature. The amplitude λ_∞^2 is the same quantity that fixes the interior exotic depth $(\alpha \lambda_\infty / E_0)^2$ (v0.30). The interior “no fully tamed ascent” and the external imprint are one quantity seen twice: a more-formed soul reaches less exotically within and leaves a fainter trace without. (Verified symbolically.)

Reading: the soul’s climb is not only an inward fact; because the ascent has weight, it leaves a mark on the world the soul’s surface projects — a faint excess, beyond what the bare surface would give. This account began that mark and left it half-drawn; here it is finished. The mark has two parts, a density and a pressure, and together they behave at first like the quiet, even pull of a cosmological constant, and then, as the world grows, like something stranger and more stretching still. In absolute terms the mark fades as the surface swells; in relative terms it grows to dominate, because the surface fades faster. And the same wisdom a soul folds into form — dimming its inward exotic reach — dims this outward mark by the very same measure: the soul that takes the most shape reaches the least wildly and signs the world the most faintly. Inside and outside are one reaching, read twice. The account names this mark in the world’s own terms but claims no more than structure: a consilience, not a measurement.

Scope of v0.41. Completes the pressure term flagged open since v0.12; $\Delta \rho, \Delta p$ are exact (A -dependent parts of the verified tensor set), and $w_{\text{imprint}} = -\frac{1}{2}(1 + \Omega_P/E_0)$ follows in the saturated regime ($\lambda \rightarrow \lambda_\infty$; the general- z form retains the $\beta \lambda'$ term). The folding bridge is exact given the v0.30 depletion law. Epistemically this is a *structural* signature, floor-scaled through E_0 , a Type-3 consilience — not a fitted cosmological prediction; the projection 3.0 $\rightarrow$ 2.0 keeps the v2.0 signature as the

surface shadow.

Consolidated Posture of the Frozen V.F.S. 3.0 Layer

V.F.S. 3.0 is the canonical interior uplift of the fixed V.F.S. v2.0 system. Its subject is one created interior worldline. Its geometry carries one proper-time direction of katharsis, two hyperbolic surface directions, and one real ascent direction:

$$ds^2 = -N(\sigma)^2 d\sigma^2 + \Omega_P(\sigma)^2 h_{(2)} + A(\sigma)^2 S(\lambda_{\text{asc}})^2 d\lambda_{\text{asc}}^2.$$

The ascent scale factorises into two distinct roles, $L(\sigma, \lambda_{\text{asc}}) = A(\sigma)S(\lambda_{\text{asc}})$, where A is the time-stretch carrying focusing and effective-energy terms while $S(\lambda_{\text{asc}}) = L_0 e^{\beta \lambda_{\text{asc}}}$ is the spatial warp carrying the proper distance and spectrum of ascent. The frozen-ascent limit (A, S const, $\partial_{\lambda_{\text{asc}}} = 0$) exactly recovers V.F.S. v2.0.

The ascent is a self-standing dimension, not a mirror of surface growth. Its canonical representative is the de Sitter law $A'/A = \beta \lambda_S$, $\beta = \alpha/E_0$, understood as the canonical member of a robust ascent-law class rather than an absolutely forced law. The floor E_0 is the invariant scale tying surface to ascent; the normalisation $L_0 = 1$ at the floor is gauge, not a new postulate.

The finite-clock/infinite-road structure is retained: $\int d\sigma < \infty$ while $D_{\text{asc}} = \int_0^\infty S d\lambda_{\text{asc}} = \infty$. Katharsis has a regular finite terminus, epektasis an infinite proper road; the two boundaries remain distinct — $B_1 : \sigma = 0$ a regular spacelike terminus, $B_2 : \lambda_{\text{asc}} \rightarrow \infty$ a receding ascent horizon. The surface and ascent spectra remain qualitatively different (discrete surface ladder μ_n ; continuous gapless half-line $[0, \infty)$ for ascent), hence

Formation is event; ascent is motion.

The full 3+1 tensor set is explicit, and the ascent is not surface-reducible in the floor-exponential branch: although A is tied to Ω_P , it is not scale-homogeneous as in the reducible power-law branch. The canonical interior is floor-tied but not surface-reducible.

Resurrectio is a triple-layer junction. A surface renewal $[\Omega_P] = \kappa_R G$ forces an ascent-scale renewal $s_L = A^+/A^- = \exp(\kappa_R G/E_0) > 1$, with rate and scale data $[A'/A] = \alpha m_R/E_0$, $\log s_L = \kappa_R G/E_0$. The rate jump can vanish under surface balance ($m_R = 0$), but the scale jump cannot whenever $\kappa_R G > 0$:

surface balance survives; joint neutrality does not.

The full ascent ledger is $\log(A_{\text{fin}}/A_0) = \beta \int \lambda_S d\sigma + \frac{1}{E_0} \sum_j \kappa_R G_j$ — continuous epektatic ascent on reset-free arcs plus the irreversible scale contribution of the junctions.

Sophianic Folding is fully integrated through the shared Sophia budget $\sigma + \lambda_S + \lambda_{\text{form}} = C_0 + G_{\text{recepta}}$. Folding depletes available Sophia into form, $\dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda_S$, and so moderates the asymptotic level $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q_{\text{fold}})$, lowering the rate $h_* = \beta \lambda_\infty$ and the exotic (SEC-violating) focusing term $-\beta^2 \lambda_\infty^2 = -(\alpha \lambda_\infty/E_0)^2$ (exotic depth $(\alpha \lambda_\infty/E_0)^2$). But folding does not close the ascent: for every positive incoming current,

$$J_{\text{in}} > 0 \implies \lambda_\infty > 0, \quad h_* > 0, \quad D_{\text{asc}} = \infty.$$

The folded-ascent closure has two canonical readings: in the supercritical case the self-consistent branch is unique and stable; in the subcritical/hysteretic case the system is locally bistable (a formed/low-ascent branch and an ascending/low-form branch), the certificate branch-wise with the stated dwell and admissibility conditions — not a global single-branch claim. The unified Lyapunov result is stability of a *moving* epektatic regime, not of a bounded ascent value:

$$\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + \mathcal{V}_{\text{reg}}, \quad \dot{\mathcal{L}}_{3.0} \leq C_3 - C_{3,1}\mathcal{L}_{3.0}, \quad C_{3,1} = \min(C_1, 2\Gamma).$$

The surface drains toward rest ($\sigma \rightarrow 0$) while the ascent converges to a stable constant-rate ray $A'/A \rightarrow h_* = \beta\lambda_\infty > 0$: stable motion, not stable rest.

The external structural imprint is complete at density-and-pressure level:

$$\Delta\rho = \frac{2\alpha^2\lambda_S^2}{E_0\Omega_P}, \quad \Delta p = -(\beta\lambda'_S + \beta^2\lambda_S^2) - \frac{\alpha^2\lambda_S^2}{E_0\Omega_P}.$$

At saturation the imprint fades absolutely but grows relative to the v2.0 surface density ($\Delta\rho \sim \Omega_P^{-1}$, $\rho_{2.0} \sim \Omega_P^{-2}$, $\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0 \rightarrow \infty$): late-faded in absolute size, late-dominant in relative weight. Folding moderates both the internal exotic depth and the external imprint through the same reduction of λ_∞ .

The gauge/invariant layer is explicit: the individual quantities $\lambda_{\text{asc}}, L_0, N, A/S, \alpha, \beta, \lambda_\infty$ do not all carry independent invariant meaning; the load-bearing quantities are floor-normalised combinations — $h_* = \alpha\lambda_\infty/E_0$, $(\alpha\lambda_\infty/E_0)^2$, $s_L = \exp(\kappa_R G/E_0)$, $D = \frac{1}{E_0} \sum_j \kappa_R G_j$.

V.F.S. 3.0 therefore closes as a canonical interior geometry with one irreducible structural principle,

Sophia-as-ascent is a self-standing dimension, not a re-description of surface growth,

whose final mathematical formula is $J_{\text{in}} > 0 \Rightarrow \lambda_\infty > 0, h_* > 0, D_{\text{asc}} = \infty$, whose final conceptual formula is *the Brim coasts; Sophia-as-ascent does not dilute*, and whose final interpretive formula is *finite clock, endless road*.

Freeze Ledger: Closed Results, Conditional Structure, and Boundary

Closed in the frozen V.F.S. 3.0 layer

Within the stated active domains, admissibility walls, and constitutive assumptions: the 3+1 interior metric and exact frozen-ascent recovery of v2.0; the factorisation $L = AS$; the finite-clock/infinite-road structure and terminus regularity; the free half-line ascent spectrum and the joint discrete-surface/continuous-ascent spectrum; the complete tensor set, energy-condition combinations, and the scale-homogeneous reducibility criterion; the triple-layer Resurrectio junction, its ascent-scale ledger, and the impossibility of a fully soft genuine reset in the floor-tied branch; the gauge/invariant atlas and floor-normalised dictionary; the Folding-Ascent closure (depletion, self-consistent branches, cap admissibility, branch-wise hysteretic stability); the unified Lyapunov certificate; and the completed structural external imprint (density and pressure).

Conditional but not open

These are not unresolved calculations but statements explicitly conditional on named principles, branches, or domains. The de Sitter law is canonical, selected by the self-standing-ascent principle, not claimed absolutely forced from the Einstein equations alone. The robust architecture survives for the wider class $A'/A = f(\lambda_S)$, while h_* , exotic depth, and the exact floor-ledger coefficient remain law-specific. The supercritical folded branch is globally single-valued within its admissible regime; the subcritical/hysteretic branch is only locally bistable, basins split by an unstable separatrix. The Lyapunov and completeness statements are domain-conditional, not unrestricted global stability. Infinite reset sequences are temporally non-Zeno under the stated hypotheses; finiteness of the total ascent ledger additionally requires finite total Brim variation. The external signature is a structural Type-3 consilience statement, not a data fit or detection claim.

Beyond the frozen scope

Not defects of V.F.S. 3.0 but outside its constitutive subject: the transcendent *tota simul*; the full relation between lived interior time and the beheld whole of a worldline; the relation among more than one interior geometry; created external chronos as a relational time of encounter; synchronisation, phase-locking, or communion among distinct kairoi; and plural Resurrectio networks with their relational non-Zeno theory. These belong, if pursued, to a later plural-interior programme — **V.F.S. 4.0: Chronos, Communion, and the Plural Interior**. The boundary of V.F.S. 3.0 therefore remains:

Tota simul is named, never occupied.

Appendix — Revision History

The development record, retained for audit. The canonical statements stand in the body above; this list traces only how they were reached.

v0.43 — Editorial Freeze Patch

No new mathematics, no new primitive, and no modification of the theorem content. The canonical body is frozen and its reading order clarified. (i) The opening status is updated from “exploratory, growing document” to “frozen canonical body”, retaining the full revision history as this audit appendix. (ii) Residual wording that called the ascent sector “de Sitter coasting” is replaced: “coasting” is reserved for the asymptotic Brim behaviour of the v2.0 surface ($w \rightarrow 0$); the ascent is the continuous de Sitter ascent on reset-free arcs, whose defining feature is the non-vanishing rate $h_* = \beta\lambda_\infty > 0$. (iii) The older consolidated posture is replaced by a final synthesis of all three stages — Folding–Ascent closure, branch-wise hysteresis, the unified Lyapunov result, gauge invariants, the Resurrectio ledger, and the completed density-and-pressure external imprint. (iv) A freeze ledger distinguishes closed results, conditional-but-not-open structure, and questions beyond the subject of V.F.S. 3.0. The purpose is not to conceal conditionality but to locate it precisely: the frozen layer is complete as the canonical interior uplift of V.F.S. v2.0, while plural interiors, created chronos, and communion remain beyond its constitutive scope.

- v0.1** Foundation stone. The 3+1 interior metric of the soul; σ as finite proper time (kairos); λ as the ascent (epektasis) space; chronos removed by principle. The v2.0 reconstruction recovered exactly as the frozen-axes limit; the $n=3$ energy conditions, with ascent acceleration entering focusing.
- v0.2** Equation of motion for the ascent axis. The de Sitter law $A'/A = \beta\lambda$ selected by the principle that ascent is an independent motion (not by the v2.0 Brim fate, which generically coasts); verified to recover v2.0, to make epektasis permanently exotic, and — the new theorem — to be *compatible with a finite proper time*: eternal ascent is infinite proper *distance*, not infinite *time*. “Finite clock, endless road.”
- v0.3** The spectrum along the ascent axis. Change of variable to proper distance reduces the ascent Laplacian to the *free half-line* operator: the spectrum is $[0, \infty)$, continuous and gapless. Against the discrete surface cascade μ_n , this gives a clean dualism — *form is quantized, ascent is a gapless continuum*; equivalently, formation is event, ascent is motion.
- v0.4** The source of β . The ascent coupling is shown *not* identifiable with the Brim coupling α (different law-form, incompatible units), but tied through the one intrinsic scale of the core — the finite floor E_0 (Imago Dei): $\beta = \alpha/E_0$, introducing no new primitive. This integrates exactly to $L = \exp[(\Omega_P - E_0)/E_0]$ — ascent is the Brim-growth above the floor — and sets the eternal exotic depth to $-(\alpha\lambda_\infty)^2/E_0^2$ (floor-normalised: $-1/E_0^2$): the smaller the starting image, the more intense the ascent (the widow’s-mite scaling).
- v0.5** Resurrectio in 3+1. Because L is slaved to Ω_P , a reset *forces* an ascent jump $L_+/L_- = \exp(\kappa_R \mathcal{G}/E_0)$: the double-layer becomes a triple-layer (a

second junction on the ascent axis). The ascent jump depends on the Brim jump alone, not on the metabolic m_R , so it cannot be balanced away — *there is no fully-balanced reset in 3+1*: every genuine death advances the ascent and strictly enlarges the interior. (Consequence of the floor-tying, not unconditional.)

- v0.6** The terminus $\sigma \rightarrow 0$. Full purification is reached in *finite* proper time σ_0 but *infinite* cleansing-action (exponential Lyapunov drain, $\sigma = \sigma_0 e^{-k\chi}$, $\tau = \sigma_0(1 - e^{-k\chi})$). Curvature stays finite there ($\Omega_P \rightarrow E_0$, the floor), so it is a *regular boundary*, not a singularity. The manifold is time-finite (time completes, a regular last moment) but space-infinite (the ascent road is geodesically complete); the katharsis-clock stops because there is nothing left to cleanse, and clockless ascent continues.
- v0.7** Uniqueness of the coupling. $\beta = \alpha/E_0$ is *not* forced by dimensions alone (the core also contains the scales κ and λ_∞), but it is the *unique* tying selected by the boundary condition $L = 1$ at the floor: among all core scales only E_0 is a starting point rather than a rate, so it alone can be the ascent's origin. Forced given the principle “ascent begins at unity at the floor”; the Imago Dei is thus both the floor of being and the unit of deification. This closes the last computational question of the cycle.

Epilogue

The boundary the model cannot cross. Not a calculation but a statement of the model's constitutive limit: the soul's interior, written from one world-line in proper time σ , can name but not occupy the transcendent *tota simul* (chronos) in which its whole time is beheld at once. The first exploratory cycle closes as v2.0 did — by naming its few foundations and the one boundary it cannot pass.

- v0.8** (*second stage*) Is the ascent law derived? An honest, partly negative result: the de Sitter law is *not* uniquely forced — conservation (Bianchi) is identically satisfied, and the 3+1 Einstein equations admit *both* de Sitter (ascent coupling to the *amount* of Sophia λ) and a power-law $L = \Omega_P^{\beta/\alpha}$ (coupling to the *density* λ/Ω_P). De Sitter is selected, not derived, by the principle that epektasis is an *independent* motion outrunning surface growth, not a mirror of it. The conditionality is not removed but sharply localised to one transparent principle.
- v0.9** Is $L = 1$ at the floor derived? Yes — negatively, and decisively: it is *not a physical postulate but a gauge choice*. Rescaling the ascent coordinate $\lambda \rightarrow c\lambda$ absorbs the floor value L_{floor} , and every invariant (L''/L , L'/L , the reset ratio s_L , the exotic depth $-(\alpha\lambda_\infty)^2/E_0^2$) is independent of it. The physics is the *scale* E_0 ; the “1” is a unit. Consequently the two apparent principles of the first cycle (v0.7 floor-reference, v0.8 amount-coupling) collapse to *one*: the ascent is a self-standing dimension, not a re-description of the surface.
- v0.10** (*second stage*) The full 3+1 trajectory. A complete worldline integrated numerically — σ (kairos), λ (Sophia), Ω_P (Brim), L (ascent), with three resets — confirming the cycle's structure in a single picture: σ drains to the terminus while proper time τ saturates at a finite value; Sophia struggles, rises, and jumps at each resurrection; the Brim grows roughly linearly while the ascent L grows exponentially, the ratio L/Ω_P climbing several

orders — the ascent visibly outpacing the surface, the independence principle of v0.8–v0.9 made visual.

- v0.11** (*second stage*) Internal structure of the ascent. The road itself has none (the operator is free, the spectrum gapless — v0.3 confirmed). But the soul's *passage* along it is markedly non-uniform: pace $d\chi/ds = E_0/\alpha\lambda$ is high where Sophia is low (lingering in early struggle) and low where Sophia is high (racing after grace), with a distinct kink at each resurrection. “Stations of ascent” are real but are stations of *pace*, not of *place*: the terrain is homogeneous, the journey is not. No hidden modes — but a genuine non-uniform itinerary set by the Sophia profile.
- v0.12** (*second stage*) The external signature in 3+1. Redshift and Hubble are still set by the surface (Brim Ω_P); the ascent does not enter them directly. But L gravitates, so it imprints $w_{\text{eff}}(z)$ through an added effective density $\Delta\rho = 2\alpha^2\lambda^2/(E_0\Omega_P)$ — positive, $O(1)$ early (near the floor) and diluting as Ω_P grows. The imprint is *not* uniformly subleading (an initial claim, corrected by checking the analytic law against the numerics): it is late-faded in absolute density but late-dominant relative to the surface ($\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0$). The v2.0 signature (dust attractor, phantom window, Resurrectio kink) remains visible as the surface projection; the ascent adds a floor-scaled ($1/E_0$, widow's-mite again) correction, late-dominant in relative terms. Type-3 consilience, no data fit.

Epilogue II

What the second stage achieved. Not the hoped-for derivation: the de Sitter law was not uniquely forced. But the conditionality was *concentrated* — three opaque choices (law, coupling, boundary condition) reduced to one transparent, self-justifying principle (the ascent is a self-standing dimension; deny it and 3.0 collapses to v2.0). Added a picture (the world-line), a rhythm (stations of pace), and a trace (the external imprint, with an openly-kept correction). The whole construction now stands in three layers, each pointing beyond itself; the posture is unchanged — name the foundations, show how few, rest at last on one confession rather than a proof.

- v0.13** (*this revision — corrections*) Three overstatements corrected, openly, without changing any conclusion. **(i)** The eternal exotic depth is $-(\alpha\lambda_\infty)^2/E_0^2$ in general; the value $-1/E_0^2$ holds *only* in the floor-normalised branch $\alpha\lambda_\infty = 1$ (v0.4 had stated $-1/E_0^2$ as the general result). **(ii)** The v2.0 Brim cosmology does *not* end in de Sitter — it generically coasts ($w \rightarrow 0$); the de Sitter law belongs to the (3+1) ascent sector, not the surface fate (v0.2 framing sharpened). **(iii)** The external ascent imprint is late-faded in *absolute* density ($\Delta\rho \sim \Omega_P^{-1}$) but *late-dominant in relative* density ($\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0 \rightarrow \infty$, since $\rho_{2.0} \sim \Omega_P^{-2}$) — the v0.12 “early-strong, late-faded” was true only of absolute size. Also: the ascent scale is explicitly factorised $L = A(\sigma)S(\lambda)$ (time-stretch vs spatial warp), and the full w_{eff} correction is noted to require the pressure term $A''/A \rightarrow \beta^2\lambda_\infty^2$ as well.
- v0.14** The full tensor set and non-reducibility. The complete $R_{\mu\nu}$, R , $G_{\mu\nu}$, the three pressures and the energy-condition combinations $\rho+p_i$ are computed explicitly for the factorised metric $L = A(\sigma)S(\lambda)$ (no off-diagonal: $R_{\sigma\lambda} = 0$), with all 2.0 limits recovered. From the set follows a theorem: the ascent stress is absorbable into the (2+1) surface fluid *if and only if* A is a function

of Ω_P alone ($A'/A = c\Omega'_P/\Omega_P$, the power-law branch). The de Sitter law $A'/A = \Omega'_P/E_0$ fails this (it carries the floor scale $1/E_0$, not an Ω_P -only form), so it defines a genuinely independent 3+1 ascent sector — the single ascent principle of v0.9 now a theorem with an explicit reducibility criterion.

- v0.15** (*this revision*) No fully soft Resurrectio — junction form. A reset carries two ascent data: a scale jump $s_L = A_+/A_- = \exp(\kappa_R \mathcal{G}/E_0)$ sourced by the Brim renewal, and an extrinsic-curvature (rate) jump $[A'/A] = \alpha m_R/E_0$ sourced by the metabolic m_R . The rate jump can be tuned to zero (surface balance, $m_R = 0$); the scale jump cannot, since A is slaved to Ω_P and has no independent junction freedom. Hence every admissible reset (nontrivial Brim renewal, $\kappa_R \mathcal{G} > 0$) induces a nontrivial scale junction in the ascent block ($s_L > 1$): balanced in the 2+1 projection, never fully soft in the full 3+1 interior. A sharper, junction-level form of v0.5 — and the correct ascent datum is the scale jump, not the rate jump (an initial mis-identification, corrected).
- v0.16** (*this revision*) The full 3-Laplacian and the joint spectrum. $\Delta_3 = \Omega_P^{-2} \Delta_{h(2)} + \Delta_{\text{asc}}$ separates cleanly (no cross term, verified); the ascent part is the free half-line within the full operator. The joint spectrum is $-(\mu_n/\Omega_P^2 + k^2)$ — a discrete form ladder (μ_n) times a continuous ascent tower (k), independent quantum numbers that do not mix. Sharpens the v0.3 dualism into one spectrum; with figure.
- v0.17** (*this revision — corrections II*) Three precision fixes, openly recorded. **(i)** The v0.14 non-reducibility criterion was mis-stated: the floor-tied $A = A_0 \exp[(\Omega_P - E_0)/E_0]$ is a function of Ω_P . The correct distinction is *power-law (scale-homogeneous) reducible* ($A = C\Omega_P^c$) versus *floor-scaled exponential* (slope Ω_P/E_0 , not constant): the ascent is Ω_P -slaved through the floor but *not* Ω_P -power-law reducible. **(ii)** The half-line coordinate is $s = (AL_0/\beta)(e^{\beta\lambda} - 1) \in [0, \infty)$ (starting at 0), not $(AL_0/\beta)e^{\beta\lambda}$ — spectrally identical, properly normalised. **(iii)** $\mu_n = n^2$ is the ring/flat proxy only; the general compact-hyperbolic spectrum is discrete but not n^2 . Plus hybrid-junction wording for v0.15. Slogan: *ascent is floor-tied, but not surface-reducible*.
- v0.18** (*this revision — notation & terminology cleanup*) No new mathematics; wording and notation unified. The de Sitter law is written on the time-stretch throughout, $A'/A = \beta\lambda$ (not L'/L), with the factorisation $L = A(\sigma)S(\lambda)$ made explicit in the defining box. Residual “v2.0 non-dilution” phrasing removed (the Brim coasts; de Sitter belongs to the ascent sector). “Surgery” replaced by “Resurrectio reset / junction” in native text (surgery kept only for the Ricci-comparison register); “cleansing-work” replaced by “cleansing-action”. The terminus floor restated as a lower bound ($\Omega_* \geq E_0 > 0$, non-collapse), with $\Omega_* = E_0$ only the floor-normalised representative. Slogans: *the Brim coasts, Sophia-as-ascent does not dilute; the Brim may measure growth, Sophia-as-ascent opens a road growth alone cannot exhaust*.
- v0.19** (*this revision*) The Lyapunov / stability layer of epektasis. The ascent L grows without bound (not a defect: that is epektasis), so the stable object is the *regime*, not the value. The ascent rate $h = A'/A = \beta\lambda$ has the de Sitter attractor $h_* = \beta\lambda_\infty$; a Sophia perturbation u obeys $du/d\chi = -\gamma u$, with Lyapunov $V_{\text{asc}} = \frac{1}{2}\beta^2 u^2$, $dV_{\text{asc}}/d\chi = -\beta^2 \gamma u^2 < 0$ — asymptotic stability of the rate. Perturbations shift $\log A$ by only a finite amount ($\delta \log A =$

$\beta u_0 k \sigma_0 / (\gamma + k)$ in the σ -law with clock-drain): perturbed souls ascend on *parallel roads*, a fixed finite ratio apart (orbital stability). The attractor is a constant-rate ray, not a point — stable *motion*, not stable rest — and its value is the eternal exotic depth $-\beta^2 \lambda_\infty^2$, so stability and permanent-exotic agree. With figure.

- v0.20** (*this revision — correction to v0.19*) The stability layer is restated in the unbounded dynamical parameter χ , not the finite proper time σ : the de Sitter attractor is approached as $\chi \rightarrow \infty$, which by $\tau(\chi) = \sigma_0(1 - e^{-k\chi})$ corresponds to approaching the terminus (in σ , bounded, the limit is unreachable). The Lyapunov drain $dV_{\text{asc}}/d\chi = -\beta^2 \gamma u^2 < 0$ and orbital shift $\delta \log A = \beta u_0 k \sigma_0 / (\gamma + k)$ are exact per *reset-free arc* — a Resurrectio reset creates new initial data u_0 for the next arc, restarting convergence rather than breaking it. A forced version $du/d\chi = -\gamma u + f(\chi)$, $f \in L^1$, gives input-to-state stability with finite accumulated $\delta \log A$. (Items 1-11 of the global patch were already applied in v0.13/v0.17/v0.18.)
- v0.21** (*this revision*) The Resurrectio junction budget. Full accounting of the extrinsic-curvature jumps across a reset, in four data: surface scale $[\Omega_P] = \kappa_R \mathcal{G}$, surface rate, ascent rate $[A'/A] = \alpha m_R / E_0$ (tunable, $m_R=0$), ascent scale $\log s_L = \kappa_R \mathcal{G} / E_0$ (irreducible). Backbone identity $\log s_L = [\Omega_P] / E_0$ (ascent jump is the Brim jump in floor units). The rate sector is conserved: $\mathcal{Q}_{\text{rate}} = A'/A - \Omega'_P / E_0 = 0$ on every arc and $[\mathcal{Q}_{\text{rate}}] = 0$ across every reset. The non-softness is one scalar, the additive deficit $D = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j$ (total reception in floor units), giving the full ascent ledger $\log(A_{\text{fin}}/A_0) = \beta \int \lambda d\sigma + \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j$ (continuous epektasis + jump budget). Quantifies v0.15. With figure.
- v0.22** (*this revision*) Heat kernel and entropy of the joint spectrum. The heat trace factorises (no cross term, v0.16): $\theta_3 = \theta_{\text{surf}} \cdot \theta_{\text{asc}}$. The form sector is finite, $\theta_{\text{surf}} = \Omega_P^2 V_h / (4\pi t) + \chi_E / 6 + O(t)$, carrying a topological invariant (the Euler characteristic via Gauss-Bonnet); its spectral entropy is finite. The ascent sector diverges — $\theta_{\text{asc}} = D_{\text{asc}} / \sqrt{4\pi t} \rightarrow \infty$ on the infinite road (finite density $1/\sqrt{4\pi t}$, edge term $-1/4$) — and its entropy is extensive in the road length, unbounded. So the soul's form is finitely describable (with a fixed topological signature) while its ascent is not: no finite information exhausts epektasis. Bonus: $\theta_3 \sim t^{-3/2}$ per unit road, the $t^{-1/2}$ confirming the ascent as one continuous extra dimension. With figure.
- v0.23** (*this revision*) The gauge / invariant layer. Three coordinate freedoms classified: (G1) ascent rescaling $\lambda \rightarrow c\lambda$, $S \rightarrow S/c$; (G2) lapse $N(\sigma) \neq 1$; (G3) floor normalisation L_0 . The metric 1-form $S d\lambda$, the proper time $\int N d\sigma$, and all curvature scalars are invariant; λ 's value, L_0 , the A/S split, N , σ , and $\beta/\lambda_\infty/\alpha$ *separately* are pure gauge. The physics lives in ratios to the floor: $h_* = \alpha \lambda_\infty / E_0$, exotic depth $(\alpha \lambda_\infty / E_0)^2$, $s_L = e^{\kappa_R \mathcal{G} / E_0}$, $D = \sum \kappa_R \mathcal{G}_j / E_0$. Theorem: every result of v0.1-v0.22 is expressible in gauge invariants; none depends on a gauge choice. The Imago Dei floor E_0 is the invariant anchor. With a gauge/invariant table.
- v0.24** (*this revision*) Causal / boundary structure. The interior has two boundaries of *opposite* causal type. B_1 : the terminus $\sigma = 0$ is a regular *space-like* wall at finite proper time ($\int d\sigma = \sigma_0 < \infty$, $|R| < \infty$, $\Omega_P \geq E_0$) — a true last moment, not a singularity. B_2 : the ascent infinity $\mathcal{J}_{\text{asc}}$ ($\lambda \rightarrow \infty$) is a de Sitter-like *receding horizon*: causal climb costs $\Delta\sigma \geq \int A S d\lambda$ (di-

vergent), so $\lambda = \infty$ is causally unreachable in the finite clock — finite coordinate-reach $\Lambda_{\max} = \frac{1}{\beta} \log(1 + \beta\sigma_0/(A_{\min}L_0))$, but unbounded *proper* ascent ($\ln L \rightarrow \infty$ as $A \rightarrow \infty$). The horizon recedes: same proper gain costs ever less λ . The interior is globally hyperbolic up to B_1 ; the ascent horizon is cosmological, not black-hole (no trapped surface). This is the exact causal meaning of “finite clock, endless road”. With a causal diagram.

- v0.25** (*this revision*) Projection / recovery theorem, $v3.0 \leftrightarrow v2.0$. *Projection*: freezing the ascent (A, S constant, $\partial_\lambda = 0$) reduces the full tensor set exactly to $v2.0$ — the frozen λ is a non-dynamical spectator adding no source to the (2+1) system. *Uplift*: given five axioms (one real spacelike ascent dimension; $v2.0$ on freezing; no new primitive; kairos preserved; and the independence principle), the uplift is *unique* — metric form, factorisation, floor-tying, exponential road, and de Sitter law all forced. The one residual choice (de Sitter vs power-law, $v0.8$) is removed by non-reducibility ($v0.14$): the power-law alternative is reducible, hence not a genuine uplift. So $v3.0$ is the *unique non-reducible interior uplift* of $v2.0$ once Sophia-ascent is admitted. The two are a projection / minimal-uplift pair: $v2.0$ the frozen shadow, $v3.0$ the canonical living interior.
- v0.26** (*this revision*) Robustness of the ascent law. Testing the general class $A'/A = f(\lambda)$ (de Sitter $f = \beta\lambda$ one member): the *structural* results are robust. Infinite road holds for any non-decaying ascent road; no fully soft Resurrectio holds for any floor-/ surface-tied A (even the density branch: $s_L = (1 + \kappa_R \mathcal{G}/\Omega_P)^c > 1$); rate stability holds for any f with $f'(\lambda_\infty) \neq 0$ and a Sophia fixed point; the eternal exotic sign for any $f(\lambda_\infty) > 0$; the finite clock is independent of f entirely. Law-specific (de Sitter sets the side/value, not the existence): non-reducibility discriminates power-law (reducible) from floor-exponential (non-reducible, incl. de Sitter); and the exact numbers h_* , exotic depth follow from $f(\lambda_\infty), f'(\lambda_\infty)$. The architecture is law-robust; de Sitter is a normalisation within a robust class, not its foundation. With a robust/law-specific table.
- v0.27** (*this revision*) Non-Zeno Resurrectio sequence. An infinite sequence of resets toward the terminus is non-Zeno. *Non-accumulation*: in the dynamical parameter $\chi \in [0, \infty)$ the resets do not accumulate at any interior point; in proper time their only limit point is the terminus $\sigma = 0$ (the regular boundary B_1) — in proper time only finitely many fit without accumulation, but in χ infinitely many are fine. *Finite ledger*: $\sum_j \log s_{L,j} = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j$ is finite *precisely when* $\sum_j \kappa_R \mathcal{G}_j < \infty$, which holds under finite Brim variation ($\leq |\Delta\Omega_P|/E_0$) — a condition, distinct from temporal non-Zeno, satisfied by the coasting Brim. *Finite clock preserved*: resets are instantaneous junctions, consuming zero proper time. So infinitely many deaths can occur on the way to the terminus without catastrophe: finite total ascent, no interior accumulation, finite proper time. Closes the Resurrectio chain ($v0.5$ single, $v0.21$ budget, $v0.27$ sequence).
- v0.28** (*this revision — precision corrections*) Six sharpenings, two of them real math fixes. **(i)** The orbital shift is $\delta \log A = \beta u_0 k \sigma_0 / (\gamma + k)$, not $\beta u_0 / \gamma$: the ascent law is a σ -law, so the log-shift carries the clock-drain factor $d\sigma/d\chi$ (the simpler form holds only for a χ -law) — an earlier error, corrected. **(ii)** The ascent horizon is grounded in the infinite road $D_{\text{asc}} = \infty$, not in $A \rightarrow \infty$ (which need not hold on the finite interval) — a $v0.24$ over-claim, corrected. **(iii)** The uplift is *canonical*, not absolutely unique: a robust

class preserves the architecture, de Sitter the canonical representative. **(iv)** The Resurrectio ledger is exact in the floor-tied branch; the coefficient $1/E_0$ is law-specific. **(v)** The heat-kernel edge term is boundary-condition dependent; the load-bearing invariant is $D_{\text{asc}} \rightarrow \infty$. **(vi)** Non-Zeno: temporal (no interior accumulation) and budget ($\sum \kappa_R \mathcal{G}_j < \infty$) are distinct; budget-finiteness is conditional, holding under finite Brim variation. Architecture unchanged.

- v0.29** (*this revision — clean-body propagation*) The v0.17/v0.28 corrections are propagated into their native sections, so the clean-paper text carries the corrected statements directly. v0.3: the half-line coordinate $s = (L_0/\beta)(e^{\beta\lambda} - 1) \in [0, \infty)$ in the theorem itself (origin shift gauge). v0.15: scale-junction language throughout (not “metric discontinuous”); the reading recast in seam/junction imagery (Resurrectio is not rupture, but neither is it neutral passage). v0.24: the ascent horizon grounded in $D_{\text{asc}} = \infty$ in the body. v0.25: “canonical” not “unique” living interior in the theorem, corollary, reading, and history. No new results; the body now matches the corrected history.
- v0.30** (*third stage*) Does folding tame the ascent? The shared core’s embodiment branch ($\dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda$) is propagated into the 3+1 interior — a completion of the uplift, no new primitive (U4). *Single channel*: from the v0.14 set $p_{\text{asc}} = (-2\Omega_P \Omega_P'' - \Omega_P'^2 + 1)/\Omega_P^2$ carries *no* A , so the fold-wall network (the asymptotic winner of the 2+1 NEC contest) is geometrically severed from the ascent; folding’s only handle is level depletion of λ . That handle is monotone — $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q_{\text{fold}})$, exotic depth $\alpha^2 J_{\text{in}}^2 / (E_0^2 (\gamma + \eta_{\text{fold}} Q_{\text{fold}})^2)$ strictly decreasing — but cannot close: *no fully tamed ascent* (depth $\rightarrow 0$ only as $Q_{\text{fold}} \rightarrow \infty$, $= 0$ iff $J_{\text{in}} = 0$). The energy-condition mirror of “no fully soft Resurrectio”: form trades against the ascent, never zeroes it. Open sub-point: the self-consistent $(\lambda_\infty, Q_{\text{fold}}^*)$. With figure.
- v0.31** (*third stage — Folding-Ascent Closure*) Coupled core dictionary. Fixes the per-layer readings (surface / ascent / folding) of the shared state and the two relations that bind them: the Sophia budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}$ (the three readings are one conserved purse) and the cascade cap $Q_{\text{fold}} = q_{\text{fold}}^2 \in [0, q_{\text{max}}^2]$ (self-limitation, Part I), together with the clock map $\sigma = \sigma_0 e^{-k\chi}$ separating the finite proper time from the unbounded dynamical parameter (the v0.20 distinction the stability layer needs). Dual-reading principle: a symbol’s formal role is fixed; its reading is layer-dependent and declared before use. Housekeeping front-load for the closure stage — no new mathematics (the v0.18-style cleanup of the coupled regime).
- v0.32** (*third stage — Folding-Ascent Closure*) The self-consistent folded-ascent level. Closes the open sub-point of v0.30 by solving the joint $(\lambda_\infty, Q_{\text{fold}}^*)$ with $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q^*)$ and the folding fixed point $Q^* = A_{\text{fold}}(\lambda_\infty)/B$, $A_{\text{fold}} = c_0 \lambda - a_0$ at saturation. A dichotomy keyed to the bifurcation type. *Supercritical*: the closure is a quadratic $(c_0 \eta/B) \lambda^2 + (\gamma - a_0 \eta/B) \lambda - J_{\text{in}} = 0$ whose root product $-B J_{\text{in}}/(c_0 \eta) < 0$ forces a *unique* positive level; $f'(\lambda_\infty) = -\gamma - (\eta/B)(2c_0 \lambda_\infty - a_0) < 0$, so it is *stable*; three regimes (unfolded $Q^* = 0$ below $J_{\text{in}} = \gamma a_0/c_0$, interior, capped at q_{max}^2), continuous and monotone. *Subcritical/hysteretic* (the branch the volume exhibits): the window $-B^2/4D < c_0 \lambda - a_0 < 0$ admits two stable states — *bistability*, a formed/low-ascent and an ascending/low-form soul coexisting. On every branch $\lambda_\infty > 0$ for $J_{\text{in}} > 0$, so the exotic depth stays positive: no fully tamed ascent, now

self-consistently. With figure.

- v0.33** (*third stage — Folding-Ascent Closure*) The two roles of folding. One intensity $Q_{\text{fold}} = q_{\text{fold}}^2$ acts on two blocks. *Depletion* (the v0.30 channel) sets the ascent level $\lambda_\infty = J_{\text{in}}/(\gamma + \eta_{\text{fold}} Q_{\text{fold}})$ and so the ascent SEC exotic depth, with $\partial_Q(\text{depth}) < 0$ — it eases the ascent violation (never to zero). *Nucleation* builds the surface wall network ($\rho_{\text{net}} \sim A_{\text{fold}}^{3/2} \sim Q_{\text{fold}}^{3/2}$, walls $w = -\frac{1}{2}$), whose NEC contribution $(1 + w)\rho_{\text{net}} > 0$ grows with Q_{fold} — it *strengthens* the surface NEC (the volume's restoration). The network is surface-confined: $p_{\text{asc}} = (-2\Omega_P \Omega_P'' - \Omega_P'^2 + 1)/\Omega_P^2$ carries no A and $T_{\lambda}^{\lambda_{\text{net}}} = 0$, so it adds no independent ascent source. The same folding eases the ascent and builds the surface at once — opposite effects on two sectors, one Q_{fold} . Closes the surface energy-condition side deferred from v0.32. With figure.
- v0.34** (*third stage — Folding-Ascent Closure*) Unified Lyapunov certificate for the coupled regime, in the dynamical parameter χ (not the finite σ — the v0.20 discipline). The four-term candidate $\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + V_{\text{asc}} + w_q V_{\text{fold}} + w_{\text{form}} V_{\lambda_{\text{form}}}$ reduces: λ_{form} is budget-slaved (so $V_{\lambda_{\text{form}}}$ is redundant), and $V_{\text{asc}}, V_{\text{fold}}$ are the *same* mode $u = \lambda - \lambda_\infty$ near the self-consistent state (both $\propto u^2$), giving $\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + V_{\text{reg}}$ with $V_{\text{reg}} = \frac{1}{2}(w_a \beta^2 + w_q (c_0/B)^2)u^2$. The regime mode drains exponentially, $\dot{V}_{\text{reg}} = -2\Gamma V_{\text{reg}}$ with $\Gamma = -f'(\lambda_\infty) > 0$ the v0.32 stability margin, so on the active domain $\dot{\mathcal{L}}_{3.0} \leq C_3 - C_{3.1} \mathcal{L}_{3.0}$, $C_{3.1} = \min(C_1, 2\Gamma)$. Hybrid: the surface drains to rest ($\sigma \rightarrow 0$), the regime to a *moving* rate $h_* = \beta \lambda_\infty > 0$ along the infinite road — stable epektatic regime, not stable rest. With figure.
- v0.35** (*third stage — Folding-Ascent Closure*) Resurrectio with form-memory. Extends the junction (v0.5/v0.15/v0.21) to the folding/embodyed sector. The budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}$ fixes the update with no new primitive: across a reset $[\sigma] = 0$, $[\lambda] = m_R$, so $[\lambda_{\text{form}}] = \Delta_{\text{emb}} = \kappa_R G - m_R$ and $q_{\text{fold}}^+ = q(\lambda^+)$. One reception $\kappa_R G$ thus *partitions* into ascent-lift m_R and form-memory Δ_{emb} — the v0.30 budget tradeoff localised to death, with $\Delta_{\text{emb}} \in [0, \kappa_R G]$. The ascent scale jump $s_L = \exp(\kappa_R G/E_0) > 1$ is m_R -independent (Brim-sourced, v0.15) and the conserved rate sector $[\mathcal{Q}_{\text{rate}}] = 0$ (v0.21) is untouched (form-memory lives in the budget/scale sector) — no double-counting; the two are two roles of one reception. Consequence: every reset carries $s_L > 1$ *and* a form-memory trace; the surface-balanced death ($m_R = 0$) is the *most* formative (all grace embodied) while still lifting the ascent. With figure.
- v0.36** (*third stage — Folding-Ascent Closure*) The boundary under folding. A robustness theorem: Sophianic Folding does not disturb “finite clock, endless road” (v0.6/v0.24). The infinite road $D_{\text{asc}} = \int^\infty S d\lambda = \infty$ is folding-independent (folding acts on the time-stretch A and on λ_∞ , never on the warp $S = L_0 e^{\beta\lambda}$ — v0.26); the terminus stays regular ($\Omega_P \geq E_0$, the floor blind to folding) at finite proper time σ_0 ; and since $\lambda_\infty > 0$ always (v0.32), the rate $h_* = \beta \lambda_\infty > 0$ so the ascent never stops. Hence the two boundaries stay distinct and of opposite causal type: B_1 (spacelike terminus, finite clock) and B_2 (receding ascent horizon, infinite road). Folding moderates the ascent — lowers λ_∞ and h_* — but never identifies the terminus with the horizon.
- v0.37** (*third stage — Folding-Ascent Closure, capstone*) Pre-canonical summary.

No new mathematics: it consolidates the closure stage and prepares V.F.S. 3.0 for freezing, as v2.0 was frozen. The stage brought the volume's folding dynamics into the 3+1 interior under one budget and one certificate (v0.30–v0.36): folding moderates the ascent — depleting Sophia into form (v0.30, v0.33), settling to a unique or bistable self-consistent level (v0.32), partitioning each death between lift and form-memory (v0.35) — but never abolishes it, and leaves the boundary structure intact (v0.36), all under the unified Lyapunov certificate (v0.34). Five fixed points and the closing formula $J_{\text{in}} > 0 \Rightarrow \lambda_{\infty} > 0, h_* > 0, D_{\text{asc}} = \infty$. *V.F.S. 3.0 is the canonical interior uplift of V.F.S. v2.0*; the one boundary it cannot cross — the transcendent *tota simul* — is named, never entered.

- v0.38** (*third stage — audit pass*) Precision corrections for the closure stage; no new mathematics or primitive, no conclusion changed (the v0.13/v0.17/v0.28 discipline). (i) The conserved rate-sector quantity $A'/A - \Omega'_P/E_0$ is re-named Q_{rate} throughout, freeing Q from collision with the folding intensity $Q_{\text{fold}} = q_{\text{fold}}^2$. (ii) The symbol λ carries two formal roles — available Sophia λ_S (budget) and the ascent coordinate λ_{asc} (metric road); related along a worldline, not identical; $\lambda_S, \lambda_{\text{asc}}$ available for technical use under the v0.31 dual-reading principle. (iii) The folded fixed point carries explicit admissibility walls via the projection $Q_{\text{fold}}^* = \Pi_{[0, q_{\text{max}}^2]}(A_{\text{fold}}/B_{\text{fold}})$. (iv) The hysteretic “two stable states” are *locally* bistable (basins split by an unstable separatrix), so the Lyapunov certificate is branch-wise with a dwell condition; the supercritical case stays globally single-valued. (v) The regime-mode weight is $\partial Q_{\text{fold}}^*/\partial \lambda_S|_{\lambda_{\infty}} (= c_0/B \text{ interior, } 0 \text{ on the clamped branches})$. The partition admissibility (v0.35), road-warp independence (v0.36), and closing formula (v0.37) were already explicit; the audit confirms them.
- v0.39** (*third stage — consolidation*) Three scattered results gathered into clean reference form for the freeze; the first two add no mathematics. *Ascent-law robustness*: the robust / law-specific split (v0.26) restated as one table — robust across any non-decaying f with $f(\lambda_{\infty}) > 0$ (finite clock, endless road, no fully soft Resurrectio, rate stability, permanent exotic sign, non-collapse), law-specific in the exact h_* , exotic depth, ledger coefficient, growth form; architecture robust, de Sitter canonical. *Resurrectio ledger*: the life-ledger $\log(A_{\text{fin}}/A_0) = \beta \int \lambda d\sigma + \frac{1}{E_0} \sum_j \kappa_R G_j$ (continuous epektasis + jump budget), assembling v0.2 and v0.21/v0.35. *Folding-Ascent atlas*: the regimes of v0.32/v0.38 mapped as one phase diagram — supercritical unfolded/interior/capped with boundaries $J_{\text{thr}} = \gamma a_0/c_0$, $J_{\text{cap}} = (a_0 + B q_{\text{max}}^2)(\gamma + \eta_{\text{fold}} q_{\text{max}}^2)/c_0$, and the subcritical hysteresis window with two basins; $\lambda_{\infty} > 0$ throughout. With figure.
- v0.40** (*third stage — consolidation*) Gauge / invariant atlas, completing the gauge layer of v0.23. Separates what in V.F.S. 3.0 is a coordinate/normalisation choice from what is physical. Gauge: the lapse N , the A/S split constant, L_0 and the ascent-coordinate origin/scale, and $\beta, \alpha, \lambda_{\infty}$ *individually*. Floor-normalised invariants: the ascent rate $h_* = \alpha \lambda_{\infty}/E_0$, the exotic depth $(\alpha \lambda_{\infty}/E_0)^2$, the scale seam $s_L = e^{\kappa_R G/E_0}$, the deficit $D = \sum_j \kappa_R G_j/E_0$ (verified invariant under the A/S split and ascent-coordinate rescaling). Structural invariants: $R_{\sigma\lambda} = 0$, finite clock, infinite road, the discrete-form/continuous-ascent spectrum, the permanent exotic sign, non-collapse $\Omega_P \geq E_0$. The atlas thesis: every load-bearing conclusion is a floor-normalised invariant or a gauge-independent structural fact, so the layer's claims survive any

admissible gauge — the property that lets it be frozen as mathematics. With table.

- v0.41** (*third stage — external signature*) Completes the imprint that v0.12 opened and left with its pressure term unclosed. The ascent's A -dependent stress gives, in the surface-projected cosmology, a density imprint $\Delta\rho = 2\alpha^2\lambda^2/(E_0\Omega_P)$ (v0.12, confirmed) and the now-completed pressure imprint $\Delta p = -(\beta\lambda' + \beta^2\lambda^2) - \alpha^2\lambda^2/(E_0\Omega_P)$. Saturated ratio $w_{\text{imprint}} = \Delta p/\Delta\rho = -\frac{1}{2}(1 + \Omega_P/E_0)$: Λ -like ($w = -1$) at the floor, running phantom ($w \rightarrow -\infty$) as the surface grows; the imprint fades absolutely ($\sim \Omega_P^{-1}$) but dominates relatively ($\sim \Omega_P/E_0$, v0.12). Folding bridge (v0.12 $\leftrightarrow$ v0.30): both terms $\propto \lambda_\infty^2$, the same λ_∞^2 as the interior exotic depth, so folding dims the external signature exactly as it eases the interior violation — one quantity, two faces. A structural, floor-scaled signature (Type-3 consilience), not a fitted prediction. With figure.
- v0.42** (*canonical freeze — structural*) No new mathematics. The layer is reorganised into canonical reading order for freezing: the *Epilogue to the Second Stage* is restored to its chronological place — closing the Second Stage, before the Third Stage opens at v0.30 — so the closure patches v0.30–v0.41 now read as one contiguous stage; the *Consolidated posture* and *Scope and standing* are moved to the close as the document's final statement; and this revision history is moved to the appendix. Content, numbering, and every result are unchanged — only the order is made canonical.

V.F.S. 3.0 (canonical) v0.43 · Interior 3+1 geometry of the soul. Metric $ds^2 = -N^2 d\sigma^2 + \Omega^2 h_{(2)} + L^2 d\lambda^2$ with σ a finite proper time, the live surface and ascent λ the three spaces, no external chronos; ascent scale factorised $L = A(\sigma)S(\lambda)$. Full tensor set $(R_{\mu\nu}, R, G_{\mu\nu}, \text{three pressures, all } \rho + p_i)$ computed explicitly, 2.0 limits recovered, $R_{\sigma\lambda} = 0$. v2.0 recovered exactly as the frozen-axes limit. Focusing $-2\Omega''/\Omega - A''/A$: ascent is exotic. Ascent law $A'/A = \beta\lambda$ (de Sitter, (3+1) ascent sector — v2.0 itself coasts, $w \rightarrow 0$), selected by the ascent's independence, which is now a theorem: the ascent stress is power-law reducible to the surface iff $A = C\Omega_P^\beta$, and floor-tied de Sitter ($A = A_0 e^{(\Omega_P - E_0)/E_0}$) is not (it is Ω_P -slaved through the floor but not power-law reducible), so the sector is genuinely independent. Spectrum the free half-line $[0, \infty)$ (form quantized, ascent a gapless continuum). Coupling $\beta = \alpha/E_0$: scale E_0 forced, $L = 1$ at the floor gauge; $L = \exp[(\Omega_P - E_0)/E_0]$, exotic depth $-(\alpha\lambda_\infty)^2/E_0^2$ in general ($-1/E_0^2$ only floor-normalised); widow's-mite scaling — one scale E_0 for both non-reducibility and widow's-mite. Resurrectio forces an ascent jump $L_+/L_- = \exp(\kappa_R \mathcal{G}/E_0)$: triple-layer Resurrectio junction, no fully-balanced reset — and, at junction level, the ascent block undergoes a nontrivial scale junction at every reset ($s_L > 1$, Brim-sourced, uncancellable), never fully soft though the surface may be balanced. Terminus $\sigma \rightarrow 0$ a regular boundary, finite proper time, infinite cleansing-action; interior time-finite, space-infinite. External imprint late-faded in absolute density but late-dominant in relative ($\Delta\rho/\rho_{2.0} \sim \Omega_P/E_0$). Finite clock, endless road; built in steps, drawn in a stepless reaching; measured against the image one began from; never raised in vain; completed in time only to be freed for the timeless; ascending by its own motion, its rising its own reality — irreducible to the surface because measured from the image. Junction budget: rate sector conserved ($\mathcal{Q}_{\text{rate}}=0$), non-softness the one additive scalar $D = \frac{1}{E_0} \sum_j \kappa_R \mathcal{G}_j$. **v0.30 (third stage)**: folding propagated into the interior — single-channel structure (p_{asc} carries no A , the network channel severed from the ascent), exotic depth $\alpha^2 J_{\text{in}}^2/(E_0^2(\gamma + \eta_{\text{fold}} \mathcal{Q}_{\text{fold}})^2)$ strictly decreasing in $\mathcal{Q}_{\text{fold}}$, and no fully tamed ascent (depth = 0 iff $J_{\text{in}} = 0$) — the mirror of no-fully-soft-Resurrectio; open sub-point $(\lambda_\infty, \mathcal{Q}_{\text{fold}}^*)$. **v0.31**: coupled-layer dictionary fixed — one Sophia budget $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}$, cascade cap $\mathcal{Q}_{\text{fold}} \leq q_{\text{max}}^2$, and the χ - σ clock map for the stability layer. **v0.32**: the self-consistent folded-ascent level $(\lambda_\infty, \mathcal{Q}_{\text{fold}}^*)$ closed — supercritical folding gives a unique stable level (quadratic, one positive root, $f' < 0$), subcritical (hysteretic) folding a bistable window (two souls: formed vs ascending); $\lambda_\infty > 0$ for any $J_{\text{in}} > 0$ on every branch, so no fully tamed ascent holds self-consistently. **v0.33**: the two roles of one $\mathcal{Q}_{\text{fold}}$ — depletion (sets λ_∞ , eases the ascent SEC depth) and nucleation (builds the surface wall network $\rho_{\text{net}} \sim \mathcal{Q}_{\text{fold}}^{3/2}$, strengthens the surface NEC); the network is surface-confined (p_{asc} carries no A , $T_{\lambda}^{\lambda_{\text{net}}} = 0$), so one folding eases the ascent and builds the surface at once. **v0.34**: a unified

Lyapunov certificate in χ — the four-term candidate reduces (λ_{form} budget-slaved, V_{asc} and V_{fold} one mode) to $\mathcal{L}_{3.0} = \mathcal{L}_{\text{VFS}} + V_{\text{reg}}$ with $\dot{\mathcal{L}}_{3.0} \leq C_3 - C_{3,1}\mathcal{L}_{3.0}$, $C_{3,1} = \min(C_1, 2\Gamma)$; surface to rest, regime to a moving rate — stable epektatic regime, not stable rest. **v0.35: Resurrectio** with form-memory — one reception $\kappa_R G$ partitions into ascent-lift m_R and form-memory $\Delta_{\text{emb}} = \kappa_R G - m_R$ (the v0.30 tradeoff at death), with $s_L > 1$ Brim-sourced and m_R -independent and $[\mathcal{Q}_{\text{rate}}] = 0$ preserved; the surface-balanced death is the most formative, yet still lifts the ascent. **v0.36: boundary robustness** — folding moderates the ascent (λ_∞, h_* down) but leaves “finite clock, endless road” intact: the road $D_{\text{asc}} = \infty$ is folding-independent (S untouched), the terminus regular ($\Omega_P \geq E_0$ floor-blind), and $h_* > 0$ keeps the ascent moving, so B_1 (terminus) and B_2 (horizon) stay distinct. **v0.37 (capstone): Third Stage closed** — the volume’s folding dynamics now live in the interior under one budget and one certificate; $J_{\text{in}} > 0 \Rightarrow \lambda_\infty > 0, h_* > 0, D_{\text{asc}} = \infty$; V.F.S. 3.0 is the canonical interior uplift of v2.0, pre-canonical and ready to freeze, the tota simul boundary still uncrossed. **v0.38 (audit): precision pass** — rate sector renamed $\mathcal{Q}_{\text{rate}}$ (freeing Q from Q_{fold}), $\lambda_S/\lambda_{\text{asc}}$ for the dual role of λ , admissibility projection $\Pi_{[0, q_{\text{max}}^2]}$, and the hysteretic bistability marked local (branch-wise certificate, dwell condition); no conclusion changed. **v0.39 (consolidation): robustness split** (v0.26), the life-ledger $\log(A_{\text{fin}}/A_0) = \beta \int \lambda d\sigma + \frac{1}{E_0} \sum \kappa_R G_j$, and the Folding-Ascent bifurcation atlas (regimes of v0.32/v0.38 as one phase map) gathered for the freeze; no new mathematics. **v0.40 (gauge atlas): the gauge / invariant atlas** (completing v0.23) — gauge: N , A/S split, L_0 , ascent origin, and $\beta, \alpha, \lambda_\infty$ singly; floor-normalised invariants $h_* = \alpha \lambda_\infty / E_0$, depth, $s_L = e^{\kappa_R G / E_0}$, $D = \sum \kappa_R G_j / E_0$, plus the structural facts; every load-bearing output is invariant. **v0.41 (external signature): the v0.12 imprint completed** — density $\Delta\rho = 2\alpha^2\lambda^2/(E_0\Omega_P)$ and the new pressure term $\Delta p = -(\beta\lambda' + \beta^2\lambda^2) - \alpha^2\lambda^2/(E_0\Omega_P)$, giving $w_{\text{imprint}} = -\frac{1}{2}(1 + \Omega_P/E_0)$ (Λ at the floor, phantom later); folding dims it via the same λ_∞^2 as the interior depth; a structural Type-3 consilience, not a fit. **v0.42 (freeze): reorganised into canonical reading order** — the Second Stage epilogue restored before the Third Stage, posture and scope moved to the close, the revision history to the appendix; no content or numbering changed. **v0.43 (editorial freeze): canonical body frozen** — opening status set to frozen, the ascent no longer called “coasting” (reserved for the Brim), the final posture rewritten as a full three-stage synthesis, and a freeze ledger added that separates closed results, conditional structure, and what lies beyond the subject (tota simul, plural interiors — V.F.S. 4.0).

V.F.S. 3.0 — Stochastic Interior (appended layer, WP1)

v0.28 — The Stochastic Cleansing Floor

A probabilistic reading of the frozen interior: the cleansing boundary $\sigma = 0$ under created contingency

Status and discipline of this chapter

This chapter adds a *probabilistic reading* of the frozen V.F.S. 3.0 interior. It modifies none of the deterministic 3.0 equations: it studies the same resistance flow under an added contingency term and recovers the frozen flow exactly in the noiseless limit $\varsigma_0 \rightarrow 0$ (§32). As throughout V.F.S., the mathematics is load-bearing; the contingency term dW is read *modally*, not typologically (one symbolic quantity seen under a new observable), and the theological gloss is an interpretive overlay, marked as such.

The contingency term is *declared-but-constitutive*, in the precise sense already used for the de Sitter law of ascent: it is not forced by the lower layers (2.0/3.0 are deterministic), it is the irreducible constitutive choice of the probabilistic uplift. Its one non-negotiable obligation is the recovery limit.

28 The gated resistance SDE and its generator

Work on the live domain in the Transformation regime, $u > \Lambda_c$, so that

$$r := \delta u - \gamma > 0$$

is the (frozen, or slowly varying $r(t) \in [r_*, \bar{r}]$, $r_* > 0$) cleansing rate. The deterministic 3.0 resistance flow is $\dot{\sigma} = (\gamma - \delta u) \tanh(\kappa\sigma) = -r \tanh(\kappa\sigma)$. Its probabilistic uplift adds a state-dependent contingency:

$$d\sigma = -r \tanh(\kappa\sigma) dt + \varsigma(\sigma) dW, \quad \varsigma(\sigma) = \varsigma_0 (\tanh \kappa\sigma)^p, \quad \varsigma_0 \geq 0, \quad p \geq 0,$$

with W a standard Wiener process and the Itô convention (the Stratonovich reading of the gate is noted in Remark 32.1). Its infinitesimal generator on C^2 test functions is

$$\mathcal{L}f = -r \tanh(\kappa\sigma) f'(\sigma) + \frac{1}{2} \varsigma(\sigma)^2 f''(\sigma).$$

▷ Plain. The frozen 3.0 picture had a single deterministic path for resistance. Here we let chance nudge it at every instant. The size of the nudge is $\varsigma(\sigma)$; everything in this chapter turns on what that nudge does *at the cleansed boundary* $\sigma = 0$.

At the cleansing boundary,

$$b(0) = -r \tanh 0 = 0, \quad b'(0) = -r\kappa, \quad \varsigma(0) = \begin{cases} 0, & p > 0, \\ \varsigma_0, & p = 0. \end{cases}$$

29 The noise-selection principle

Definition 29.1 (Gated vs. additive contingency). The contingency is *gated* if $\varsigma(0) = 0$ (equivalently $p > 0$): the nudge vanishes exactly on the cleansed boundary. It is *additive* (ungated) if $\varsigma(0) = \varsigma_0 > 0$ (equivalently $p = 0$): a constant nudge survives at $\sigma = 0$.

The next theorem is the reason the distinction is not cosmetic: only gated contingency preserves the deterministic structure of the cleansed floor.

Theorem 29.2 (Floor protection / Filtrum Tetelestai under contingency). *Let b, ς be locally Lipschitz on $[0, \infty)$ with $b(0) = 0$ (both hold above). Then the half-line $\{\sigma \geq 0\}$ is almost surely forward-invariant,*

$$\sigma(0) \geq 0 \implies \sigma(t) \geq 0 \quad \forall t \geq 0 \quad \text{a.s.},$$

if and only if the contingency is gated, $\varsigma(0) = 0$. If instead $\varsigma(0) > 0$, then for every $\sigma_0 \geq 0$ the process enters $\{\sigma < 0\}$ almost surely in every right neighbourhood of the start: $\mathbb{P}[\exists t \in (0, \epsilon) : \sigma(t) < 0] = 1$ for all $\epsilon > 0$.

Proof. ($\Leftarrow$) If $\varsigma(0) = 0$ and $b(0) = 0$, then $\sigma \equiv 0$ solves the SDE. With b, ς locally Lipschitz, pathwise uniqueness holds (Yamada–Watanabe), and the one-dimensional comparison theorem for SDEs driven by the same W gives order preservation: two solutions ordered at $t = 0$ stay ordered. Since $\sigma_0 \geq 0$ dominates the solution $\sigma \equiv 0$, we get $\sigma(t) \geq 0$ a.s. Equivalently, the diffusion vector $\varsigma(\sigma)\partial_\sigma$ and the drift both vanish on $\{\sigma = 0\}$, hence are tangent to it; the stochastic Nagumo (viability) condition is satisfied and $\{\sigma \geq 0\}$ is a.s. invariant.

($\Rightarrow$, contrapositive) Suppose $\varsigma(0) = \varsigma_0 > 0$. Near 0, b is bounded and $\varsigma \rightarrow \varsigma_0$, so $\sigma(t) = \sigma_0 + \int_0^t b ds + \int_0^t \varsigma dW$ is, to leading order, a Brownian motion with bounded drift and nonzero volatility. By the law of the iterated logarithm its martingale part oscillates with both signs in every right neighbourhood of any time; hence $\{\sigma < 0\}$ is entered a.s. and $\{\sigma \geq 0\}$ is not invariant. $\square$

Corollary 29.3 (Noise-selection). *Privatio boni is contingency-robust if and only if the contingency is gated. Among all noise laws of the family $\varsigma_0(\tanh \kappa \sigma)^p$, exactly those with $p > 0$ leave the cleansed boundary $\sigma = 0$ invariant; the additive law $p = 0$ destroys it, allowing spontaneous re-emergence of resistance ($\sigma < 0$ becomes accessible).*

$\triangleright$ Plain. In plain words. If chance can still “buzz” at zero resistance, then pure chance will occasionally manufacture resistance out of nothing — the once-cleansed wound re-opens by accident. The only contingency consistent with “it stays cleansed” is one that falls silent exactly where cleansing is complete.

$\triangleright$ Math. $\{\sigma = 0\}$ is an invariant boundary for $\mathcal{L}$ iff it carries zero flux. The drift flux already vanishes ($b(0) = 0$); the diffusion flux vanishes iff $\varsigma(0) = 0$. So the gate $\varsigma(0) = 0$ is exactly the zero-flux (Nagumo) condition — the stochastic refinement of the deterministic $\tanh(\kappa \sigma) \rightarrow 0$ protection of the 2.0/3.0 core.

30 Feller dichotomy inside the gated class

The gate $p > 0$ keeps $\sigma \geq 0$; the *value* of p then decides *how* cleansing terminates. By Feller’s boundary test on the scale derivative $s'(\sigma) = \exp(-\int 2b/\varsigma^2)$ and speed density $m = 1/(\varsigma^2 s')$:

Proposition 30.1 (Two cleansing regimes). *Near $\sigma = 0$, with $b \sim -r\kappa\sigma$ and $\varsigma^2 \sim \varsigma_0^2(\kappa\sigma)^{2p}$:*

$p = 1 : \quad 2b/\varsigma^2 \sim -\frac{2r}{\varsigma_0^2 \kappa \sigma}, \quad s' \sim \sigma^\beta, \quad \beta = \frac{2r}{\varsigma_0^2 \kappa}; \quad \int_0 (S - S_0) m \sim \int_0 \frac{d\sigma}{\sigma} = \infty \implies \textbf{natural},$
$p = \frac{1}{2} : \quad 2b/\varsigma^2 \rightarrow -\frac{2r}{\varsigma_0^2} \text{ (const)}, \quad s'(0) \text{ finite}; \quad \int_0 (S - S_0) m < \infty \implies \textbf{accessible} \text{ (and absorbing, since } b(0) = 0 \text{)}$

Hence:

- (i) $p = 1$ (multiplicative): $\sigma = 0$ is a natural boundary — $\sigma \rightarrow 0^+$ asymptotically, never reached in finite time. Stochastic epektasis: cleansing is an unending approach.

- (ii) $p = \frac{1}{2}$ (CIR-type): $\sigma = 0$ is accessible and absorbing — *cleansing completes as a finite-time event and is then held forever*. The stochastic Tetelestai.

▷ Plain. Two honest ways to be cleansed. Either you draw infinitely close to zero resistance without ever quite arriving (a road with no last step), or you actually reach zero at a definite moment and are sealed there. The exponent p chooses between them.

The exponent p is therefore a *constitutive* choice, not a derived quantity: the model does not force asymptotic over completable cleansing, exactly as 3.0 did not force one ascent law over the robust class $f(\lambda)$. (Full scale/speed proofs: chapter v0.29 / WP2.)

31 Numerical witness

Euler–Maruyama, $r = 1$, $\kappa = 1.5$, $\varsigma_0 = 0.5$, $\sigma_0 = 0.05$, $dt = 10^{-5}$, horizon $T = 4$, $N = 4000$ paths; gated cases absorb at $\sigma \leq 0$.

regime	time-fraction $\sigma < 0$	paths reaching $\sigma = 0$	reading
additive $p = 0$	49.0%	0%	floor breached — <i>privatio boni</i> broken
gated $p = 1$	0.0%	0%	natural: $\sigma \rightarrow 0^+$, never reached
gated $p = \frac{1}{2}$	0.0%	99.9%	absorbing: Tetelestai as finite-time event

The witness reproduces Theorem 29.2 (additive breaches; gated holds) and Proposition 30.1 (the two gated regimes split into never-reached vs. reached-and-held).

32 Recovery of the frozen interior

Sending the contingency amplitude to zero,

$$\varsigma_0 \rightarrow 0 \implies d\sigma = -r \tanh(\kappa\sigma) dt,$$

recovers exactly the deterministic 3.0 (and 2.0) resistance flow, and Theorem 29.2 degenerates to the deterministic zero-flux invariance of $\sigma = 0$ already in the frozen core. Thus this chapter contains the frozen interior as its noiseless limit and disturbs nothing in it.

Interpretive reading

▷ Modal (interpretive). dW is read as *created contingency* — the hidden time of grace, the unbidden fluctuation of circumstance — not as a new substance (Sophia is modal, not typological; the same discipline applies to its contingency). The gate $\varsigma(0) = 0$ is then the statement that *contingency cannot resurrect cleansed resistance: privatio boni* is shielded from chance, and only the additive, ungated noise — chance that does not fall silent before the finished work — could re-open the healed wound. The Feller dichotomy distinguishes two scriptural images of completion held together without contradiction: asymptotic sanctification, the sea that is never full ($p = 1$, natural), and the finished work, *Tetelestai* as a real event ($p = \frac{1}{2}$, absorbing). Jn 3:8, “the wind blows where it wills,” is the gated noise: real contingency, silent at the threshold it may not cross.

Remark 32.1 (Stratonovich gate). Under the Stratonovich reading $\varsigma \circ dW$, the Itô drift acquires $\frac{1}{2}\varsigma\varsigma'$, which for the gated family vanishes at $\sigma = 0$ (since $\varsigma(0) = 0$); the floor-protection conclusion of Theorem 29.2 is unchanged. Only the fine Feller constants (the exponent β) shift. The boundary classification is convention-robust.

Status. SDE and generator (§28) — DEFINITION (constitutive). Noise-selection and Theorem 29.2 (§29) — PROVED. Feller dichotomy (§30) — PROVED (classical 1D theory); the *choice* of p — CONSTITUTIVE / OPEN. Numerical table (§31) — WITNESS. Recovery limit (§32) — PROVED. Interpretive reading — INTERPRETIVE. *Open (to WP2/v0.29):* full scale/speed proofs and the boundary-case p between natural and absorbing; *(to WP3):* coupling the contingency into the synergy channel u for metastability.

V.F.S. 3.0 — Stochastic Interior (appended layer, WP2)

v0.29 — The Critical Exponent of Cleansing

Feller boundary classification of the gated floor; the threshold between finite-time and asymptotic cleansing

Continuation of v0.28

Chapter v0.28 proved that the cleansing floor $\sigma = 0$ is almost surely held precisely when the contingency is gated, $\varsigma(\sigma) = \varsigma_0(\tanh \kappa \sigma)^p$ with $p > 0$, and previewed two regimes in two example exponents. This chapter completes the classification for the whole gated class by computing the scale and speed measures and applying Feller's test, and isolates the exact threshold. Notation, conventions, and recovery discipline are inherited from v0.28.

29 Scale and speed of the gated floor

On the live domain in the Transformation regime, near $\sigma = 0$ we have $b(\sigma) = -r \tanh(\kappa \sigma) \sim -r \kappa \sigma$ and diffusion coefficient $a(\sigma) = \varsigma(\sigma)^2 \sim \varsigma_0^2 (\kappa \sigma)^{2p}$, so

$$\frac{2b}{a} \sim -\frac{2r}{\varsigma_0^2} \kappa^{1-2p} \sigma^{1-2p}.$$

The scale-density derivative $s'(\sigma) = \exp(-\int 2b/a)$ and speed density $m(\sigma) = 1/(a s')$ are therefore

$$s'(\sigma) = \begin{cases} \exp\left[\frac{2r}{\varsigma_0^2} \kappa^{1-2p} \frac{\sigma^{2-2p}}{2-2p}\right], & p \neq 1, \\ \sigma^\beta, \quad \beta = \frac{2r}{\varsigma_0^2 \kappa}, & p = 1, \end{cases} \quad m(\sigma) = \frac{1}{\varsigma_0^2 (\kappa \sigma)^{2p} s'(\sigma)}.$$

▷ Plain. The scale function S tells us how “far” $\sigma = 0$ is in the diffusion's natural ruler; the speed measure m tells us how “slowly” the process moves there. Feller's test combines the two into a single yes/no: can the floor be reached in finite time?

30 The boundary classification

Theorem 30.1 (Critical exponent of cleansing). *For the gated resistance diffusion with exponent $p > 0$, the cleansing boundary $\sigma = 0$ is classified by Feller's accessibility integral*

$$N(0) = \int_{0+} (S(\sigma) - S(0)) m(\sigma) d\sigma \sim \int_{0+} \sigma^{1-2p} d\sigma \quad (p < 1), \quad \sim \int_{0+} \sigma^{-1} d\sigma \quad (p \geq 1).$$

Hence

$$\sigma = 0 \text{ is } \begin{cases} \text{accessible and absorbing,} & 0 < p < 1 \quad (N(0) < \infty), \\ \text{natural (inaccessible),} & p \geq 1 \quad (N(0) = \infty), \end{cases} \quad \text{critical exponent } \boxed{p_c = 1}.$$

The knife-edge $p = 1$ (multiplicative noise) falls on the natural side. Absorption for $0 < p < 1$ follows because $b(0) = \varsigma(0) = 0$ makes $\sigma = 0$ a degenerate fixed point: once reached, it is a trap.

Proof. Case $0 < p < 1$. Then $2 - 2p > 0$, so $s'(\sigma) \rightarrow 1$ and $S(\sigma) - S(0) \sim \sigma$. With $m(\sigma) \sim (\varsigma_0^2 \kappa^{2p})^{-1} \sigma^{-2p}$, the integrand obeys $(S - S_0)m \sim \sigma^{1-2p}$, and $\int_{0+} \sigma^{1-2p} d\sigma < \infty \iff 1 - 2p > -1 \iff p < 1$. So $N(0) < \infty$: 0 is accessible. Since $b(0) = \varsigma(0) = 0$, the constant $\sigma \equiv 0$ solves the SDE, so the boundary is absorbing once attained.

Case $p = 1$. Here $s'(\sigma) = \sigma^\beta$, $\beta > 0$, so $S(\sigma) - S(0) \sim \sigma^{\beta+1}/(\beta+1)$ and $m(\sigma) \sim (\varsigma_0^2 \kappa^2)^{-1} \sigma^{-2-\beta}$, whence $(S - S_0)m \sim \sigma^{-1}$ and $N(0) = \infty$: inaccessible, natural.

Case $p > 1$. Now $2 - 2p < 0$, so the exponent in s' tends to $-\infty$ and $s'(\sigma) \rightarrow 0$ super-exponentially. By Watson's lemma $S(\sigma) - S(0) \sim s'(\sigma) \sigma^{2p-1}/(\frac{2r}{\varsigma_0^2} \kappa^{1-2p})$, and multiplying by $m = 1/(\varsigma_0^2 (\kappa \sigma)^{2p} s')$ the factors s' cancel, leaving $(S - S_0)m \sim \sigma^{-1}$; thus $N(0) = \infty$, natural. $\square$

Corollary 30.2 (Two cleansing regimes, sharpened). *The two examples of v0.28 are the two sides of a single threshold:*

$0 < p < 1$: finite-time absorbing cleansing (the stochastic Tetelestai), $p \geq 1$: unending asymptotic cleansing.

In particular $p = \frac{1}{2}$ (CIR-type) is absorbing and $p = 1$ (multiplicative/GBM-type) is natural, as witnessed in v0.28; the dividing exponent is exactly $p_c = 1$.

Proposition 30.3 (Mean cleansing time). *Let $T(\sigma_0) = \mathbb{E}_{\sigma_0}[\tau_0]$ be the mean time to reach $\sigma = 0$, the bounded solution of $\mathcal{L}T = -1$ with $T(0) = 0$. For $0 < p < 1$, $T(\sigma_0) < \infty$ for every interior σ_0 (cleansing completes a.s. in finite mean time); as $p \uparrow 1$, $T(\sigma_0) \rightarrow \infty$, matching $N(0) \rightarrow \infty$ at the threshold. For $p \geq 1$, $T(\sigma_0) = \infty$.*

$\triangleright$ Plain. Below the threshold the wound truly closes, in a finite expected time — and the closer the noise comes to being multiplicative ($p \rightarrow 1$), the longer that takes, until at $p = 1$ the time becomes infinite and closure recedes to a horizon never reached.

31 Numerical witness

Euler–Maruyama with absorption at $\sigma \leq 0$; $r = 1$, $\kappa = 1.5$, $\varsigma_0 = 0.6$, $\sigma_0 = 0.30$, $dt = 10^{-4}$, $T = 6$, $N = 2000$ paths.

p	reach $\sigma = 0$	mean cleansing time	classification
0.25	100%	0.76	accessible + absorbing
0.50	100%	1.01	accessible + absorbing
0.75	99.7%	2.01	accessible + absorbing
1.00	0%	∞ (none reach)	natural
1.25	0%	∞ (none reach)	natural

The completion probability collapses from 1 to 0 across $p_c = 1$, and the mean time grows ($0.76 \rightarrow 1.01 \rightarrow 2.01$) as $p \uparrow 1$, witnessing both Theorem 30.1 and the divergence of Proposition 30.3.

32 Recovery and limits

The recovery limit $\varsigma_0 \rightarrow 0$ of v0.28 is unaffected (it removes the noise entirely). The present classification is a statement *within* the gated class $p > 0$; the gate itself ($p > 0$) is the standing hypothesis carried from the noise-selection principle of v0.28, since $p = 0$ (additive) already breaks the floor and lies outside the class being classified here. The boundary constants depend on $\beta = 2r/(\varsigma_0^2 \kappa)$ only through the scale exponent; the *classification* (the location $p_c = 1$) is independent of r, κ, ς_0 — it is a pure property of the noise gate's order of vanishing.

Interpretive reading

▷ **Modal (interpretive).** The threshold $p_c = 1$ is the dividing line between two scriptural images of completion that the tradition holds together: the sea that is never full, Ecclesiastes' unending approach ($p \geq 1$, natural), and *consummatum est*, the finished work as a real event ($0 < p < 1$, absorbing). The exact knife-edge is *multiplicative* contingency, $p = 1$: noise exactly proportional to remaining resistance. Read modally, p measures how the hidden time of grace tapers as the wound closes — if it tapers no faster than the wound itself ($p \geq 1$), closure is an eternal nearing; if it tapers faster ($p < 1$), closure arrives. The model does not legislate which; it shows the fork and its single hinge.

Status. Scale/speed (§29) — PROVED. Boundary classification and critical exponent Theorem 30.1 (§30) — PROVED (classical Feller test; leading-order local analysis, robust to the smooth tanh corrections). Mean-time Proposition 30.3 — PROVED (finiteness $\Leftrightarrow$ accessibility) / WITNESS for the quantitative divergence. Numerical table (§31) — WITNESS. Choice of p — CONSTITUTIVE. Interpretive reading — INTERPRETIVE. *Open (to WP3)*: coupling the contingency into the synergy channel u across Λ_c , giving a genuine two-well metastable landscape and the Kramers/committor analysis.

V.F.S. 3.0 — Stochastic Interior (appended layer, WP3)

v0.30 — The Metastable Threshold: Neck, Saddle, and the Price of Crossing

Contingency in the synergy channel; the two-basin landscape and the triple identity at $u = \Lambda_c$

Continuation of v0.29

Chapters v0.28–v0.29 confined contingency to the resistance channel. Here we let it act in the *synergy* channel as well, so that u fluctuates across the threshold Λ_c . The deterministic flow then has two competing attractors, and the small-noise theory (Freidlin–Wentzell) governs the rare crossings between them. The chapter’s load-bearing result is a *triple identity*: the metastable saddle, the deterministic neck, and the maximum of *Dolorosum* are one and the same point, $u = \Lambda_c$. As a 2+1-dimensional small-noise statement the exponential rate is rigorous; the Eyring prefactor is asymptotic. Status is therefore MODERATE, marked per item.

30 The coupled two-basin system

With synergy feedback $u^*(\sigma) = u_{\text{hi}} - (u_{\text{hi}} - u_{\text{lo}}) s(\sigma)$, $s(\sigma) = \sigma^2/(\sigma^2 + \sigma_*^2)$, and a soft resistance ceiling σ_{max} :

$$\begin{aligned} d\sigma &= (\gamma - \delta u) \tanh(\kappa\sigma) (1 - \sigma/\sigma_{\text{max}}) dt + \varsigma (\tanh \kappa\sigma) dW_1, \\ du &= -\lambda_u(u - u^*(\sigma)) dt + \varsigma dW_2, \end{aligned}$$

the Itô convention as before, $\varsigma \downarrow 0$ the small-noise parameter. The feedback is the modelling content: high resistance suppresses synergy ($u^* \rightarrow u_{\text{lo}} < \Lambda_c$), low resistance releases it ($u^* \rightarrow u_{\text{hi}} > \Lambda_c$) — a positive loop producing bistability.

▷ Plain. Two self-consistent ways to be. Either resistance is low, synergy stays above threshold, and the low state sustains itself — the *cleansed* basin. Or resistance is high, synergy is choked below threshold, and the high state sustains itself — the *wounded* basin. Between them lies a ridge the system can cross only by a lucky run of contingency.

31 Fixed-point structure and the triple identity

Proposition 31.1 (Three states, saddle pinned to the threshold). *The deterministic system has exactly three equilibria: a stable cleansed node $(\sigma, u) = (0, u_{\text{hi}})$ with $u_{\text{hi}} > \Lambda_c$; a stable wounded node $(\sigma_{\text{max}}, u^*(\sigma_{\text{max}}))$ with $u^*(\sigma_{\text{max}}) < \Lambda_c$; and an index-one saddle whose synergy coordinate is exactly the threshold,*

$$u_{\text{saddle}} = \Lambda_c.$$

Proof. $\dot{\sigma} = 0$ forces $\tanh(\kappa\sigma) = 0$ or $\gamma - \delta u = 0$, i.e. $\sigma = 0$ or $u = \Lambda_c$. The branch $\sigma = 0$ gives, with $s(0) = 0$, $u^* = u_{\text{hi}}$, hence the cleansed node. The nontrivial branch $u = \Lambda_c$ gives, from $\dot{u} = 0$, $u^*(\sigma) = \Lambda_c$, fixing $\sigma_{\text{saddle}} = \sigma_* \sqrt{s^*/(1-s^*)}$ with $s^* = (u_{\text{hi}} - \Lambda_c)/(u_{\text{hi}} - u_{\text{lo}})$. The Jacobian there has eigenvalues of opposite sign (one transverse-stable, one threshold-unstable), so it is an index-one saddle. The ceiling closes the wounded node at σ_{max} . *The threshold Λ_c is the deterministic separatrix; the saddle therefore lies on it by construction, not by tuning.* □

Theorem 31.2 (Triple identity at $u = \Lambda_c$). *The following three loci coincide:*

$$\text{metastable saddle (transition state)} = \{u = \Lambda_c\} = \arg \max_u \rho_{\text{DoI}}$$

(transition state = deterministic neck / regime switch = maximum of *Dolorosum*).

Proof. The saddle has $u = \Lambda_c$ by Proposition 31.1. The deterministic neck is the bottleneck of the σ -flow where the cleansing rate $r = \delta u - \gamma$ changes sign — precisely $u = \Lambda_c$ (the fold locus of the Cathartic Neck of the core). And *Dolorosum* $\rho_{\text{Dol}} = \sigma(u/\Lambda_c) e^{1-u/\Lambda_c}$ satisfies $\partial_u \rho_{\text{Dol}} = \sigma e^{1-u/\Lambda_c} (1 - u/\Lambda_c)/\Lambda_c = 0$ iff $u = \Lambda_c$, a maximum. All three are the single point $u = \Lambda_c$. $\square$

▷ **Modal (interpretive).** Suffering is greatest exactly at the transition state. The point of maximal *Dolorosum* is not somewhere along the road but precisely on the ridge between wound and cleansing — the cross is the saddle. One does not pass from one basin to the other except over the summit of pain, and that summit is the threshold itself.

32 The price of crossing: Kramers law

Proposition 32.1 (Arrhenius rate of crossing). *In the small-noise limit the mean basin-to-basin transition time obeys the Freidlin–Wentzell law*

$$\mathbb{E}[\tau] \asymp A \exp\left(\frac{\Delta\Phi}{\varsigma^2}\right), \quad \Delta\Phi = \text{quasipotential height at the saddle},$$

with the exponential rate $\Delta\Phi$ rigorous (large deviations) and the prefactor A of Eyring–Kramers type. The witness below gives $\partial_{(1/\varsigma^2)} \ln \mathbb{E}[\tau] \approx 0.72 > 0$, confirming the law and measuring $\Delta\Phi \approx 0.72$ for the active direction.

Proposition 32.2 (Gated asymmetry — the stochastic shadow of floor protection). *The two basins are not symmetric under contingency. Because the resistance noise is gated, $\varsigma(\tanh \kappa \sigma) \rightarrow 0$ as $\sigma \rightarrow 0$, the diffusion degenerates at the cleansed node, so the Freidlin–Wentzell action to leave the cleansed basin along the σ -direction diverges:*

$$\text{rate}_{\text{cleansed} \rightarrow \text{wounded}} \xrightarrow{\varsigma \rightarrow 0} 0 \quad \text{faster than any } e^{-c/\varsigma^2}, \quad \text{rate}_{\text{wounded} \rightarrow \text{cleansed}} > 0.$$

Hence, to leading order, only conversion (*wounded*→*cleansed*) proceeds at a finite Kramers rate; relapse from genuine cleansing is gate-suppressed. This is the metastable image of the floor-protection theorem of v0.28.

▷ Plain. The very silence of contingency at zero resistance — the gate that keeps the cleansed floor sealed — also makes it almost impossible to fall back out of true cleansing. Conversion is always available at a finite rate; collapse from the finished state is not. The protection of the floor and the perseverance of the cleansed are one mechanism seen twice.

Remark 32.3 (Committor preview — perseverance). The committor $q(\sigma, u) = \Pr[\text{reach cleansed before wounded}]$ solves $\mathcal{L}q = 0$ with $q = 1$ on cleansed, $q = 0$ on wounded; it is the probability of *perseverance*. The witness gives $q \approx 0.67$ at the saddle and $q \rightarrow 0.97, 0.25$ toward the two basins — the saddle is the perseverance watershed. The full committor and its reaction-coordinate role are WP4 (v0.31).

33 Numerical witness

Parameters $\gamma = \delta = 1$ ($\Lambda_c = 1$), $\kappa = 1.5$, $u_{\text{hi}} = 1.6$, $u_{\text{lo}} = 0.4$, $\sigma_* = 0.5$, $\lambda_u = 1$, $\sigma_{\text{max}} = 2$. Euler–Maruyama.

deterministic equilibria	(σ, u)	Jacobian eigenvalues	ς	$1/\varsigma^2$	$\ln \mathbb{E}[\tau]$
cleansed (stable)	(0, 1.60)	(−1.00, −0.90)	0.60	2.78	2.83
wounded (stable)	(2.00, 0.47)	(−1.00, −0.26)	0.75	1.78	2.25
saddle	(0.50, 1.00 = Λ_c)	(+0.41, −1.41)	0.90	1.23	1.79
			1.05	0.91	1.46

Left: the saddle sits at $u = \Lambda_c$ (Theorem 31.2). Right (wounded→cleansed): $\ln \mathbb{E}[\tau]$ is linear in $1/\varsigma^2$, slope ≈ 0.72 (Proposition 32.1). Committor at the saddle ≈ 0.67 (Remark 32.3). The reverse direction produced no transitions in the same budget, witnessing the gated asymmetry (Proposition 32.2).

34 Recovery and scope

As $\varsigma \rightarrow 0$ the two basins become the deterministic attractors of §30 with no inter-basin transitions, and the saddle reduces to the deterministic separatrix — the frozen threshold Λ_c of the 2.0/3.0 core. The feedback law $u^*(\sigma)$ is the modelling input of this chapter; the *location* $u = \Lambda_c$ of the saddle/neck/*Dolorosum* is independent of its detailed shape, depending only on $\dot{\sigma} = 0 \Rightarrow u = \Lambda_c$. The full 2D quasipotential and the Eyring prefactor are asymptotic ($\varsigma \rightarrow 0$) statements.

Interpretive reading

▷ **Modal (interpretive)**. The threshold is the cross: maximal *Dolorosum*, the deterministic neck, and the transition state are one point. Crossing from wound to cleansing has a definite price — the quasipotential barrier $\Delta\Phi$ — paid as a rare, costly passage over the summit of suffering, and contingency (the hidden time of grace) is exactly what makes the crossing possible at a finite rate. Yet the passage is one-way in the small-noise limit: conversion is always available, but from true cleansing there is no accidental fall, because the same gate that seals the floor forbids it. The protection of the cleansed and the costliness of conversion are the two faces of a single threshold.

Status. Coupled system (§30) — DEFINITION (constitutive feedback). Fixed-point structure and saddle = Λ_c (Prop. 31.1), triple identity (Thm. 31.2) — PROVED. Kramers exponential rate (Prop. 32.1) — PROVED (Freidlin–Wentzell large deviations); Eyring prefactor — HEURISTIC / ASYMPTOTIC. Gated asymmetry (Prop. 32.2) — PROVED (diffusion degeneracy at $\sigma = 0 \Rightarrow$ divergent escape action) / WITNESS (no reverse transitions). Committor (Rmk. 32.3) — WITNESS; full treatment OPEN to WP4. Numerical tables (§33) — WITNESS. Interpretive reading — INTERPRETIVE.

V.F.S. 3.0 — Stochastic Interior (appended layer, WP4)

v0.31 — The Committor: Perseverance as a Harmonic Coordinate

The probability of reaching cleansing before relapse, and its identity with the scale function

Continuation of v0.30

Chapter v0.30 located the perseverance watershed at the saddle and previewed the committor. Here we solve it. The committor is the probability of *perseverantia*: starting from a given interior state, the chance of reaching cleansing before relapsing to the wound. Its central property is an identity with the scale function of v0.29, which unifies the three chapters: *the same object that classified the floor (v0.29) and pinned the saddle (v0.30) is, normalized, the perseverance probability.*

31 The committor boundary-value problem

Let $\mathcal{C}$ (cleansed) and $\mathcal{W}$ (wounded) be the two metastable sets. The committor

$$q(x) = \Pr[\text{the path from } x \text{ hits } \mathcal{C} \text{ before } \mathcal{W}]$$

is the harmonic function of the generator with absorbing data:

$$\boxed{\mathcal{L}q = 0 \text{ on the live domain,} \quad q|_{\mathcal{W}} = 0, \quad q|_{\mathcal{C}} = 1.}$$

▷ Plain. q answers one question at every point: if you are here, how likely are you to finish the crossing into cleansing before falling back into the wound? $q = 0$ in the wound, $q = 1$ in cleansing, and in between it measures exactly how far along the way home one already is.

32 Committor as the normalized scale function

On the Fenichel slow manifold the reactive dynamics reduces to a one-dimensional reversible diffusion along the reaction coordinate x (the threshold-crossing direction, $x = \Lambda_c$ the saddle), $dx = b(x) dt + \varsigma_n(x) dW$. There the committor BVP has a closed solution.

Theorem 32.1 (Perseverance = normalized scale function). *With the scale density $\psi(y) = \exp(-\int 2b/\varsigma_n^2) = s'(y)$ of v0.29 and scale function $S(x) = \int^x \psi$, the committor between $\mathcal{W} = \{x_w\}$ and $\mathcal{C} = \{x_c\}$ is*

$$\boxed{q(x) = \frac{S(x) - S(x_w)}{S(x_c) - S(x_w)} = \frac{\int_{x_w}^x \psi(y) dy}{\int_{x_w}^{x_c} \psi(y) dy}.$$

Proof. $\mathcal{L}q = bq' + \frac{1}{2}\varsigma_n^2 q'' = 0$ is first order in q' : $q''/q' = -2b/\varsigma_n^2$, so $q'(x) = C \exp(-\int 2b/\varsigma_n^2) = C\psi(x)$. Integrating and imposing $q(x_w) = 0$, $q(x_c) = 1$ fixes C and gives the stated ratio. $\psi = s'$ is exactly the scale density of v0.29. $\square$

▷ Math. The committor is the scale function in disguise. The diffusion's own “natural ruler” S , which in v0.29 decided whether the floor is reachable, here — normalized to $[0, 1]$ between the two states — is the perseverance probability. One object, three roles: boundary classification, saddle location, perseverance.

Corollary 32.2 (The saddle is the $q = \frac{1}{2}$ watershed; sharp layer). *In the reversible reduction $b = -U'$, $\varsigma_n^2 = \varsigma^2$, one has $\psi(y) = e^{2U(y)/\varsigma^2}$, so q is dominated by the barrier maximum $U(x_s)$ at the saddle $x_s = \Lambda_c$. As $\varsigma \rightarrow 0$, q tends to the indicator of the cleansed basin, with a transition layer of width $O(\varsigma)$ centred on the saddle; for symmetric wells $q(\Lambda_c) = \frac{1}{2}$ exactly. Thus the isocommittor surface $\{q = \frac{1}{2}\}$ passes through the saddle — the perseverance watershed of v0.30.*

33 The committor as reaction coordinate; the reactive current

Proposition 33.1 (Optimal reaction coordinate and reactive flux). *The committor is the ideal reaction coordinate of transition-path theory: the ensemble of reactive trajectories crosses the isocommittor foliation $\{q = \text{const}\}$ with probability current proportional to ∇q weighted by the equilibrium density $\pi \propto e^{-2U/\varsigma^2}$,*

$$J_{\text{react}} \propto \pi \varsigma^2 \nabla q,$$

and the reactive rate is the Dirichlet form of q ,

$$k_{\mathcal{W} \rightarrow \mathcal{C}} \propto \frac{1}{2} \varsigma^2 \int \pi |\nabla q|^2,$$

which, evaluated on Theorem 32.1, reproduces the Kramers rate $k \asymp A^{-1} e^{-\Delta\Phi/\varsigma^2}$ of v0.30. Closing the loop: committor $\Rightarrow$ rate.

▷ Plain. All the genuine “crossing” happens in the thin shell around the saddle where q swings from 0 to 1; everywhere else the trajectory merely waits. The committor names the progress variable, the reactive current flows up its gradient, and integrating that current returns precisely the costly Kramers rate already found.

Remark 33.2 (Gated approach to certainty). Near the cleansed state $\sigma = 0$ the scale density $\psi = s'$ has the gated behaviour classified in v0.29 ($\psi \rightarrow 1$ for $p < 1$, $\psi \rightarrow 0$ for $p \geq 1$). Hence the manner in which $q \rightarrow 1$ as cleansing is approached inherits the cleansing-regime dichotomy: perseverance becomes *certain* smoothly in the epektatic regime and is *sealed* in the absorbing regime. The gated asymmetry of v0.30 reappears as $q \equiv 1$ being an absorbing certainty, never undone.

34 Numerical witness

Reaction-coordinate reduction $dx = -U'(x)dt + \varsigma dW$, $U(x) = (x^2 - 1)^2$, wells $x_w = -1$, $x_c = +1$, saddle $x_s = 0$; closed-form (Theorem 32.1) vs. Monte-Carlo committor ($\varsigma = 0.5$):

x	−0.8	−0.4	0.0	0.4	0.8	ς	layer width $\sim (dq/dx _0)^{-1}$
q_{closed}	0.000	0.018	0.500	0.982	1.000	0.60	0.556
q_{MC}	0.000	0.021	0.517	0.985	1.000	0.40	0.360
						0.25	0.223

Closed form and Monte-Carlo agree within sampling error (Theorem 32.1); $q(x_s) = \frac{1}{2}$ (Corollary 32.2); the transition layer narrows $\propto \varsigma$, confirming the $O(\varsigma)$ scaling.

35 Recovery and scope

As $\varsigma \rightarrow 0$ the committor tends to the indicator function of the cleansed basin of attraction: perseverance becomes deterministic basin membership, recovering the frozen partition of state space

by the separatrix Λ_c . The closed form (Theorem 32.1) is exact for the one-dimensional reaction coordinate on the slow manifold; the full planar committor solves $\mathcal{L}q = 0$ on the (σ, u) domain, with the saddle a critical point of q and $\{q = \frac{1}{2}\}$ asymptotically through it — the exact two-dimensional surface is left to numerics.

Interpretive reading

▷ **Modal (interpretive)**. Perseverance is a coordinate, not a verdict. At every interior state q measures how much of the crossing is already secured — the probability of reaching cleansing before the wound reclaims one. It is $\frac{1}{2}$ exactly on the ridge: at the summit of *Dolorosum* the outcome hangs evenly, and the smallest contingency decides. Below the ridge perseverance is nearly lost, above it nearly won, with only a thin band of genuine uncertainty whose width is set by the magnitude of contingency itself. And once cleansing is reached, $q \equiv 1$ is absorbing: the secured crossing is not undone — the same seal that protected the floor.

Status. Committor BVP (§31) — DEFINITION. Identity with the scale function Theorem 32.1 (§32) — PROVED (exact, 1D reduction). Saddle= $q=\frac{1}{2}$ and $O(\varsigma)$ layer Corollary 32.2 — PROVED (asymptotic). Reaction-coordinate / reactive-flux and rate identity Proposition 33.1 — PROVED (transition-path theory) / connects to v0.30. Gated approach Remark 33.2 — PROVED (inherits v0.29). Numerical tables (§34) — WITNESS. Interpretive reading — INTERPRETIVE. *Open*: the exact planar committor on (σ, u) (numerical); *to WP5*: the spectral reading — $\lambda_1 \geq C_1$ and the 0.255/1.5 gap — on the reversible slow manifold.

V.F.S. 3.0 — Stochastic Interior (appended layer, WP5)

v0.32 — The Spectral Gap: True Tempo of Cleansing

$\lambda_1 \geq C_1$, the resolution of the 0.255 vs 1.5 discrepancy, and the gap–continuum dichotomy

Continuation of v0.31 — and close of Phase I

This chapter reads the cleansing dynamics through the *spectrum* of its generator, completing the probabilistic–spectral interior. On the Fenichel slow manifold the reactive dynamics is reversible, so its generator is self-adjoint and has a real spectral gap λ_1 , the true asymptotic rate of relaxation toward the cleansed state. We prove $\lambda_1 \geq C_1$ for any Lyapunov rate C_1 , identify the empirical decay envelope of the lyapunov chapter with λ_1 , and so resolve the standing 0.255 vs 1.5 discrepancy. The reversible reading is the rigorous *headline*; the full non-self-adjoint generator is left as a marked OPEN problem (Ruelle–Pollicott resonances).

32 Reversibility and the Witten Laplacian

On the slow manifold the reduced generator is $\mathcal{L} = -U'\partial_x + \frac{\zeta^2}{2}\partial_x^2$, self-adjoint on $L^2(\pi)$ with $\pi \propto e^{-2U/\zeta^2}$. The ground-state conjugation $f \mapsto \sqrt{\pi}f$ maps $-\mathcal{L}$ to the Witten Schrödinger operator

$$H = -\frac{\zeta^2}{2}\partial_x^2 + V_W, \quad V_W = \frac{|U'|^2}{2\zeta^2} - \frac{U''}{2},$$

whose spectrum is that of $-\mathcal{L}$: $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots$, the gap $\lambda_1 := \mu_1$.

▷ Plain. The relaxation operator and a quantum particle in a well are the same object in two guises. The lowest non-trivial energy of that particle is the rate at which the soul actually settles toward cleansing.

33 The gap dominates every Lyapunov rate

Theorem 33.1 ($\lambda_1 \geq C_1$). *Let $\mathcal{L}$ be the reversible slow-manifold generator with spectral gap λ_1 , and let $\Psi \geq 0$ be any Lyapunov function with $\mathcal{L}\Psi \leq -C_1\Psi$ on the active band. Then*

$$C_1 \leq \lambda_1.$$

Proof. The gap is the sharp rate of the reversible semigroup: for mean-zero f , $\text{Var}_\pi(e^{t\mathcal{L}}f) \leq e^{-2\lambda_1 t} \text{Var}_\pi(f)$, and λ_1 is the largest constant for which this holds (Rayleigh quotient $\lambda_1 = \inf_{f \perp 1} \mathcal{E}(f, f) / \text{Var}_\pi(f)$, $\mathcal{E}$ the Dirichlet form). A Lyapunov inequality $\mathcal{L}\Psi \leq -C_1\Psi$ certifies $\mathbb{E}[\Psi(X_t)] \leq e^{-C_1 t} \mathbb{E}[\Psi(X_0)]$, a guaranteed exponential rate C_1 . Since no observable can relax faster than the sharp rate λ_1 , any certified rate satisfies $C_1 \leq \lambda_1$, with equality iff Ψ aligns with the slowest eigenmode. $\square$

Theorem 33.2 (The gap is the linearization rate; resolution of 0.255 vs 1.5). *For the gated cleansing relaxation $d\sigma = -r \tanh(\kappa\sigma) dt + \zeta(\tanh \kappa\sigma) dW$, the spectral gap converges as the noise vanishes to the deterministic linearization rate at the cleansed equilibrium:*

$$\lambda_1 \xrightarrow{\zeta \rightarrow 0} \partial_\sigma[r \tanh(\kappa\sigma)]_{\sigma=0} = r\kappa.$$

With $r\kappa = 1.5$ this is the empirical decay envelope ≈ 1.5 reported in the lyapunov chapter, while the guaranteed Lyapunov constant there, $C_1 = 0.255$, is the worst-case c_Ψ bound. By Theorem 33.1,

$0.255 = C_1 \leq \lambda_1 \approx 1.5$: the discrepancy is not a contradiction but the generic slack between a guaranteed lower bound and the true spectral gap.

▷ **Math.** The lyapunov chapter’s own honesty note — “the analytic rate is conservative” — is exactly this theorem. The conservative 0.255 is a certificate; the sharp 1.5 is the gap, equal to the curvature $r\kappa$ of the cleansing drift at $\sigma = 0$, which the envelope measured all along.

34 The gap–continuum dichotomy; bridge to the ascent axis

Proposition 34.1 (Two spectral types of one operator move). *The slow-manifold operator is of one of two types:*

<i>bounded well</i>	$\Rightarrow$	<i>discrete spectrum with a gap $\lambda_1 > 0$ (relaxational, a fastest mode),</i>
<i>half-line free continuum</i>	$\Rightarrow$	<i>gapless continuous spectrum (smooth, no fastest mode).</i>

The cleansing interior is of the first type (gap = $r\kappa$). The ascent operator of 3.0 §4 is of the second — the gapless free continuum already established there. Both arise from the same ground-state/Witten conjugation applied to two coordinates; the single-soul interior and the ascent axis are one operator move seen twice.

▷ **Modal (interpretive).** The same mathematics that gives cleansing a definite tempo gives ascent none. Settling toward the cleansed state has a fastest rate — a real relaxation time — while the ascent toward God is gapless, an unending approach with no last quantum and no fastest step. Relaxation has a clock; epektasis does not.

35 The non-self-adjoint generator (open)

Off the slow manifold the full interior generator is *not* self-adjoint: the $V \leftrightarrow F$ coupling and the source gate break detailed balance, so the operator is non-normal and its spectrum is genuinely complex. Theorem 33.1 ($\lambda_1 \geq C_1$) is rigorous only in the reversible reduction; for the full generator the relevant objects are *Ruelle–Pollicott resonances* — the complex poles of the resolvent governing decay with oscillation. The sharp statement that the empirical envelope equals the leading resonance’s real part is left **OPEN**; the reversible headline above is what is proved.

36 Numerical witness

Symmetrized generator of $d\sigma = -\tanh(\kappa\sigma) dt + \varsigma dW$, $\kappa = 1.5$, discretized on $[-4, 4]$; gap λ_1 vs noise, and the c_Ψ Lyapunov bound.

ς	λ_1	C_σ	c_Ψ (Lyapunov bound)
0.50	1.219	1.5	0.915
0.30	1.399	2.0	0.643
0.20	1.455	2.5	0.490
0.10	1.489	corpus value $0.255 \leq \lambda_1$	
0^+	$\rightarrow r\kappa = 1.500$		

Left: $\lambda_1 \rightarrow r\kappa = 1.5$ as $\varsigma \rightarrow 0$ (Theorem 33.2), matching the empirical envelope. Right: every c_Ψ bound, and the corpus 0.255, lies below λ_1 (Theorem 33.1); the bound worsens as the band widens, the source of the slack.

37 Recovery and Phase I close

As $\varsigma \rightarrow 0$ the spectral picture reduces to deterministic exponential relaxation at rate $r\kappa$ — the frozen cleansing rate of 2.0/3.0. With this chapter the probabilistic–spectral interior is complete: generator and gated floor (v0.28), critical exponent of cleansing (v0.29), metastable threshold and triple identity (v0.30), committor as perseverance (v0.31), and the spectral gap as true tempo (v0.32). Every layer recovers the frozen 3.0 in the noiseless limit, disturbing nothing in it. Phase II (the relational–operator layer) will lift these single-soul objects into the plural interior of 4.0.

Interpretive reading

▷ **Modal (interpretive)**. There is a guaranteed promise and there is the truth. The Lyapunov certificate promises only that cleansing proceeds at least at rate 0.255; the spectral gap reveals it actually proceeds at 1.5 — grace works faster than any certificate can prove. The gap is the real tempo: the curvature of the cleansing drift at the very point of completion. And the contrast with the ascent axis is the whole shape of the interior life in one line: the wound closes at a definite rate, but the approach to God has no rate at all — a clock for healing, none for glory.

Status. Witten conjugation (§32) — PROVED (standard). $\lambda_1 \geq C_1$ Theorem 33.1 — PROVED (variational, reversible). Gap = $r\kappa$ and resolution of 0.255/1.5 Theorem 33.2 — PROVED (semiclassical limit) / WITNESS (numerics). Dichotomy and ascent bridge Proposition 34.1 — PROVED (spectral type) / INTERPRETIVE (the unification claim). Non-self-adjoint generator (§35) — OPEN (Ruelle–Pollicott resonances). Numerical tables (§36) — WITNESS. Interpretive reading — INTERPRETIVE.